Political incentives and municipal performance: An ex ante evaluation of Brazil's proposed *Lei de Responsabilidade Gerencial*

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Contents and conventions

This document contains the details that the article summarises: the full description of the proposal, the derivations of the model, the data and their checks, the methods behind each estimate, the specification of the simulation and the results that the article reports only in part. Its eleven sections follow the order of the article. Section S1, "The proposal and its codification", supports Section 2 of the article; Section S2, "The analytical model: derivations and propositions", supports Section 3 of the article; Section S3, "Data: sources, checks and replication", supports Section 4.1 of the article and the replication of every result; Section S4, "What the indicators measure", supports Sections 4.2 to 4.4 of the article; Section S5, "Money and politics today: further results", supports Section 5 of the article; Section S6, "Behavioural evidence and calibration", supports Section 6 of the article; Section S7, "The simulation model", supports Section 7 of the article; Section S8, "Additional simulation results", supports Section 8 of the article; Section S9, "Tests of the sanctions, robustness and sensitivity", supports Section 9 of the article; Section S10, "Legal feasibility in detail", supports Section 10 of the article; and Section S11, "Learning whether the law works", supports Section 11.2 of the article. A list of the references cited in this document closes it.

Sections, tables, figures and equations carry the prefix S followed by the number of the section to which they belong, so that Table S8.4 is the fourth table of Section S8 and equation (S7.2) the second equation of Section S7. Sections, tables, figures and equations without the prefix are those of the article, and they are always cited as such (for example, "Table 5 of the article").

The conventions of the simulation results are those of the article. The LRG as proposed is the only policy evaluated, and every other design switches its instruments on and off, or changes its sanctions, to learn where its effects come from. Unless stated otherwise, results are medians over the 400 calibrated draws, with 5th to 95th percentiles in brackets; they are differences from the status quo in the same draw, averaged over 2029 to 2036 and over municipalities, and expressed in standard deviations (SD) of each indicator's true, noise-free value between municipalities of the same state

(Section 7.1 of the article). Families of scenarios use the first 120 calibrated draws with the same random shocks as the main run.

S1. The proposal and its codification

Section 2 of the article reconstructs the *Lei de Responsabilidade Gerencial* from three sources: the government plan's chapter on the law (Partido Missão, 2026a, pp. 104-121), the condensed plan (2026b, chs. III-IV) and the statements of the party's president and presidential pre-candidate, Renan Santos, in interviews and live broadcasts during the 2026 pre-campaign (Santos, 2026). This section gives the full description of each pillar on which that reconstruction rests (Section S1.1) and the complete codification used in the simulation, with its variants and calendar (Section S1.2).

S1.1. The five pillars and the reform of the federation

The diagnosis behind the proposal has a fiscal side that the article mentions only briefly. According to the condensed plan, most municipalities cannot pay for their own administration, almost 90% of them spend more on payroll than they collect in own taxes, and, by the plan's count, the R$ 196 billion distributed through the FPM in the last year exceeded federal spending on Bolsa Família (2026b, ch. III). Together with the patrimonial diagnosis described in Section 2.1 of the article, this fiscal picture explains why the proposal combines rules on performance with a reform of the federation.

The first pillar, measurement and information, provides the grades on which everything else depends. A federal agency (an *autarquia*), the Comissariado Federal de Gestão Pública, linked to the Ministry of Finance with technical autonomy and fixed mandates, would run the Marco Gerencial Nacional, a system of floors common to all municipalities of a population group ("metas-patamar") and of individual improvement paths over the mandate ("metas-trajetória"). The government plan's own example uses IDEB, personnel spending, child vaccination and timely answers to information requests, whereas the public statements announce the ten headline indicators listed in Section 2.2 of the article. Economic activity would carry the most weight, results would be public on dashboards, and the best performers would receive presidential awards.

The second pillar turns the grades into sanctions. Each year a municipality would receive a colour. Yellow, assigned below a floor or off the path, triggers a public alert and a recovery plan within 90 days. Orange triggers managerial tutelage, in which voluntary federal transfers need the commissioner's co-signature and contracts, credit, new posts and new works are subject to the commissioner's veto. Red, assigned far below the floor without recovery, leads the commissioner to file for the mayor's supervening ineligibility (*inelegibilidade superveniente*) for eight years, a new ground for ineligibility in the Ineligibility Law (LC 64/1990), and grave and intentional failure can lead to removal by the state court. The public statements add a simpler rule, by which a mayor who "did not

improve anything" over the mandate becomes ineligible, and so does a mayor who obstructs the commissioner. An anti-mafia clause modelled on Italy, finally, would let the Superior Court of Justice dissolve administrations captured by organised crime.

The third pillar ties federal money to the same grades. Municipalities that improve their indicators would receive more federal money, and parliamentary amendments could only reach mayors who are "working well". Amendments would also have to fund "policies that work"; the proposal's own example is a regional sanitation consortium in place of a square named after a relative. The fourth pillar addresses capacity. Every municipality that depends on federal money would receive a federal commissioner, who would both advise and supervise in real time, bring ready-made policies, such as a literacy programme, an intermunicipal sanitation consortium or security cameras, on the premise that small municipalities lack the capacity to design them, and take over the functions of a failing mayor.

The fifth pillar extends the logic of results to the parties. The Fundo Partidário, today distributed according to votes for the Chamber of Deputies, would be distributed only according to the performance of each party's mayors, weighted by the population they govern, and half of the electoral fund (FEFC) would follow the same rule. Parties with mayors in the red band or removed from office would lose funds, and a "circuit breaker" would put a party in compulsory recovery if more than a quarter of its mayors were removed in a cycle. The stated purpose is to make parties compete for good managers instead of celebrity candidates.

Alongside the five pillars, the condensed plan proposes a reform of the federation. Under the heading of a "Great Municipal Consolidation", it cites a study that would reduce the number of municipalities from 5,570 to about 1,656, and it adopts as a viability test the criterion of the 2019 constitutional amendment proposal PEC 188: own taxes of at least 10% of total revenue. The public statements add unpaid councillors in municipalities below 60,000 to 70,000 inhabitants, freedom to reallocate the minimum shares of revenue that the Constitution reserves for education and health, federal intervention in municipalities that do not improve and, for states that do not become viable within eight years, a return to the status of federal territory. Section S5.4 measures the fiscal base on which this reform rests, and Section S10 the constitutional obstacles it would meet.

S1.2. The codification used in the simulation

A proposal stated in plans and interviews leaves several values open: the floors, the pace of the paths, the way indicators are combined, the size of the performance money and what exactly "did not improve anything" means. The simulation therefore fixes a baseline codification that follows the text where it is explicit and, where it is not, takes the reading most favourable to the proposal's aims, and each choice is then varied (Table S1.1). Three choices matter most, and their reasons deserve to be stated. The simulated scorecard uses the six indicators that can be measured (Section 4.2 of the article). The money pillar is modelled as a tranche of 10% of the FPM redistributed in proportion to each

municipality's points and its own FPM, which keeps the performance rule from also changing the equalising structure of the FPM; the variant distributes the tranche by points and population, the weighting that the proposal uses for party funds. And municipalities in tutelage keep their amendments but receive them through regional consortia, an instrument the proposal favours, whereas the variant redirects them to qualified municipalities, which is the literal reading of amendments reserved for mayors who are "working well".

Table S1.1. How the proposal is codified in the simulation.

Element	Baseline codification ("as proposed")	Variants
Indicators scored	Six measurable headline indicators: private formal jobs per 100 inhabitants, in logs (K1); share of value added outside public administration (K2); share of second graders literate (K3); enrolment aged 0 to 5 per 100 children (K4); share of the population with sewage collection (K6); homicides per 100,000 (K7). Publication lags of one year (two for K2)	None
Floors and paths	Floor: national 20th percentile by population group (up to 20k, 20k to 100k, above 100k), moving with the national trend; path: keep the national pace above the floor, reach the floor in four years below it	No tolerance (literal)
Bands and stages	Yellow: below a floor or off a path (tolerance of one standard deviation of the noise of a change); orange: two floors missed, or a floor missed and off-path; tutelage: orange, or yellow twice; red: one SD below a floor and off-path in two consecutive readings; ban: red while in tutelage	Red at 1.5 SD
"Did not improve anything"	At the last evaluation of the mandate, no indicator improved by more than its tolerance: ban	Rule removed
Points	Weighted improvement over the mandate baseline in SD units (economy 40%, the other four 15% each), capped at ±2 SD	None
Federal money	10% of the FPM paid back in proportion to points × own FPM	Points × population
Amendments	R$ 32 billion a year reach municipalities; 70% tied to approved programmes; municipalities in tutelage receive theirs through regional consortia	Redirected to qualified municipalities
Commissioners	One per municipality, all deployed in 2028	1,500 hired a year
Party funds	Fundo Partidário by mayors' points × population; half of the FEFC by the same rule	By colour bands
Sanctions	Alerts, tutelage (co-signature, supervision, extra support, slower execution) and eight-year bans for results, as above	No bans for results; no performance sanctions; bans threatened but never executed; bans executed but not anticipated; weight of a ban from ×0 to ×8
All five instruments	Switched on together	Each switched on and off: the 32 combinations

Notes: K5 (favelas) and K10 (waiting times) have no annual municipal series, K8 (robberies) has one for only ten states, and K9 (care for aggression against women) produces counts too small to be reliable at municipal level (Table 2 of the article); none of them is simulated. SD: standard deviation of the indicator's true value between municipalities of the same state (Section 7.1 of the article). The rules are implemented as described in Section S7.1.

The codification also needs a calendar. The simulation assumes that the law is enacted in 2027. The first scorecard grades 2027 data early in 2028; alerts, tutelage, the first FPM tranche, the conditions on amendments and the new party-fund rule apply from 2028, when commissioners are also deployed; the first bans can occur from 2029, after two consecutive red readings; and municipal elections take place in 2028, 2032 and 2036. Effects are therefore reported over the two full mandates from 2029 to 2036, the period in which every instrument is in force.

S2. The analytical model: derivations and propositions

Section 3 of the article presents the model that organises the evaluation in a compact form, through equations (1) to (11) of the article: how outcomes and their measurement are generated, what the mayor values, the marginal expected incentive, the effort rule and the implementation of programmes. This section develops two parts of that argument that the article only summarises. Section S2.1 derives the effort rule and what it implies for the strength of incentives, by comparing two benchmark mayors and by recalling two results of contract theory. Section S2.2 states the ten propositions of Section 3.5 of the article and proves each of them from the model; where a proposition rests on evidence more than on the model, an argument takes the place of the proof. The equations of this section are numbered (S2.1) to (S2.23). The derivations of Section S2.1 and the mathematical steps of the proofs of Section S2.2 are also written in Lean 4, a language in which a computer checks every step of a proof. The folder lean/ of the replication package holds them, with a table that links each equation and proposition to its formal statement and the record of the build that checked them (Section S3.3).

S2.1. The effort rule, two benchmark mayors and the strength of incentives

The terms of equation (3) of the article deserve a more detailed reading before the comparison. Re-election responds to measured performance with a slope β and to clientelist mobilisation with a slope $\gamma v_i D_i$, which grows with the vulnerability of voters v_i and with the discretionary money D_i that the mayor controls. Each term of the marginal expected incentive (equation 4 of the article) then varies across municipalities in its own way: the money term grows with the share of revenue at stake, the party term grows with the municipality's weight in the party's score, and the electoral term is small wherever voters reward performance weakly, which Section 6.2 of the article shows to be the Brazilian case. The sanction term is the most uneven of all. Consider two municipalities subject to the same law: in the first the reported value sits near a floor and the indicator is measured precisely, so the sanction term is large; in the second the value is far from any floor or the indicator is noisy, so the same sanction creates almost no incentive.

The effort rule follows from equating the marginal cost of effort, equation (6) of the article, with the marginal expected incentive I_{ik}. Multiplying both sides by $1 - a/(4\delta)$ gives

$$\frac{a_{ik}^{*}}{\delta}\left(1 + \frac{I_{ik}}{4}\right) = I_{ik} \tag{S2.1}$$

whose solution is the effort rule, equation (7) of the article. The rule has a simple reading. For small incentives, effort is small relative to the ceiling 4δ and its cost is approximately quadratic,

$$C(a) \approx \frac{a^{2}}{2\delta} \tag{S2.2}$$

and effort is proportional to the incentive,

$$a_{ik}^{*} \approx \delta I_{ik} \tag{S2.3}$$

which is the textbook result in which the gain in the indicator equals $\alpha^{2}b/k$ for a reward b per unit, a productivity of effort α and a cost parameter $k = 1/\delta$. As the incentive grows, effort approaches its ceiling 4δ and each additional real of reward buys less.

The effort rule already allows a comparison that clarifies what performance pay can and cannot do. Consider a mayor paid a fixed salary and another who also receives a bonus b for each standard deviation of improvement in a score, so that her marginal expected incentive rises by b. In the linear region of the effort rule, equations (8) and (9) of the article give the gain in true quality and in the reported score:

$$\Delta q_{ik} = c_{i}\left(1 + \varphi\theta_{j}\right)\delta b \tag{S2.4}$$

$$\Delta \hat{y}_{ik} = \Delta q_{ik} + \frac{b}{\kappa_{g}} \tag{S2.5}$$

The cosmetic share of the measured gain is therefore

$$\frac{b/\kappa_{g}}{\Delta \hat{y}_{ik}} = \frac{1}{1 + c_{i}\left(1 + \varphi\theta_{j}\right)\delta\kappa_{g}} \tag{S2.6}$$

which, for a mayor of average competence ($\theta_{j} = 0$), equals $1/(1 + c_{i}\delta\kappa_{g})$. It is highest where capacity is lowest; in other words, weak administrations respond to rewards more on paper than in services.

Two results from contract theory then limit how strong the incentive should be. The first concerns partial measurement. When only some tasks are measured, effort moves towards them; with two tasks that are substitutes in cost, the optimal weight on the measured task is

$$\alpha_1 = \frac{B_1 - B_2 C_{12}/C_{22}}{1 + r\sigma_1^2\left(C_{11} - C_{12}^2/C_{22}\right)} \tag{S2.7}$$

where B_1 and B_2 are the marginal values of the two tasks, C_{11}, C_{12} and C_{22} the second derivatives of the agent's cost, r the agent's risk aversion and σ_1^2 the noise of the measure (Holmström and Milgrom, 1991, equation 7). For a risk-averse agent and a task worth rewarding, the weight falls as the measure becomes noisier. Substitution acts through two terms. It lowers the numerator whenever the unmeasured task has value, so that for a risk-neutral agent the weight falls with C_{12}; it also lowers the risk term of the denominator, so that for a risk-averse agent the net effect depends on the parameters. When an essential task cannot be measured at all, the optimal weight is zero (Holmström and Milgrom, 1991, Proposition 1). The second result concerns distortion. When the measure responds to effort differently from welfare, the optimal piece rate is

$$b^* = \frac{\rho\sigma_V\sigma_P + 1}{\sigma_P^2 + 1} \tag{S2.8}$$

where σ_V and σ_P measure how much the effects of effort on welfare and on the measure vary across circumstances, each around a mean of one, and ρ is their correlation (Baker, 1992, equation 8). The rate is below one whenever $\rho\sigma_V < \sigma_P$, even for a risk-neutral agent; when the two effects are equally dispersed, it is below one unless they are perfectly correlated. Both results point in the same direction: noisy, partial and partly self-reported municipal indicators call for muted, smoothed incentives, the opposite of sharp thresholds.

Career incentives work through the same logic. In Bertrand et al. (2020), a bureaucrat's effort is proportional to how much performance moves the probability of the final promotion, multiplied by the value of that promotion, and officers whose age capped their careers performed worse; removing the cap raised their effectiveness ratings by about 0.3 standard deviations. For a mayor, the prize is another mandate or a higher office. A mayor in her second consecutive mandate, a lame duck, therefore has no electoral prize at stake, and a ban for results imposes the same cap on anyone who crosses the line. The LRG, finally, does not pay mayors personally. It pays municipalities, whose money mayors value at ω_i, and parties, and it acts on careers through the sanctions. Whether a personal bonus would work better is examined as a theoretical benchmark in Section S8.9; such a bonus would need a constitutional amendment, because mayors are paid a single fixed *subsídio* that cannot carry any bonus or prize (art. 39 §4).

S2.2. The ten propositions and their proofs

Each proposition below is stated in its heading and followed by its proof, which uses the equations of the article and those of Section S2.1, and by a note on where the article tests it. Two definitions are used throughout. First, a sanction follows when the reported value $\hat{y}_{ik}$ falls below a threshold τ_k; with normal noise of standard deviation σ_{ik}, its probability is

$$\Pr\left(\hat{y}_{ik} < \tau_k\right) = \Phi(z_{ik}) \tag{S2.9}$$

where z_{ik} is the expected distance below the threshold, in units of noise:

$$z_{ik} = \frac{\tau_k - \mathrm{E}\left[\hat{y}_{ik}\right]}{\sigma_{ik}} \tag{S2.10}$$

Second, with quadratic costs for manipulation and clientelism, clientelist mobilisation is proportional to the vulnerability of voters, to the discretionary money of the mayor and to the share of amendments left untied,

$$x_i^* \propto \gamma v_i (1 - t) D_i \tag{S2.11}$$

where t is the share of amendments tied to programmes.

P1. The sanction lands on places more than on persons

Proof. By equations (1) and (2) of the article, the expected reported value is the sum of the place s_{ik}, the mayor's contribution $\lambda_k\theta_j$ and the effects of effort, money and manipulation. Differentiating (S2.9) shows that the probability of a sanction responds to the place and to the mayor with the same factor $\varphi(z_{ik})/\sigma_{ik}$ per unit of their contribution to the reported value:

$$\frac{\partial \Pr\left(\hat{y}_{ik} < \tau_k\right)}{\partial s_{ik}} = -\frac{\varphi(z_{ik})}{\sigma_{ik}} \tag{S2.12}$$

$$\frac{\partial \Pr\left(\hat{y}_{ik} < \tau_k\right)}{\partial \theta_j} = -\lambda_k \frac{\varphi(z_{ik})}{\sigma_{ik}} \tag{S2.13}$$

Their contributions to who is sanctioned are therefore scaled by their dispersions across municipalities, σ_s for places and $\lambda_k\sigma_\theta$ for mayors. To bound what a sanction reveals about the mayor, write $\hat{y}_{ik} = \lambda_k\theta_j + u_{ik}$, where u_{ik} collects the place, the other terms and the noise, and treat u_{ik} as independent of θ_j. Then

$$\operatorname{corr}\left(\hat{y}_{ik}, \theta_j\right) = \frac{\lambda_k \sigma_\theta}{\sigma_{\hat{y}}} \tag{S2.14}$$

A sanction is a function of $\hat{y}_{ik}$ alone and, for jointly normal variables, no function of $\hat{y}_{ik}$ is more correlated with θ_j than $\hat{y}_{ik}$ itself, so the correlation between a sanction and the mayor's competence is bounded by $\lambda_k\sigma_\theta/\sigma_{\hat{y}}$. When the mayor's component is small relative to the place and to the noise, a sanction therefore carries little information about the mayor. ■

Where the article tests it: Section 4.3 of the article measures these dispersions, and Section 8.1 of the article counts how many bans would remain with an average mayor in office.

P2. Noise dilutes the incentive and misdirects the sanction

Proof. The sanction term of the marginal expected incentive, the last term of equation (4) of the article, is $L_i\varphi(z_{ik})/\sigma_{ik}$. It is largest at the threshold, where $z_{ik} = 0$, and there it equals

$$\frac{L_i \varphi(0)}{\sigma_{ik}} = \frac{L_i}{\sqrt{2\pi}\,\sigma_{ik}} \tag{S2.15}$$

which falls as the noise σ_{ik} rises; away from the threshold it falls further, with $\varphi(z_{ik})$. Small municipalities, where indicators are dominated by noise, are thus less motivated by the sanction. They are also more often sanctioned by mistake. A mayor whose true performance is Δ above the threshold is sanctioned when the noise falls below $-\Delta$, which happens with probability

$$\Pr(\varepsilon_{ik} < -\Delta) = \Phi\left(-\frac{\Delta}{\sigma_{ik}}\right) \tag{S2.16}$$

a probability that rises with σ_{ik} (Kane and Staiger, 2002). Finally, with K tests, each failed with probability p independently of the others, the probability of failing at least one is

$$1 - (1 - p)^K \tag{S2.17}$$

which approaches one as K grows, so a scorecard with several tests flags almost everyone at least once. ■

Where the article tests it: Section 4.3 of the article measures the noise of each indicator by municipal size, and Section 8.1 of the article shows the multiple-testing problem at national scale.

P3. Capacity is a complement, and its absence breeds gaming

Proof. By equations (8) and (9) of the article, an incentive I_{ik} raises true quality by $c_i(1 + \varphi\theta_j)a^*_{ik}$ and the reported number, through manipulation, by I_{ik}/κ_g. The cosmetic share of the measured response is therefore

$$\frac{g^*_{ik}}{\Delta q_{ik} + g^*_{ik}} = \frac{1}{1 + \kappa_g c_i\left(1 + \varphi\theta_j\right) a^*_{ik}/I_{ik}} \tag{S2.18}$$

which falls as capacity c_i rises, because real effort scales with capacity while manipulation does not; in the linear region of the effort rule it reduces to (S2.6). Sanctions on weak administrations thus mostly produce numbers, whereas assistance that raises capacity raises the real response. ■

Evidence and tests: In Ceará, technical assistance roughly doubled the effect of a results-based transfer (Lautharte, de Oliveira and Loureiro, 2021; Ponne, 2023). Section 8.2 of the article reports the cosmetic share of the measured gain indicator by indicator.

P4. What the money rewards depends on what it is paid for

Proof. Write the reported value of a municipality in two consecutive years as the place plus the mayor plus noise that is independent across years. A transfer paid on levels pays in proportion to $\hat{y}_{it}$, whose variation across municipalities is dominated by the place, so it rewards the place. A transfer paid on raw changes pays in proportion to $\Delta\hat{y}_{it} = \hat{y}_{it} - \hat{y}_{i,t-1}$, which contains the previous year's noise with a negative sign:

$$\operatorname{Cov}\left(\Delta\hat{y}_{it}, \varepsilon_{i,t-1}\right) = -\sigma^2_{\varepsilon} \tag{S2.19}$$

A municipality with bad luck in one year is therefore paid in the next for the rebound that follows (Chay, McEwan and Urquiola, 2005). Only a transfer paid on improvement against an expected path, shrunk by its reliability, rewards effort. A further consequence follows when money is worth more where coverage is lower, that is, when the value of money is concave in coverage. Moving a real from a municipality with low coverage to one with high coverage then adds less where it arrives than it subtracts where it leaves, so moving money towards better-performing places lowers aggregate outcomes unless the incentive response compensates. ■

Where the article tests it: Section 8.4 of the article and Section S8.4 compare the tranche paid in proportion to each municipality's own FPM with the variant paid in proportion to population.

P5. Party funds reward parties for owning good mayors more than for making them

Proof. Under the proposed rule, as codified in Section S1.2, the share of the Fundo Partidário of party p is proportional to the positive points of its mayors weighted by the population they govern:

$$F_p \propto \sum_{i \in p} n_i \max(P_i, 0) \tag{S2.20}$$

where n_i is the population and P_i the points of municipality i. On day one, the sum is dominated by the parties that hold the most mayors in the largest municipalities. A party can raise its sum in two ways, by improving the points of its own mayors or by recruiting a mayor whose points are already high. Because Brazilian mayors can change party without losing office, recruitment is the cheaper route. The rule improves selection only if parties can observe competence before elections and if the risk of a noise-driven ban does not deter competent candidates, in the way that better pay attracts abler ones (Dal Bó, Finan and Rossi, 2013; Gagliarducci and Nannicini, 2013). ■

Where the article tests it: Section 5.2 of the article computes the shares on day one, and Section 8.5 of the article follows them, and the competence of mayors, until 2036.

P6. Clientelism responds to money quickly and to emancipation slowly

Proof. By (S2.11), clientelist mobilisation is proportional to $(1 - t)v_i D_i$. Tying a share t of amendments to programmes reduces it at once, in proportion to the discretionary money removed:

$$\frac{\partial x_i^*}{\partial t} \propto -\gamma v_i D_i \tag{S2.21}$$

Reducing vulnerability v_i, by contrast, requires jobs and incomes that mayors influence little and that change slowly (Bobonis et al., 2022). ■

Where the article tests it: Section 8.5 of the article reports the discretionary money and the share of vulnerable households in 2036.

P7. Information matters when it is clear and local

Argument. Publishing scores changes votes only if voters update their beliefs, that is, only if the published score adds to what they already know about the mayor. In the simulation, this addition is the parameter β_{info} of equation (S7.2), the effect of the published score on re-election beyond that of measured performance. The evidence sets it close to zero for routine information: voters did update when audit reports of corruption were released before elections (Ferraz and Finan, 2008), but not, on average, when routine performance information was distributed (Dunning et al., 2019).

Where the article tests it: Section 6.2 of the article estimates how much voters reward measured performance, and Table 6 of the article gives the effect of publishing the scorecard alone.

P8. Conditioning money adds an incentive only at the margin

Proof. Compare a results-based transfer $T(\hat{y})$ with an unconditional transfer of the same expected value. Both raise the mayor's expected utility, equation (3) of the article, by the same amount,

ω_i times their common expected value. Only the first, however, enters the marginal expected incentive, equation (4) of the article, through the term

$$\omega_i \frac{\partial T}{\partial \hat{y}_{ik}} \tag{S2.22}$$

which is the whole difference between them. It is small where the mayor values municipal money little, that is, where ω_i is low, and where the reward is spread over many indicators, since each of them then carries only part of it. ■

Evidence and tests: In Indonesia, conditioning a fifth of a block grant on twelve indicators added 0.04 standard deviations after eighteen months (Olken, Onishi and Wong, 2014). Section 8.4 of the article and Section S8.4 separate the part of the tranche's effect that comes from the condition from the part that comes from the money.

P9. Career sanctions bite only where there is a career to lose, and only near the line

Proof. The electoral term of the marginal expected incentive, $R_i \partial\pi/\partial\hat{y}_{ik}$ in equation (4) of the article, is zero for a mayor who cannot run again ($R_i = 0$, a lame duck; de Janvry, Finan and Sadoulet, 2012) and small wherever voters reward performance weakly. The ban term, $L_i \varphi(z_{ik})/\sigma_{ik}$, reaches its maximum (S2.15) at the threshold and vanishes far from it. The ban term can therefore exceed the electoral term by an order of magnitude for a mayor close to a threshold and be zero for every mayor far from one, while applying to the measured indicator instead of to welfare. ■

Where the article tests it: Section 6.1 of the article bounds the electoral incentive with the close races of 2020, and Section S8.2 reports the incentive created by each source by municipal size.

P10. Sanctions can help implementation, do nothing for it or hurt it

Proof. By equation (10) of the article, implementation quality rises with the total incentive I_{ik} and with supervision S_i when $\eta > 0$, at a rate that falls as $I_{ik} + sS_i$ grows:

$$\frac{\partial \iota_{ik}}{\partial I_{ik}} = \frac{\eta}{I^*}\left(1 - \frac{c_i}{2}\right)\exp\left(-\frac{I_{ik} + s\,S_i}{I^*}\right)e^{-\theta_j/4} \tag{S2.23}$$

The threat of a ban adds to I_{ik}, so it raises cooperation like any other incentive, and the gain is largest where money, party funds, elections and supervision provide little, since (S2.23) is largest when $I_{ik} + sS_i$ is small; where they already provide much, its marginal value is small. A ban that is executed works in the opposite direction. It sets the banned mayor's incentive to zero for the rest of the mandate, which lowers ι_{ik} to $1 - \eta(1 - c_i/2)\exp(-sS_i/I^* - \theta_j/4)$, and it forces a change of government. Which effect dominates depends on η and on the rest of the design. ■

Where the article tests it: Section 9.1 of the article and Section S9.2 separate the threat of a ban from its execution, including when implementation depends on incentives.

S3. Data: sources, checks and replication

Every step of the evaluation rests on a national dataset of the 5,570 municipalities assembled from public sources, which Section 4.1 of the article describes in a single paragraph. This section gives the details: the variables, producers, years and coverage of each source (Section S3.1), the repository from which each file was copied and the check applied to it (Section S3.2), and the replication package that rebuilds every number, table and figure (Section S3.3).

S3.1. Data sources

The dataset has three parts, which correspond to the three kinds of question that the article asks of real data (Table S3.1). The first describes the starting point that the scorecard would grade: population (2022 Census and IBGE estimates), the economy (municipal GDP and value added by sector), formal jobs (RAIS, 2006 to 2024), Bolsa Família coverage (2024 and 2025), sewage and water (2022 Census and the providers' reports to SNIS, 1995 to 2022), literacy in the second grade (INEP, 2023 and 2024), daycare and preschool enrolment (School Census, 2015 to 2025), homicides (Atlas da Violência, 1996 to 2016 and 2022), favelas (2022 Census) and the IDEB used in the government plan's own example (2015 to 2025). The second describes the money and politics on which the law would act: the FPM, computed for every municipality from its legal formula; the 13,177 action plans of the 2025 "transferências especiais" (12,663 of them for municipalities), with their authors and policy areas; current revenue and own taxes for 3,243 municipalities in 2024 (SICONFI); and the mayors elected in 2016, 2020 and 2024, with their parties. The third covers the parties themselves: Chamber seats and votes in 2022, and the 2025 Fundo Partidário and 2024 electoral-fund allocations.

Table S3.1. Data sources.

Variable	Source	Years	Municipalities
Population; age structure; households; sewer and water connection; formality; schooling	IBGE: Censuses (2000, 2010, 2022), 2007 Count, annual estimates	2000 to 2025	5,570
GDP and value added by sector	IBGE, PIB dos Municípios	2010 to 2023 (sectors to 2021)	5,570
Formal jobs by sector	RAIS (Ministry of Labour)	2006 to 2024	5,570
Bolsa Família families	Ministry of Development and Social Assistance	2024, 2025	5,570
Literacy at the end of 2nd grade	INEP, Indicador Criança Alfabetizada	2023, 2024	5,448

Variable	Source	Years	Municipalities
Daycare and preschool enrolment	INEP, School Census	2015 to 2025	5,570
IDEB, early years, municipal networks	INEP	2015 to 2025	5,375
Sewage and water served (self-reported)	SNIS (Ministry of Cities)	1995 to 2022	5,549
Homicide rates	Atlas da Violência (SIM/DATASUS)	1996 to 2016, 2022	5,570
Population in favelas	IBGE, 2022 Census	2022	656 with favelas
Current revenue, own taxes (2024); personnel spending (2020 to 2024)	National Treasury, SICONFI (RREO, RGF)	2020 to 2024	3,243; 3,812
FPM	Computed from the legal formula (DL 1,881/1981; Res. TCU 242/1990)	2024	5,569
Discretionary amendments (transferências especiais)	Transferegov, compiled by Transparência Brasil	2025	4,302 recipients
Mayors, parties and votes; who could run again (winners linked across elections by tax identifier)	TSE (Superior Electoral Court)	1996 to 2024	5,320 to 5,569
Chamber seats and votes; party funds	TSE	2022, 2024, 2025	n.a.
Municipal boundaries	IBGE (older vintage, without the six municipalities installed since 2009)	n.a.	5,564

Notes: All files are copies of official data in open repositories, checked against official national totals (Table S3.2). n.a.: not applicable.

S3.2. Repositories and checks

Government data portals could not be reached from the environment in which the study was produced. Every file was therefore obtained from an open repository that republishes official data, and each was checked against official national totals. Table S3.2 lists, for each part of the dataset, the original producer, the repository from which the copy was taken and the check applied to it. The files themselves are in the folder data/raw/ of the replication package, where each source has a SOURCES.md file that gives the repository, the URL, the original producer and the checks.

Table S3.2. Data sources, repositories and checks.

Data	Original producer and coverage	Repository of the copy used	Check
Population	IBGE: counts (2000, 2007, 2010, 2022) and intercensal estimates (2001 to 2025)	brpop package (rfsaldanha/brpop); IBGE municipal indicators snapshot (peterwkdev-creator/numeros-publicos)	203.1 million inhabitants in 2022
GDP and value added	IBGE, PIB dos Municípios workbook, 2010 to 2023	DanielNego/projeto-bi-pib-municipios	Municipal totals add up to national GDP

Data	Original producer and coverage	Repository of the copy used	Check
Formal jobs	RAIS establishment files, aggregated by municipality and sector	rafapolo/rais	46.7 million (2019) and 57.8 million (2024), as in published RAIS
Bolsa Família	Payroll, December 2025	jorgel-mendes/folha-pagamento-municipios	18.5 million families
Revenue, own taxes and personnel spending	National Treasury, SICONFI RREO and RGF reports, 2020 to 2024, retrieved through the Treasury API	peterwkdev-creator/painel-fiscal-ne	3,243 municipalities with 2024 revenue reports and 3,812 with personnel spending in at least one year; coverage nearly complete above 50,000 inhabitants and about 40% below 5,000
Literacy	INEP, Indicador Criança Alfabetizada	tuanyfortunato/predicao-alfabetiza-brasil	National public-network rates of 55.9% (2023) and 59.2% (2024), as published by INEP
School Census and IDEB	INEP spreadsheets	realdanielelias/dashboard-publico	Official national totals
Sewage and water	SNIS municipal table, 1995 to 2022	juniorcarvalho/painel-saneamento	Population served by sewage collection in 2022 adds up to 55.5% (official 56.0%), and water to 84.2% (official service index 84.9%)
Homicides	Atlas da Violência export, 1996 to 2016; IPEA rates for 2022	vgs2/Socioeconomic-analysis-of-the-brazilian-cities; FelipeCassimiro37/PainelSocioAmbiental (which also provides SINISA 2024 and robberies in ten states)	Implied totals match official counts
Favelas	IBGE, 2022 Census	brunopantaleao/favela_dados	656 municipalities with favelas
Mayors and parties, 2024	TSE, official candidate file	pedrovitorsilva/eleicoes-2024	PSD elected 886 mayors in the ordinary election, MDB 859, PP 747, União 588 and PL 515
Mayors, 2020	TSE, first-round results, completed for runoffs	marcofaga/eleicoes2020	Official national totals
Winners of 1996 to 2016 and candidates of 2020, with tax identifiers	TSE	pindograma/materias	Used only to link mayors across elections and to tell lame ducks from mayors who can run again
Correspondence of TSE and IBGE municipal codes	TSE and IBGE	betafcc/Municipios-Brasileiros-TSE	n.a.
Chamber of Deputies, 2022	TSE, election results	kalebefukuda/brazil-election-results-2026 (despite its name, it holds the 2022 results)	Seats add up to 513

Data	Original producer and coverage	Repository of the copy used	Check
Discretionary amendments, 2025	Transferegov transferências especiais, compiled by Transparência Brasil	Transparencia-Brasil/emendas-parlamentares-transferencias-especiais	Authors' parties matched to the 2022 and 2018 elections for 91.5% of the value
Municipal boundaries	IBGE (older vintage)	tbrugz/geodata-br	5,564 municipalities (the six installed since 2009 are missing)
FPM	Computed from the legal coefficients by population bracket, the fixed state shares of Resolution TCU 242/1990, the 10% share of capitals and the 3.6% reserve	Computed in this study	Scaled to the R$ 216 billion distributed in 2024
Party funds	TSE allocations of the 2025 Fundo Partidário and the 2024 FEFC	TSE	R$ 1.229 billion (including fines) and R$ 4.96 billion

Notes: n.a.: a correspondence table used only to link sources, with no total to check. The file with candidates' tax identifiers is used only to link mayors across elections; its derived products in the package keep no tax identifiers.

Two features of these sources matter for reading the article. The first is that the Treasury data are incomplete for small municipalities, so own taxes are imputed for the municipalities that do not report them (Section 5.3 of the article and Section S5.4). The second is that some series are short or end early: the homicide panel ends in 2016, with 2022 rates added, and literacy has two editions, 2023 and 2024. Section 11.3 of the article discusses how these limits qualify its conclusions.

S3.3. The replication package

The replication package contains the code and the data that rebuild every number, table and figure of the article and of this document. The folder code/ holds the scripts listed in Table S3.3. The folder data/raw/ holds the raw files described in Section S3.2, and data/derived/ the municipal dataset and the indicator panels. The folder output/ holds the simulation results, and its file output/numbers.json contains every simulated number cited in the text. The folder tables/ holds CSV versions of the real-data tables. The folder lean/ holds the formal proofs of Section S2 in Lean 4: compiling it with the Mathlib library checks every proof by machine, and lean/results/ keeps the record of the build of 9 October 2026, in which every proof was accepted. The scripts read and write paths relative to the root of the package, and all random draws are seeded, so every result can be reproduced exactly: re-running the simulation engine reproduces the stored results of 9 designs without a single difference in their 705 draw-level arrays, and re-running the calibration from its seeds reproduces its 400 posterior draws of 47 parameters. The README of the package gives the requirements and the running time of each script.

Table S3.3. Scripts of the replication package and the results they reproduce.

Script	What it does	Results in the article	Results in this document
01_build_national.py	Builds the national dataset from the raw files	Section 4.1	Table S3.1
02_indicator_properties.py	Variance decomposition, reliability and validity of the indicators	Sections 4.2 to 4.4; Tables 2 and 3	Sections S4.2 to S4.4
04_amendments.py	The 2025 transferências especiais: areas, size and party alignment	Section 5.1; Figure 4	Section S5.2
05_federalism.py	FPM, own taxes, the PEC 188 test, mergers, councillors and the cost of commissioners	Sections 5.1, 5.3 and 8.4	Sections S5.1, S5.4 and S8.5
06_party_funds.py	The party funds on day one	Section 5.2; Figure 5	Section S5.3
16_incumbency.py	Who governed, who could run again and the open seats, 2012 to 2024	Section 6.1	Section S6.1
17_incentive_estimates.py	Close races with and without the incumbent, robustness, bounds on the electoral incentive, panel and electoral returns	Sections 6.1 and 6.2; Table 4; Figure 6	Sections S6.1 and S6.2
18_fpm_rd.py	Bunching at the FPM brackets and the discontinuity on unconditional money	Section 6.3	Section S6.3
19_calibrate.py	Calibration by sampling-importance-resampling, its variants and the saturation variants	Sections 6.5 and 9.2	Sections S6.5, S6.6, S7.6 and S9.3
03_run_designs.py, 03b_merge_runs.py, 03c_merge_v5.py	National simulation: 400 calibrated draws of 66 designs, and the scenario runs	Sections 7 to 9	Sections S8 and S9
22_scenarios.py	Summary of the scenario runs	Section 9; Tables 7 and 8	Section S9.4
23_v5_tests.py	Tests of the sanctions, complementarity, effects by indicator, plausibility of larger responses and technical checks	Sections 8.2, 9.1 and 9.2	Sections S8.3, S9.1 to S9.3 and S9.6
07_sobol.py	Variance-based sensitivity analysis	Section 9.3	Section S9.5
08_chain.py	Illustrative chain from municipal outcomes to the Chamber	Sections 8.5 and 10.1	Section S7.7
09_rtm.py and 20_pilot_power.py	Regression to the mean in real data; power of a factorial pilot	Section 11.2	Section S11
10_figures_data.py, 11_figures_diagrams.py, 12_figures_sim.py, 21_figures_behaviour.py	Figures from real data, diagrams, simulation results and behavioural estimates	All figures	All figures
13_numbers.py	Collects every simulated number into output/numbers.json	All numbers in the text	All numbers in the text

Script	What it does	Results in the article	Results in this document
15_verdict.py	Assessment of each causal link of the proposal	Table 10; Figure 13	Section S8.10
lrg2_sim.py	The simulation engine, used by the scripts above	Section 7	Section S7
lean/ (Lean 4 project)	Formal statements of the derivations and propositions of the model, with proofs checked by machine	Sections 3.3 and 3.5	Section S2

S4. What the indicators measure

Sections 4.2 to 4.4 of the article ask what the scorecard can see: which indicators can be measured every year for every municipality, how much of their variation reflects places, administrations and noise, and whether official measures of the same thing agree. This section gives the details behind those answers: the weaknesses of each indicator (Section S4.1), the method of the decomposition (Section S4.2), the reliability of changes and the upper bound on the mayor's share (Section S4.3), the comparison of official measures (Section S4.4) and the indicators of the government plan's own example (Section S4.5).

S4.1. The audit of the ten indicators in detail

Four of the ten announced indicators fail the first requirement of a sanctioning rule, an annual municipal series. Favelas and precarious housing (K5) are measured only by the Census, every ten years. Robberies (K8) are recorded by state police with incompatible definitions, and a municipal series exists for only ten states. There is no national series of waiting times for specialist appointments and exams (K10). Hospital care for aggression against women (K9) exists, but it produces counts of a few cases a year in most municipalities and, as the pre-candidate himself acknowledged, rises with better reporting.

The remaining six indicators can be measured, but each has a weakness that matters for a rule that rewards and punishes. Value added (K2) is published with a two-year lag, so the first two evaluations of a mandate grade the predecessor and the last two arrive after the election. Formal employment and value added respond to commodity prices, harvests and single plants, and both can be gamed in ways that hurt citizens: public hiring counts as formal employment in the announced indicator, cutting public payroll (teachers and health staff included) raises the private share of value added, and an indicator that penalises dependence on transfers rewards removing eligible families from Bolsa Família. Literacy can be raised by teaching to the test and by selective participation, homicides by reclassifying deaths as of undetermined intent, and daycare coverage by part-time or low-quality places and by disputes over the population denominator.

S4.2. Estimating places, mayors and noise

The decomposition behind Table 3 of the article answers a simple question: of the differences between municipalities of the same state, how much is permanent, how much changes with the mayoral term and how much is noise? To answer it, each indicator is first residualised on state × year means, which removes common trends and the differences between states. Then, separately for six population classes, the covariance across municipalities is computed for every pair of years. These covariances are fitted, by robust least squares weighted by the number of municipalities in each pair, to

$$\mathrm{Cov}\left(y_{it}, y_{is}\right) = \sigma_f^2 + \sigma_a^2 \rho^{|t-s|} + \sigma_b^2 \cdot \mathbf{1}[\text{same term}] + \sigma_\varepsilon^2 \cdot \mathbf{1}[t = s] \qquad \text{(S4.1)}$$

where the term indicator equals one when both years fall in the same mayoral mandate (2001 to 2004, 2005 to 2008, and so on up to 2021 to 2024). Each component of equation (S4.1) has a direct reading. σ_f^2 is the persistent component, common to all years; $\sigma_a^2\rho^{|t-s|}$ is a drift that fades with the distance between years; σ_b^2 is the component shared by years of the same mandate, that is, the component of the mayoral term; and σ_ε^2 is the noise, which appears only when a year is compared with itself. The term component is identified because pairs of years at the same distance can fall either within one mandate or across two: if administrations matter, the first kind of pair covaries more than the second. Because effects that outlast a mandate are absorbed into the persistent component, the term share is a lower bound on the mayor's influence.

Some indicators need specific treatment. Formal jobs enter in logs. Homicide rates use the years 2004 to 2016, and the Poisson variance implied by population and rates is compared with the estimated noise; in municipalities under 100,000 inhabitants the noise is almost entirely Poisson counting noise. Literacy has only two editions, both in one mandate, so test-retest reliability is reported for it instead of a decomposition. The intervals for the term share come from 40 cluster-bootstrap replications over municipalities.

S4.3. Noise, reliability and the upper bound on the mayor's share

Noise depends on the indicator and on the size of the municipality. Administrative counts, such as jobs, enrolments and value added, are measured almost without noise, whereas test-based and count-based indicators are not. In municipalities under 5,000 inhabitants, noise accounts for 35% of the variance of IDEB, 67% of literacy and 89% of homicide rates, and the reliability of a change is close to zero for homicides below 20,000 inhabitants and at most 0.15 for IDEB below 50,000. Homicides illustrate the mechanism well. In small municipalities their transitory variation is almost exactly what Poisson counting predicts: a town of 3,000 people with an average homicide rate expects one death every three years, and a single death moves its rate by 33 per 100,000.

The upper bound on the mayor's share of change comes from decomposing four-year changes between the last years of consecutive mandates into three parts: a persistent trend, measured by the covariance of successive changes; noise, equal to twice the noise variance; and a remainder that includes both the mayor and local shocks. Because the remainder mixes the two, it bounds the mayor's share from above. The bound is informative only where noise dominates: in municipalities under 5,000 inhabitants, at most 1% of the variance of four-year changes in homicide rates and 11% of that in IDEB could reflect anything other than noise. For administrative indicators it is not informative, since 70% to 100% of the variance of four-year changes in jobs, value added, daycare and sewage is neither noise nor a common trend and could reflect administrations as well as local shocks. This is why Section 8.1 of the article asks what happens to the sanctions if the mayor's component is five times larger than estimated.

S4.4. Two official measures of sewage, and the error of population denominators

Section 4.4 of the article reports two checks of measurement validity, which Figure S4.1 draws. Panel (a) compares, municipality by municipality, the sewage coverage that providers report to the national system (SNIS) with the share of households that the 2022 Census found connected to a network, for the 2,900 municipalities that report sewage. The two measures differ by a median of 9 percentage points, and by more than 10 points in 46% of these municipalities. In addition, 679 municipalities where the Census found more than a fifth of households connected report no sewage service at all. Part of the gap is conceptual, because the Census also counts connections to rainwater drains. Panel (b) compares IBGE's 2021 municipal population estimates with the 2022 Census, by population class. The median error is 6.6% per municipality and exceeds 10% in a third of them, an error that every per-capita indicator inherits until the next census.

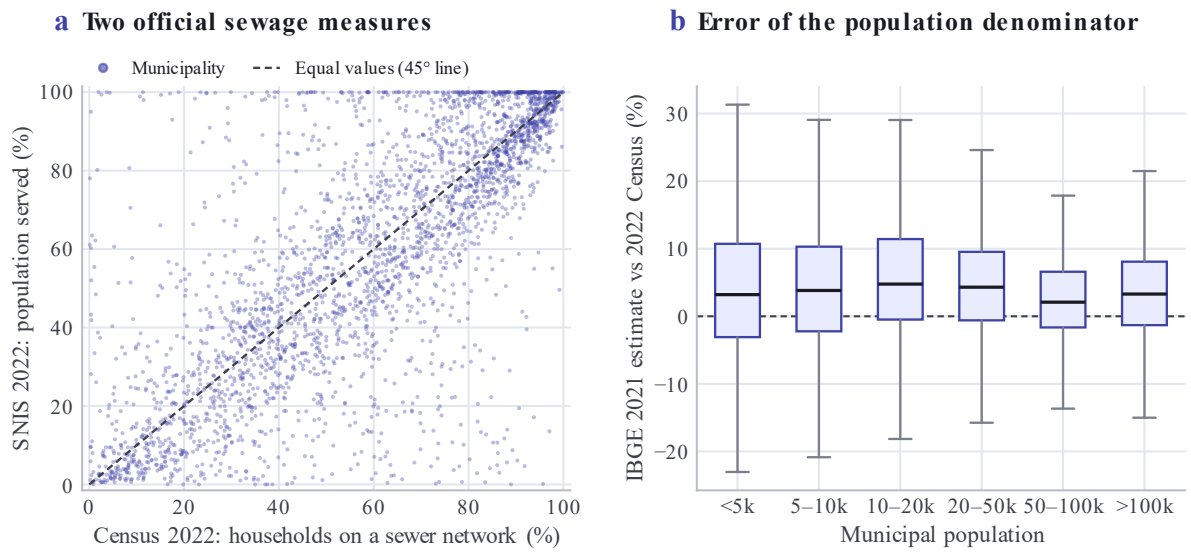


Figure S4.1. Measurement validity: two official sewage measures, and the error of population denominators. **(a)** Each dot is a municipality reporting sewage to SNIS in 2022; the dashed line is equality. Part of the gap is conceptual (the Census counts connections to rainwater drains). **(b)** IBGE's 2021 municipal estimates against the 2022 Census, by population class; boxes show the interquartile range and whiskers 1.5 times it.

S4.5. The government plan's own indicators

The government plan's own example used different measures from the ten headline indicators: IDEB, personnel spending, child vaccination and timely answers to requests for information. Except for IDEB, these measure administrative process instead of outcomes, and mayors control them more directly, so they are better candidates for sanctions than the headline indicators. They are not, however, what the public statements promise to reward, and the data assembled here do not cover them for long enough to estimate how much they change with each administration, since personnel spending is available only from 2020.

S5. Money and politics today: further results

Section 5 of the article measures, with real data and before any behavioural response, the flows of money and politics on which the LRG would act. This section completes that picture with the fiscal base of the municipalities by size (Section S5.1), the content and politics of the discretionary amendments (Section S5.2), the arithmetic of the party funds on day one (Section S5.3) and the viability test and merger exercise behind the proposal's reform of the federation (Section S5.4).

S5.1. The FPM and the fiscal base by municipal size

The FPM distributed about R$ 216 billion in 2024 before the deduction for FUNDEB (R$ 177 billion net). Its coefficients rise in steps with population, but much less than proportionally, so the transfer per inhabitant falls steeply with size (Table S5.1). This schedule rewarded the creation of small municipalities after 1988, until Constitutional Amendment 15/1996 tightened the rules for creating them, and it explains why a performance rule tied to population and one tied to each municipality's own FPM produce very different winners (Section 8.4 of the article). The same table shows how own taxes, the share of municipalities below the viability line of PEC 188 and the amendments per inhabitant vary with size.

Table S5.1. The fiscal base by municipal size.

Population	Municipalities	FPM per inhabitant (R$)	Own taxes / revenue (median)	Below 10% (share)	Amendments per inhabitant, 2025 (R$)
< 5,000	1,288	5,101	5.5%	89%	218
5,000 to 10,000	1,178	2,452	6.6%	77%	137
10,000 to 20,000	1,357	1,982	6.9%	72%	96
20,000 to 50,000	1,072	1,432	9.1%	56%	62
50,000 to 100,000	339	1,033	13.8%	34%	33
> 100,000	336	522	20.9%	7%	11

Notes: Population classes by 2024 estimates. FPM: gross 2024, computed from the legal formula. Own taxes: SICONFI 2024 (municipal taxes, including income tax withheld on municipal payroll, over current revenue), imputed for non-reporters (Section S5.4); medians use point imputations, and the shares below 10% are medians over 500 imputation draws. FPM per inhabitant excludes Brasília, which receives none. Amendments: 2025 transferências especiais.

S5.2. What the discretionary amendments buy, where they go and to whom

The 2025 "transferências especiais" are individual amendments transferred to municipalities with no programme attached, and they are the most discretionary instrument in the federal budget. Of their R$ 8.2 billion in 12,663 action plans for 4,302 municipalities, urban works such as paving and squares take 40% of the money, agriculture 12%, health 11% and sport and leisure 7%, whereas education receives 4% and sanitation 2% (Figure 4a of the article). Across municipalities, amendments per inhabitant are uncorrelated with literacy (r = 0.00) and only weakly correlated with Bolsa Família coverage (0.10), so the allocation barely tracks need. The over-representation of the deputies' own parties described in Section 5.1 of the article, 31.6% of the money against the 13.8% expected, holds for every large party (Figure 4c of the article), and a fifth of the value was blocked by technical impediments.

S5.3. The party funds on day one

Applying the LRG formula to the mayors elected in 2024, weighted by population governed and before any quality weight, the six parties with most mayors (PSD, MDB, PP, União Brasil, PL and Republicanos) would see their share of the Fundo Partidário rise from 57% to 75%. The broader set of parties without a firm ideological line known as the centrão, which comprises the six plus Podemos, Avante, Solidariedade, PRD and five small parties, would rise from 68% to 83%. In money, PSD and MDB would each gain about R$ 125 to 130 million a year, whereas PT would lose R$ 96 million, PL R$ 56 million and PSOL, which elected no mayor, all of its R$ 42 million.

The rule would also change the conditions of entry for new parties. A new party with 3% of the vote, 15 deputies and no mayors passes the constitutional threshold for receiving the fund, which rises to 3% of the vote or 15 deputies from 2030 (Constitutional Amendment 97/2017). Under the LRG, such a party would go from about R$ 38 million a year to nothing, and to recover its share through mayors it would need about 560 mayors of median-sized municipalities. The concentration also runs across municipalities: São Paulo's mayor alone would carry 5.7% of the fund, and 3.7% of the municipalities with a mayor, the largest 204, would carry half of it.

Two facts bear on the expectation, expressed by the pre-candidate, that the rule would requalify the parties that gain from it. The parties that would gain funds held 225 of the 513 seats elected in 2022, and those that would lose held 288, so an ordinary law changing the funds would have to pass with votes from parties that pay for it. And Brazilian mayors switch parties freely, because holders of majoritarian offices do not lose their mandates for leaving their party (ADI 5081, 2015): 49% of the mayors re-elected in 2024 had changed party since 2020 (after accounting for mergers), and open seats stayed with the incumbent's party only 24% of the time.

S5.4. Fiscal viability and the merger exercise

The condensed plan adopts the viability test of PEC 188, own taxes of at least 10% of revenue. In 2024, 65% of the 3,231 municipalities whose reports to the Treasury include their tax revenue fell below that line. For the other 2,339, namely the 2,327 for which the copy of the Treasury data used here has no 2024 report and 12 with inconsistent tax records, own taxes are imputed from a model of size, income, economic structure and region ($R^2 = 0.73$ on the logit scale), and the shares below 10% reported in Table S5.1 are medians over 500 imputation draws. With the imputations, about 3,760 municipalities (68%) would fail the test, and in fact more would, because the measure of own taxes used here includes the income tax withheld on municipal payrolls, which the PEC 188 test excluded. Figure S5.1a shows the resulting distribution by population class, separating reported from imputed values; the map of own taxes in Figure 2 of the article uses the same imputations.

The PEC 188 itself limited incorporation to municipalities under 5,000 inhabitants. By that criterion, about 1,170 to 1,250 municipalities would be absorbed, depending on how the missing data are imputed, which is close to the 1,217 counted by the National Confederation of Municipalities in 2019. Contrary to the image of an unviable Northeast, most of them are in the South (423) and Southeast (372), where tiny municipalities are most common (Figure S5.1b). Nor are they the worst performers: on an index of five scorecard indicators and IDEB they score above the national average, and only 10% are in its bottom fifth.

A merger exercise makes the trade-offs concrete. If each of the 1,252 municipalities that fail the PEC 188 test joined its most populous contiguous neighbour in the same state, 4.3 million people would change municipality, 1,252 mayors and 11,268 council seats would disappear, and the councillors' pay bill would fall by at most R$ 1 billion a year (at the maximum seats and pay allowed by art. 29, as amended by Constitutional Amendments 25/2000 and 58/2009). Because the FPM coefficients are concave in population, the merged areas would lose about 22% of their FPM (R$ 12 billion a year) to the rest of their state, unless the transition that the plan promises compensated them. The evidence on mergers is mixed (Reingewertz, 2012; Allers and Geertsema, 2016; Blesse and Baskaran, 2016; Blom-Hansen et al., 2016). In Brazil, the creation of new municipalities raised local public employment and services in the districts that split (Dahis and Szerman, 2026), at the cost of economies of scale (Lima and Silveira Neto, 2018).

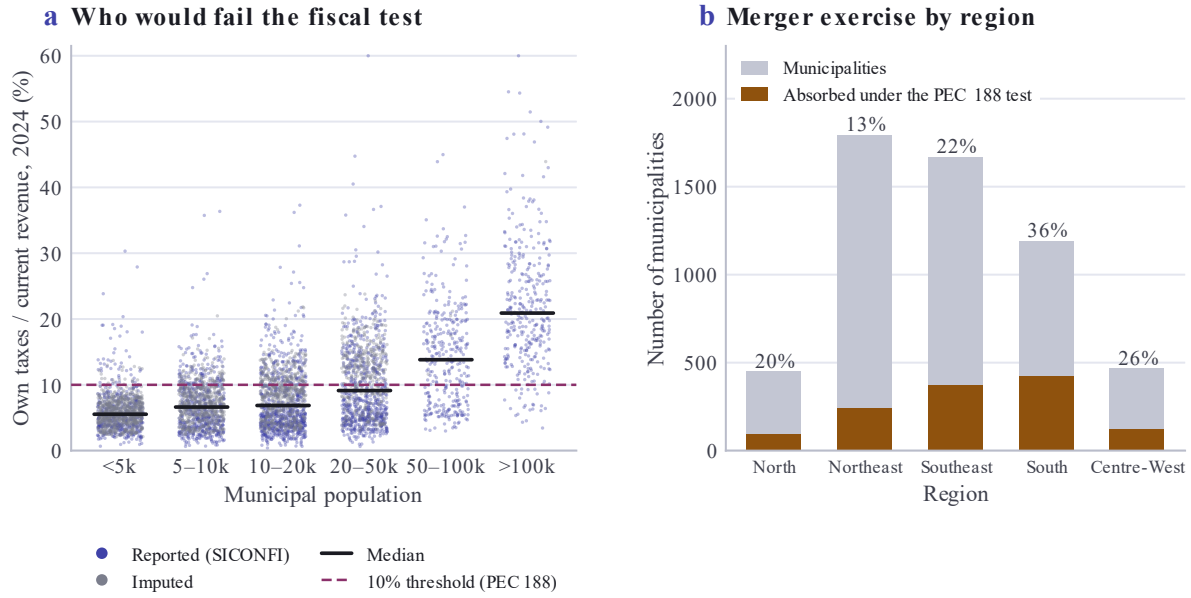


Figure S5.1. Fiscal viability and the merger exercise. **(a)** Own taxes as a share of current revenue in 2024; blue: reported to SICONFI; grey: imputed; black bars: medians. **(b)** Municipalities by region and those absorbed in the merger exercise (under 5,000 inhabitants and own taxes below 10%; each joins its most populous contiguous neighbour in the same state).

S6. Behavioural evidence and calibration

Section 6 of the article builds, from evidence, the number that the simulation needs: how strongly mayors would respond to the incentives of equation (4) of the article. This section reports the evidence in full and explains how it enters the model. Section S6.1 gives the complete table of evidence and the details of the close-race designs, Section S6.2 the electoral returns, Section S6.3 the evidence on money, Section S6.4 the evidence from other settings, Section S6.5 the way the model reproduces each calibration target, and Section S6.6 the fit of the calibration, its variants and its out-of-sample checks.

S6.1. The evidence in full, and the close-race designs

Table S6.1 summarises the evidence on how public officials respond to incentives and the use that the model makes of each piece; Table 4 of the article is a condensed version of it.

Table S6.1. Evidence on how public officials respond to incentives, and its use in the model.

Evidence	Incentive	Estimate	Use in the model
Close 2020 races in which the incumbent ran (this article)	Joint effect: same mayor and party, and no prospect of re-election	Lame duck minus new first-term mayor, 2021 to 2024: daycare +0.09 SD [−0.08, 0.34], IDEB +0.20 [−0.03, 0.46], formal jobs +0.03 [−0.04, 0.12]; sewage and value added ≈ 0; stable over 9 specifications	Calibration target (M3), as a joint effect
Close 2020 races for open seats (this article)	Continuity of the ruling party; a new eligible mayor on both sides	Outgoing party kept minus lost, 2021 to 2024: IDEB +0.37 [0.03, 0.71] (+0.27 controlling for the previous trend); other indicators ≈ 0	Calibration target (M3o); bounds on the electoral incentive
Three mandates, 2013 to 2024 (this article)	Career, within municipality	Lame duck minus first-term: daycare −0.10 (SE 0.02), IDEB +0.06 (0.02); jobs, sewage, value added ≈ 0	Validation check (not used)
Re-election of incumbents who ran, 2020 and 2024 (this article)	Voters' reward for performance (association)	+3.9 and +1.2 percentage points of re-election per SD of a composite of the indicators	Calibration target (M4); scenarios from zero to four times
Ceará, State Law 14,023/2007 (Lautharte, de Oliveira and Loureiro, 2021; Ponne, 2023)	18% of the municipal ICMS share paid by learning results (1.7% to 2.0% of current revenue)	Rule alone: +0.04 to +0.06 pupil SD in 5th grade after one year (significant at 10% only in Portuguese), +0.15 in 9th; with technical assistance (PAIC): +0.37 to +0.43 in 2013 to 2017	Calibration targets (M1, M2); time path as a check
FPM population brackets, 2014 to 2019 (this article)	Unconditional money	4.5% more FPM at the brackets; daycare −0.12 SD (SE 0.24) and formal jobs −0.15 SD (0.14) per R$ 100 per inhabitant a year	Calibration target (M5), weak

Evidence	Incentive	Estimate	Use in the model
FPM brackets in census years (this article)	Money that depends on a count	2.4, 4.5, 6.2 and 3.4 times more municipalities just above a bracket than just below (2000, 2007, 2010, 2022); about 1 in intercensal estimates	Evidence on manipulation; scenarios
Generasi, Indonesia (Olken, Onishi and Wong, 2014)	20% of a block grant paid by relative performance on 12 indicators	+0.04 SD on targeted health indicators after 18 months, +0.03 after 30; education ≈ 0	Check
Punjab tax collectors (Khan, Khwaja and Olken, 2016)	Personal performance pay	+9 log points of revenue (+15 with the revenue-only scheme); bribes +32%	Personal bonus benchmark (Section S8.9)
Italian mayors (Gagliarducci and Nannicini, 2013); Brazilian councillors (Ferraz and Finan, 2009); FPM and corruption (Brollo et al., 2013)	Pay for office; discretionary money	+33% pay: better-educated mayors and more efficient administration, through selection; +10% FPM: +7 pp in severe corruption, −2.7 pp college-educated challengers	Selection priors
Bolsa Escola (de Janvry, Finan and Sadoulet, 2012); Indian civil service (Bertrand et al., 2020)	Career incentives and implementation	First-term mayors: 36% larger programme impact; lifting a career cap: +0.3 SD in effectiveness ratings	Implementation scenarios (Section 3.4 of the article)

Notes: SD: standard deviation between municipalities of the same state (Section 7.1 of the article), except where stated as pupil SD (between students). Brackets: robust bias-corrected 95% intervals. The estimates of this article are reproduced by scripts 17, 18 and 19 of the replication package (Section S3.3).

The first design uses the 2020 races in which the 2016 winner ran again (3,137 municipalities, linked across elections by the candidates' tax identifiers) and a regression discontinuity on the incumbent's margin (Calonico, Cattaneo and Titiunik, 2014). Over 2021 to 2024, municipalities where the incumbent barely won changed by +0.09 SD in daycare and preschool coverage [−0.08, 0.34], +0.03 in formal jobs [−0.04, 0.12], +0.02 in sewage and +0.01 in value added relative to those where a new mayor took over. Literacy in 2024 was 0.32 SD lower, but not significantly [−0.93, 0.20], and learning measured by IDEB improved by 0.20 SD more (robust p = 0.08). These estimates hold across nine specifications (half to twice the optimal bandwidth, a local quadratic, the coverage-error-optimal bandwidth, baseline covariates and a donut), ranging from 0.14 to 0.23 SD for learning and from 0.01 to 0.12 for daycare, and controlling for each municipality's change over the previous mandate gives 0.02 for daycare and 0.19 for learning.

The second design uses the 2020 races for open seats in which the outgoing mayor's party ran (875 municipalities; density test p = 0.80). Where the outgoing party barely kept the office, learning improved by 0.37 SD more than where it barely lost [0.03, 0.71]. The placebo for learning over 2017 to 2020 is −0.30 (p = 0.09), so part of this difference may predate the election; controlling for the previous trend it is 0.27 [−0.09, 0.63], and 0.32 if continuity is defined by the outgoing coalition.

Section 6.1 of the article combines the two designs to bound the effect of being able to run again, E, as C + P minus the first estimate, where C is the continuity of the party estimated by the second design and P the continuity of the person, assumed to lie between zero and the largest party-continuity effect that the open-seat estimates do not reject. Without sampling error, E lies between 0.07 and 0.64 SD for learning and between −0.06 and 0.22 for daycare; with it, between −0.28 and 0.99 and between −0.35 and 0.50.

S6.2. How much voters reward performance

Among incumbents who ran, the change during the mandate in a composite of the indicators (learning, daycare, jobs and, in 2024, literacy) is related to re-election by a linear probability model with state fixed effects and population. A change one standard deviation better was associated with a 3.9 percentage-point higher chance of re-election in 2020 (SE 1.0) and 1.2 in 2024 (SE 0.9), about 0.14 on the logit scale. Daycare was the most rewarded indicator in 2020 (+6.0 points per SD) and learning in 2024 (+2.8), while literacy, published for the first time a few months before the 2024 election, was not rewarded at all. The bias of these associations can run either way: better-performing mayors may also be better campaigners, which inflates the estimate, whereas noise in the indicators and voters who reward what the indicators miss deflate it.

S6.3. Money: Ceará, gaming at the FPM brackets and unconditional money

Ceará distributes 18% of the municipal share of its value-added tax according to an index of literacy at the end of second grade and of fifth-grade learning, in levels and improvements, computed by a state agency from a state test (Lautharte, de Oliveira and Loureiro, 2021). Comparing Ceará's municipalities with their neighbours across the state border, the rule alone raised fifth-grade test scores by 0.04 to 0.06 pupil standard deviations in its first year (significant at 10% only in Portuguese) and ninth-grade scores by 0.15, and once the state extended its literacy programme (PAIC) to these grades the cumulative gain reached 0.37 to 0.43 in the years 2013 to 2017. A synthetic-control comparison with other states finds 1.8 to 6.6 test-score points for the rule alone and 12.7 to 18.5 points with assistance (Ponne, 2023).

The FPM coefficient of each municipality jumps at seven population brackets (10,189; 13,585; 16,981; 23,773; 30,565; 37,357; and 44,149 inhabitants), so a municipality just above a bracket receives more money than one just below. If the population counts that reset the coefficients can be influenced, municipalities should bunch just above the brackets, and Figure S6.1 tests this prediction by comparing, year by year, the number of interior municipalities within 1% above a bracket with the number within 1% below. In the years of the Censuses and of the 2007 Count the ratio is well above one: 2.4, 4.5, 6.2 and 3.4 times more municipalities lay just above a bracket than just below in 2000, 2007, 2010 and

2022; in 2010, for example, 246 interior municipalities lay within 1% above a bracket and only 40 within 1% below. In the intercensal estimates, which IBGE produces with a demographic model that municipalities can contest but cannot count, the two numbers are about equal (103 and 101 in 2019). Panel (b) shows the same pattern in the distribution of distances to the nearest bracket.

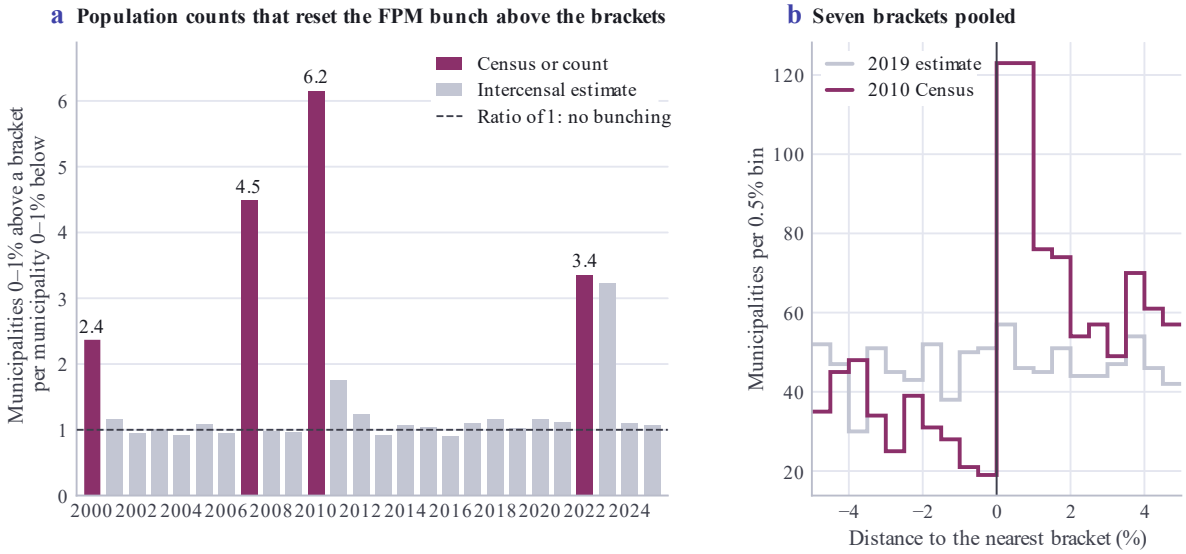


Figure S6.1. Gaming at the FPM population brackets. Interior municipalities within 1% above an FPM population bracket per municipality within 1% below, by year. **(a)** Red: Census or population count; grey: IBGE intercensal estimate (the 2001, 2011 and 2023 estimates inherit the preceding count). **(b)** Distribution of the distance to the nearest bracket, seven brackets pooled. Source: IBGE counts and estimates.

Because they are free of bunching, the intercensal estimates also allow a regression discontinuity on the FPM brackets of the kind that Brollo et al. (2013) used for 2001 to 2008. Being just above a bracket in the 2013 estimate raised the FPM received over 2014 to 2019 by only 4.5%, because growing municipalities just below the bracket crossed it within a year or two; at 2024 values, this is about R$ 49 per inhabitant a year. Per R$ 100 per inhabitant a year, daycare coverage changes by −0.12 SD (SE 0.24) and formal jobs by −0.15 SD (SE 0.14). These estimates rule out very large returns to unconditional money but are consistent with the unit costs assumed in the model, and they provide calibration target M5 with their wide standard errors; the bunching, in turn, informs the range of manipulated shares that the model draws (Table S7.1).

S6.4. Evidence from other settings

No Brazilian mayor has ever been paid for results, so the personal bonus examined in Section S8.9 rests on evidence from other officials. This evidence shows, first, that personal performance pay can be powerful when the agent controls the outcome: Punjab's property-tax collectors raised revenue by 9 log points, and by 15 under the scheme that rewarded revenue alone, but bribes paid by typical

taxpayers rose by about a third (Khan, Khwaja and Olken, 2016). Pay also changes who governs. Higher salaries brought better-educated mayors and more efficient administrations in Italy, through selection rather than incentives (Gagliarducci and Nannicini, 2013), and more and better-educated candidates for Brazilian councils (Ferraz and Finan, 2009), whereas discretionary money attracts worse candidates and more rent-seeking (Brollo et al., 2013).

Consequences tied to measured results do move what is measured: No Child Left Behind raised mathematics scores in the United States (Dee and Jacob, 2011), and Chinese officials cut pollution at monitored river borders once it counted for promotion (Kahn, Li and Zhao, 2015). Grants conditioned on community performance, however, had smaller and fading effects: in Indonesia, targeted health indicators rose by 0.04 SD after eighteen months, mostly because incentivised villages achieved sooner what the others achieved later (Olken, Onishi and Wong, 2014). None of these settings is a Brazilian mayor facing the LRG, and for that reason they inform the model's priors and checks, but not its calibration targets.

S6.5. How the model reproduces the calibration targets

The calibration weights each of 120,000 prior draws of the 47 parameters by the likelihood that the model reproduces fifteen targets in five blocks, following the sampling-importance-resampling method (Rubin, 1988). Each target is reproduced with the equations of the simulation (Section S7), applied to the relevant municipalities. Targets from the same sample enter a joint normal likelihood with the bootstrap correlations of the estimates, standard errors are widened for the differences between each setting and the LRG, and pupil-level effects are converted into differences between municipalities with a factor of 1.5 to 3. The effective sample is 5,394 draws, from which 400 are resampled. The paragraphs below explain the five blocks in turn.

For Ceará (targets M1 and M2), the rule puts a stake T_{CE} of 4.5% to 7.0% of FPM-equivalent money on learning alone, six times the average weight of one indicator. The effort that it induces, net of the electoral effort that 60% of mayors already exert, is $\delta_3[0.6(f(I_{el} + I_{CE}) - f(I_{el})) + 0.4f(I_{CE})]$, with $f(I) = I/(1 + I/4)$, multiplied by the capacity of Ceará's municipalities and accumulated with persistence ρ_A. M1 is its average after one and three years, and M2 adds seven years of PAIC-like assistance, that is, β_{TA} applied to Ceará's capacity gap, with the capacity of commissioned administrations. Both are divided by 1.5 to 3 to convert differences between municipalities into pupil standard deviations; in the transportability variant, Ceará's responses are twice the national ones.

For the races in which the incumbent ran (M3), the model predicts that a lame duck's municipality changes, relative to a new mayor's, by the sum of three terms. The first is minus the electoral incentive, $-E$, where $E = \delta_k f(I_{el})(1 - \rho_A^t)$ times capacity. The second is the disruption avoided when the party stays, C, equal to $\tau\rho_{dis}^t$ in learning and half of that in daycare. The third is the continuity of the person, P, drawn uniformly between zero and the bound of Section S6.1. In these expressions, t

is the number of years between the election and the measurement. For the open seats (M3o), the model predicts C alone, and both blocks use the trend-adjusted estimates.

The two remaining blocks are simpler. M4 compares β with the pooled estimate of Section S6.2, and M5 compares the effect of R$ 100 of unconditional money per inhabitant a year with the FPM discontinuities. Within each block, the targets enter a multivariate normal likelihood with the bootstrap correlations of the estimates (0.5 between the two Ceará horizons, assumed), and standard errors are widened by 0.02 (0.15 for M2) for external validity. Draws are weighted by the product of the block likelihoods and resampled systematically. Each variant of Section S6.6 changes one element of this procedure; one of them drops the open-seat target M3o and explains M3 by the electoral incentive and the disruption alone.

S6.6. Fit of the calibration, its variants and out-of-sample checks

Table S6.2 reports the fit of the main calibration to each of its fifteen targets; its log evidence is 28.86. The last column, z, measures the distance between the posterior median and the estimate in standard errors, so values close to zero indicate a close fit.

Table S6.2. Calibration targets and fit.

Target	Block	Estimate	SE (widened)	Posterior median	5th to 95th percentile	z
M1 Ceará, rule alone (pupil SD)	CE	0.060	0.040	0.049	0.009 to 0.110	−0.26
M2 Ceará, rule + PAIC (pupil SD)	CE	0.420	0.162	0.410	0.177 to 0.649	−0.06
M3 lame duck minus first term: daycare	RD	0.016	0.082	0.070	0.010 to 0.128	0.66
M3 lame duck minus first term: learning (IDEB)	RD	0.193	0.096	0.149	0.036 to 0.264	−0.46
M3 lame duck minus first term: formal jobs	RD	0.054	0.039	0.026	0.003 to 0.052	−0.72
M3 lame duck minus first term: sewage	RD	0.026	0.071	0.049	0.005 to 0.100	0.32
M3 lame duck minus first term: value added	RD	0.011	0.043	0.033	0.006 to 0.064	0.50
M3o open seats, party kept minus lost: daycare	RDo	−0.048	0.122	0.030	0.006 to 0.087	0.64
M3o open seats, party kept minus lost: learning (IDEB)	RDo	0.267	0.154	0.087	0.019 to 0.208	−1.16
M3o open seats, party kept minus lost: formal jobs	RDo	0.025	0.060	0.000	0.000 to 0.000	−0.41
M3o open seats, party kept minus lost: sewage	RDo	0.028	0.122	0.000	0.000 to 0.000	−0.23
M3o open seats, party kept minus lost: value added	RDo	0.041	0.064	0.000	0.000 to 0.000	−0.64
M4 electoral return (logit per SD)	EL	0.142	0.040	0.140	0.075 to 0.205	−0.06
M5 FPM brackets: daycare level per R$100	FPM	−0.120	0.243	0.138	0.092 to 0.214	1.06
M5 FPM brackets: formal jobs per R$100	FPM	−0.146	0.138	0.011	0.008 to 0.017	1.14

Notes: M1 and M2: Ceará (pupil SD); M3: close 2020 races in which the incumbent ran; M3o: open seats; M4: electoral return to performance; M5: FPM population brackets. z: (posterior median − estimate)/SE.

The evidence pins down three parameters and leaves most others near their priors (the full list is in Table S7.2). The first is voters' reward for performance, 0.14 [0.08 to 0.21] logits per SD. The second is the response of learning to incentives, δ for literacy, 0.52 [0.09 to 1.59] incentive units. The third is technical assistance in a weak administration, 0.17 [0.04 to 0.33] SD a year. By contrast, the disruption that follows a change of ruling party is not pinned down (0.27 [0.06 to 0.48] SD of learning), because the learning gap of the first design can come from the continuity of the person as well as from that of the party, and the electoral and money incentives are identified only together with δ. The effect of being able to run again on learning, which Section S6.1 could only bound, has a posterior of 0.05 [0.00 to 0.19] SD over three years against a prior range up to 0.60, narrower than the bounds because the model ties the electoral incentive to voters' reward.

These posteriors depend little on the choices made in the calibration (Figure S6.2a). Leaving out the Ceará targets, the discontinuities, the electoral return or the FPM targets moves the medians of the parameters only a little, except that without the electoral return voters' reward returns to its prior; ignoring the correlations between targets changes nothing. Two variants matter more. If Ceará's municipalities respond twice as strongly as the average one, which is plausible because Ceará combined the rule with its own test and a state programme, the national response of learning falls to 0.33 and the effect of assistance to 0.08. With wider priors for 21 weakly identified parameters, the money incentive (0.88), the weight of the ban (5.4) and the disruption (0.51) take larger values. Each variant is carried through the simulation in Section 9.3 of the article.

Reproducing the targets is not a validation in itself, so the calibration was also used to predict evidence that it did not use (Figure S6.2b). Each group of targets left out is predicted by the others within its 90% interval: Ceará with assistance is predicted at 0.54 [0.14 to 1.15] pupil SD against an estimate of 0.42, and the learning discontinuity of the first design at 0.30 [−0.04 to 0.62] against 0.19. Of the checks never used, a panel of three mandates with municipality and state-by-mandate fixed effects finds lame ducks improving learning by +0.06 SD more than first-term mayors, inside the model's interval (−0.15 to 0.07), and daycare by 0.10 less, outside it, a comparison that also reflects who is re-elected; the open-seat discontinuity in literacy (+0.51) is also outside. The model, finally, builds up Ceará's gains more slowly than the synthetic control of Ponne (2023), with 0.34 [0.15 to 0.54] pupil SD after seven years against 0.62 observed, while reaching the same level after eleven years (0.50 against 0.53). If anything, then, the model understates how fast learning responds when money and help come together.

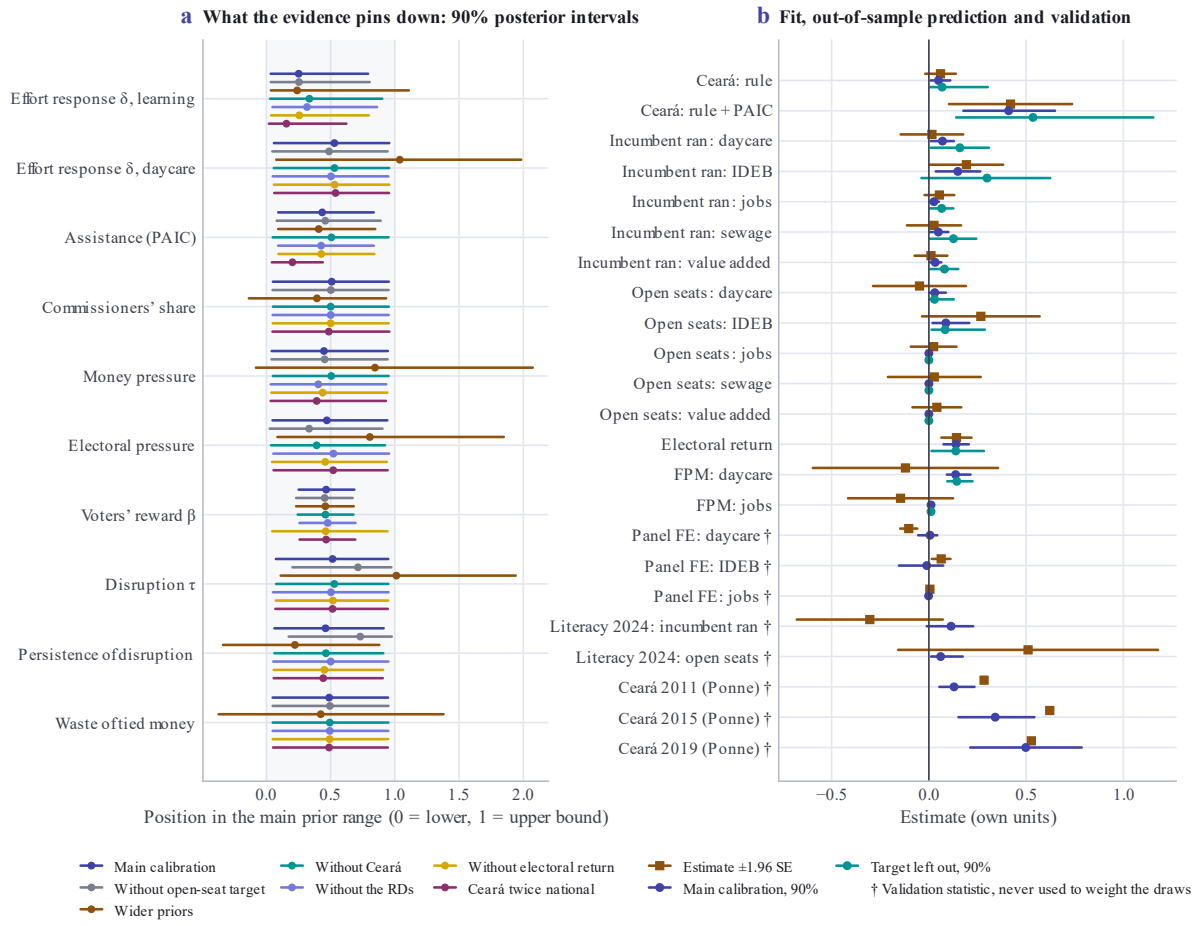


Figure S6.2. What the evidence says about the model's parameters. **(a)** 90% posterior intervals of ten parameters under the main calibration and its variants, as positions in the main prior range (0: lower bound; 1: upper bound; wider priors extend beyond it). **(b)** Each target (estimate ± 1.96 standard errors, widened for external validity), the main calibration's 90% interval and the 90% interval predicted by the calibration that leaves that group of targets out; daggers: checks not used for calibration (panel estimates, the literacy discontinuities and Ceará's time path, Ponne, 2023). Units: Ceará in pupil SD; discontinuities in SD between municipalities; electoral return in logits per SD; FPM effects in SD per R$ 100 per inhabitant a year.

S7. The simulation model

This section specifies the national microsimulation used in Sections 7 to 9 of the article. It uses the notation of the simulation code, so its symbols are defined where they appear and may differ from those of Section 3 of the article, which presents the same mechanisms in a compact form. Section S7.1 describes how quality is generated, measured and scored, Section S7.2 how mayors respond to the incentives they face, Section S7.3 how money changes outcomes and Section S7.4 how elections, parties and clientelism evolve. Section S7.5 lists the designs, scenarios and checks, Section S7.6 the 47 uncertain parameters with their priors, evidence and posteriors, and Section S7.7 the illustrative chain from municipal outcomes to the Chamber.

S7.1. Quality, measurement and the scorecard

For municipality i, indicator k and year t, true quality is the sum of several components:

$$q_{ikt} = \mu_{kt} + P_{ik} + D_{ikt} + B_{ikt} + o_k\left(A_{ikt} + A^{m}_{ikt} + A^{ta}_{ikt}\right) - o_k\tau_k\Delta_{ikt}\,SD_k \qquad \text{(S7.1)}$$

Here μ is the national trend and P the persistent level. D is an AR(1) drift, with the persistence and innovation variance estimated in Section 4.3 of the article. B is the component of the mayor in office, $B = \sigma_{b,k}(\sqrt{½}\theta_j + \sqrt{½}u_{ik})$, where $\sigma_{b,k}$ is the estimated term component and θ_j the mayor's competence. A, A^m and A^{ta} are the stocks of real improvement produced by effort, by money (net transfers and tied amendments, tracked separately) and by technical assistance. Δ is the disruption that follows a change of ruling party, equal to τ in learning and $\tau/2$ in daycare, which decays at the rate $1 - \rho_{dis}$ a year. Finally, $o_k = \pm 1$ is the orientation of the indicator, so that homicides enter with a negative sign.

The initial state is drawn from the empirical-Bayes posterior of the latent level given the latest real value of each indicator, so that the synthetic country reproduces the real one in expectation. Reported values are $y = q + o_kG + \varepsilon$, where G is manipulation and ε is normal with the size-specific variances of Table 3 of the article. Two indicators are treated differently. Homicides are Poisson counts given the true rate and population, net of manipulation. Sewage combines the true level from the 2022 Census with a persistent reporting gap, whose standard deviation of 9 points equals the median absolute gap between SNIS and the Census, plus annual noise.

The rules are those of Table S1.1. Floors are national 20th percentiles of the baseline by population group and move with the trend. Paths start from the reported value in the year before the law (2026) or in the election year that opened the mandate. The tolerance is $\sqrt{2}$ times the noise standard deviation of one reading. Points are the weighted improvement over the mandate baseline in national standard deviations, capped at ± 2.

To classify bans, the rules are also applied to two counterfactual readings. The error-free counterpart applies the same rules to true values. The "average-mayor" counterpart also removes the component B of the mayor in office from the current true values, while the baseline keeps the component of whoever was in office when it was set (for the first evaluation of a mandate, the predecessor). A ban is *wrongful* if the error-free counterpart would not trigger it, *structural* if the error-free counterpart triggers it and would also trigger it with an average mayor in office, and *attributable* to the mayor otherwise. Independently of the rules, the competence θ of each banned mayor is also recorded, which allows a test of the sanctions that does not depend on these definitions.

S7.2. Behaviour

The marginal expected incentive on indicator k (equation 4 of the article) is computed in alert units, that is, relative to the incentive created by a public alert for a mayor exactly at a floor with the median noise:

$$I_{ikt} = \kappa_{el}\frac{\beta + \beta_{info}\cdot \mathbf{1}_{law}}{\beta_{ref}}R_{it} + f_{ikt}(1 + S_2 + S_3E_{it}) \\ +\kappa_m\frac{T}{0.10}h_i\frac{w_k}{\overline{w}} + 0.5\,\kappa_m f_{ikt}e_i + \kappa_p\nu_i + \kappa_{el}\frac{b}{v\,\beta_{ref}\,p(1-p)}\frac{w_k}{\overline{w}} \tag{S7.2}$$

The first term is the electoral incentive of mayors who can run again ($R_{it} = 1$ unless the mayor is term-limited or banned). It is proportional to voters' reward for measured performance β and, with the law, to the information effect of the published score β_{info}, relative to the estimated return $\beta_{ref} = 0.14$, and it is present in every world, including the status quo. The second term is the expected consequence of the sanctions. In it, f is the normalised density of the reported value at the floor, scaled by the ratio of reference to local noise, and E equals one when the mayor is still eligible and a ban for results can follow on indicator k. E is zero in the designs without bans for results and in the designs where bans are executed but not anticipated; in the designs that vary the weight of a ban, S_3 is multiplied by a factor from 0 to 8.

The third and fourth terms come from money. T is the share of the FPM paid by points and h_i the municipality's FPM per inhabitant relative to the median, so that the money incentive is stronger where the FPM weighs more; e_i is the municipality's amendments per inhabitant relative to the median. The fifth term comes from party funds, where ν_i is the municipality's weight in party funds (population share × 1,000). The first two ratios are bounded between 0.2 and 3, and ν_i between 0.02 and 3. The last term is the personal bonus of the benchmark design: a bonus of b annual salaries per SD of the score, compared with the electoral reward of v mandates' pay times the electoral return $\beta_{ref}\cdot p(1-p)$, where $p(1-p) = 0.224$ at the re-election rate of incumbents who run.

Effort follows equation (7) of the article, $e = \delta_k I/(1 + I/4)$, and is zero for banned mayors; in a variant, banned mayors keep responding to the money, party and bonus terms. In the electoral scenarios, voters' reward β is multiplied by 0, 2 or 4. Real improvement then accumulates as

$$A_{ikt} = \rho_A A_{ik,t-1} + \left(1-\rho_A\right)e_{ikt}[1 - s_c(1 - c_{it})]\left(1 + \varphi\theta_j\right)SD_k \tag{S7.3}$$

Manipulation and the neglect of unscored tasks depend only on the effort induced by the law, $e^L = e - e^0$, where e^0 is the effort that the electoral incentive alone would produce. Manipulation is a level, $G = e^L\gamma/(1-\gamma)\cdot SD_k$, with $\gamma = \gamma_k(1.5 - c_{it})$ bounded between 0.02 and 0.9, so that it falls with capacity. Unscored tasks decay towards zero and lose 0.05ξ per unit of relative induced effort each year.

In the manipulation scenarios, γ is multiplied by 0, 0.5 or 1.5, audits remove 60% of G, and, with substitution, the effort that no longer feeds manipulation becomes real improvement.

Capacity starts at $c_{i0} = \Phi(z_i + 0.5u_i)$, where z_i is a standardised index of log population (40%), the share of adults with higher education (40%) and log income (20%). A commissioner raises capacity to $c_{i0} + \chi(1 - c_{i0})\cdot l$, where $l = \min(1, 1{,}000/\text{hires a year})^{0.3}$ discounts quality when hiring is fast. The commissioner also adds programmes worth $\beta_{TA}q_{TA}(1 - c_{i0})l$ standard deviations a year in literacy, and a share of that in daycare and sewage, decaying 5% a year; β_{TA} is calibrated on PAIC, and q_{TA} is the share of PAIC's effectiveness that a national corps reproduces. Tutelage adds capacity and slows execution.

Implementation complementarity, finally, follows equation (10) of the article. The money stocks of Section S7.3 and the assistance flow are multiplied each year by $\iota_{ikt}/\iota^{ref}_{ikt}$, where ι follows equation (10) of the article with $I^* = 1$ alert unit, $s = 1$, supervision when a commissioner is present or the municipality is in tutelage, and I the total incentive of equation (S7.2), which is zero for banned mayors. The reference ι^{ref} is the same expression for a supervised mayor facing 1.75 alert units, the median incentive under the LRG as proposed, with the same capacity and competence, so that equation (10) of the article acts relative to that reference and counts no effort twice. With $\eta = 0$ the ratio is one, and the model is the reference case.

S7.3. Money

The tranche pays back a share T of the FPM in proportion to positive points times the municipality's own FPM (or its population, in a variant). Each municipality therefore receives T × total FPM × $w_i/\Sigma w_j$ and gives up T × its own FPM. Tied amendments make a share ($w_{sq} - w_{lrg}$) of 70% of each municipality's amendments newly productive.

Productive money, $m = (1 - w_{gen})$ × net transfer + tied gain, is then spent in fixed proportions. It goes 35% to sewage, a stock equal to 100m divided by the cost per person connected, times the uncovered share; 25% to daycare, a flow of places at the cost per child-year, over 7.5% of the population aged 0 to 5; 20% to literacy programmes, a decaying stock with diminishing returns; and the rest elsewhere. Net transfers, that is, the tranche and redirected amendments but not the reallocation of amendments between uses within a municipality, also raise formal jobs through a local multiplier at the cost per formal job-year, as transfers of the FPM itself have been shown to do (Corbi, Papaioannou and Surico, 2019). The stocks bought by net transfers (the money-volume channel) and by tied amendments (the allocation channel) are tracked separately.

With the personal bonus, the mayor is paid b times her annual salary times the positive part of the standardised published points each year, computed on the LRG's own scorecard. Salaries are an assumption by size class: R$ 15,000, 18,000, 22,000 and 28,000 a month up to 10,000, 50,000 and 200,000 inhabitants and above, times 13.33.

S7.4. Elections, parties and clientelism

Eligible incumbents run with probability 0.82 and win with probability $\Lambda(\text{logit } 0.81 + \kappa_{cl}(v_i D_i/\bar{v} - 1) + \beta z^{perf}{}_i + \beta_{info} z^{pts}{}_i)$. In this expression, v is vulnerability, scaled to the 13.4% of Brazilians offered a benefit for their vote in 2010, and $\bar{v}$ its national mean at the start, so that emancipation lowers clientelist capacity. D is the ratio of discretionary money to the status quo ($1 - 0.5 \times 0.7$ when amendments are tied, minus 0.25 in tutelage); z^{perf} is the standardised change in the measured indicators since the start of the mandate, weighted as the points; and z^{pts} is the standardised published points. Together these terms approximately reproduce the 66.6% re-election rate of eligible mayors in 2024. The split between running and winning is an assumption; the match of 2020 and 2024 winners suggests values of about 0.80 and 0.83.

Open seats stay with the incumbent's party with probability 0.236, the 2020 to 2024 rate, and otherwise go to a party drawn in proportion to its regional strength in 2024 times the growth of its campaign funds to the power η_f. New mayors' competence is drawn from N(h, 1), where h combines four shifts. It adds party recruitment in proportion to the party's stake in the municipality (the full shift at about 200,000 inhabitants, proportionally less below; party pillar). It subtracts deterrence proportional to the wrongful bans observed so far (sanctions pillar). It subtracts ζ per 10% of current revenue gained in discretionary money, that is, net transfers minus tied amendments (Brollo et al., 2013). And, with the personal bonus, it adds ς times the expected bonus as a share of pay (Gagliarducci and Nannicini, 2013).

Mayors switch parties at 15% a year (48% over a mandate, against 49% observed between 2020 and 2024), plus a premium for high points under the party pillar, and destinations are drawn in proportion to party size times the growth of party funds. Vulnerability, finally, falls with formal jobs with elasticity η_{emp}.

S7.5. Designs, scenarios and checks

The evaluation compares designs, that is, versions of the world that differ only in the rules in force. All designs share the same parameter draws and random shocks, so the differences between them are measured much more precisely than their levels. The reference design is the status quo (W0), which has no law but has elections, to which mayors who can run again respond as they do today. A placebo world applies the rules of the law, but nobody responds and nothing follows from them, so it measures regression to the mean and the statistical properties of the rules. The law as proposed switches on all five instruments.

To learn what each instrument contributes, the simulation runs the full factorial of the five behavioural instruments: sanctions (S); the FPM tranche (F) and conditional, tied amendments (A), which together form the money pillar; commissioners (C); and party funds (P). The scorecard, which is the measurement pillar, is published in all 32 combinations. S comprises alerts, tutelage (with co-

signature, extra support and slower execution) and bans, and A also routes the amendments of municipalities in tutelage through consortia. From these designs the analysis computes each instrument's Shapley value, which is its average marginal contribution over all the orders in which the instruments could be added (Shapley, 1953), together with its effect on its own and its marginal effect in each of the 16 configurations of the other four. The simulation also runs single changes to the proposal (no tolerance, a red band at 1.5 instead of one standard deviation below the floor, no "no improvement" rule, slower hiring of commissioners, a tranche by population, amendments redirected away from municipalities in tutelage, party funds by colour bands, half of the tied money leaking back into discretionary channels, a mayor component with five times the estimated variance and a new party entering the system) and FPM tranches of 2.5%, 5% and 20%.

A further set of designs tests the sanctions by changing only them and keeping every other instrument of the proposal. These designs remove the bans for results while keeping alerts and tutelage; remove every performance sanction; threaten bans but never execute them, which isolates their deterrent effect; execute bans that nobody anticipates, which isolates the effects of executing them; and set the weight of a ban in mayors' objective at zero, half, twice, four and eight times its calibrated value. The same tests are repeated with the sanctions alone, on the published scorecard with no other instrument. As a theoretical benchmark only, a personal bonus of a quarter, a half and one annual salary per SD of improvement in the LRG's own scorecard measures what pure personal pay could do (Section S8.9). This makes 66 designs in all.

What the evidence of Section 6 of the article cannot settle is varied in families of scenarios, each simulated for the designs it concerns on the first 120 draws and with the same shocks. The families cover implementation complementarity η of 0.25, 0.5 and 0.75 (scenarios B, C and D; under D the full factorial is also run); voters' reward for performance at zero, twice and four times its calibrated value; manipulation absent, halved, raised by half or cut by 60% by audits, with or without the effort it absorbs going into real improvement; banned mayors who keep responding to money and party funds; and the calibration variants of Section S6.6. To ask whether the behavioural response could be larger than estimated, they also include effort that responds 1.5, 2 and 3 times as strongly to incentives and money that weighs 2 and 3 times as much (both without recalibration, so that their fit to the evidence can be checked), and effort that saturates twice as fast, twice as slowly or not at all (each recalibrated). A variance-based sensitivity analysis over the calibrated ranges, including η, shows which assumptions drive which results (Saltelli et al., 2008; Herman and Usher, 2017).

One check of the simulated world is not reported in the article. The calibration targeted the re-election rate of eligible mayors, but not the share of municipalities in which a new mayor from a different party takes office, which is 41% at each election in the simulated status quo against 45% in 2024; the real figure also counts outgoing mayors who had changed party during the mandate, which the simulated one does not.

S7.6. The parameters: priors, evidence and posteriors

Table S7.1 groups the parameters by the mechanism they govern and states, for each group, the prior, the posterior and the evidence behind them; Table S7.2 then lists all 47 parameters that the simulation treats as uncertain, with their units, priors and posteriors after the calibration of Sections S6.5 and S6.6. Most posteriors remain close to their priors, which is why the article varies the most consequential assumptions in families of scenarios (Section 9.3 of the article).

Table S7.1. The model's parameters: priors, evidence and posteriors.

Parameter	Prior	Posterior	Evidence
Response of learning to incentives (δ for literacy, SD per incentive unit)	0.02 to 2.00	0.52 [0.09 to 1.59]	Ceará's results-based ICMS rule, alone and with assistance (targets M1, M2); Ceará twice as responsive as the nation, as a variant; every δ ×1.5 to ×3 in Section 9.2 of the article
… daycare; sewage; homicides; jobs and value added	0.02 to 0.30; 0.01 to 0.10; 0.00 to 0.06; 0.00 to 0.03	prior	No Brazilian estimate: investment cycles; policing is a state function; mayoral term shares of 0% to 3% (Table 3 of the article)
Electoral pressure on mayors who can run again (alert units); voters' reward for performance (logits per SD)	0.00 to 1.00; 0.00 to 0.30	0.47 [0.05 to 0.94]; 0.14 [0.08 to 0.21]	Close races of 2020 with and without the incumbent, and re-election and performance, 2020 and 2024 (M3, M3o, M4); voters' reward from zero to four times, as scenarios
Additional effect of the published score on re-election (logits per SD)	0.00 to 0.10	prior	Routine information ≈ 0 (Dunning et al., 2019)
Pressure of a 10%-of-FPM tranche at the median FPM per inhabitant (alert units)	0.20 to 1.00	0.56 [0.23 to 0.96]	Identified only jointly with δ (M1)
Weight of tutelage and ban relative to an alert	1.50 to 3.00; 2.00 to 6.00	prior	Assumption; matters for what the bans add, not for the size of the effect; the weight of a ban is varied from zero to eight times
Manipulated share of the measured response	5% to 60% by indicator	prior	Test-score gaming (Jacob and Levitt, 2003); suspicious reported data under targets (Ghanem and Zhang, 2014; Greenstone et al., 2022); bunching at the FPM brackets (Section S6.3); none to 1.5 times, and audits, as scenarios
Implementation complementarity η (equation 10 of the article)	0 (scenario A)	not identified	0.25, 0.5, 0.75 as scenarios B to D, 0 to 0.75 in the sensitivity analysis; 36% larger impact of a programme run by mayors who could run again (de Janvry, Finan and Sadoulet, 2012) ≈ η of 0.5 to 0.75
Technical assistance in a weak administration (SD a year); share of it the commissioners reproduce	0.00 to 0.40; 0.25 to 1.00	0.17 [0.04 to 0.33]; prior	PAIC in Ceará (M2); a national corps is not a state programme: assumption
Capacity: scaling of the real response; share of the gap a commissioner closes	0.30 to 1.00; 0.10 to 0.50	prior	Management reform without incentives had no effect in India (Muralidharan and Singh, 2020)
Disruption after a change of ruling party (SD of learning); its persistence (a year)	0.02 to 0.50; 0.50 to 0.95	0.27 [0.06 to 0.48]; 0.71 [0.53 to 0.91]	Learning improves more where the incumbent or the party stays (M3, M3o); Akhtari, Moreira and Trucco (2022)
Persistence of real gains (a year); competence × effort	0.50 to 0.95; 0.00 to 0.40	0.79 [0.54 to 0.94]; prior	Speed at which Ceará's effects built up (M1, M2); Olken, Onishi and Wong (2014)

Parameter	Prior	Posterior	Evidence
Unproductive share of amendments now; when tied; of general transfers	0.40 to 0.70; 0.15 to 0.35; 0.30 to 0.60	prior	Electoral motives misallocate 26.8% of amendment funds (Finan and Mazzocco, 2021); FPM brackets (M5, weak). Tying is assumed to halve the unproductive share
Unit costs: sewer connection (R$/person); daycare place (R$/child-year); formal job (R$/job-year)	2,500 to 5,000; 7,000 to 14,000; 60,000 to 140,000	prior	Engineering and fiscal benchmarks; FPM brackets (M5, weak)
Selection: party recruitment; deterrence by wrongful bans; discretionary money; pay (SD of competence)	0.00 to 0.30; 0.00 to 0.20; 0.00 to 0.10 per 10% of revenue; 0.00 to 0.30 per 100% of pay	prior	Dal Bó, Finan and Rossi (2013); Brollo et al. (2013); Gagliarducci and Nannicini (2013)
Clientelism: effect on re-election (logits); elasticity of vulnerability to formal jobs	0.20 to 1.00; 0.20 to 1.00	prior	Bobonis et al. (2022); Avis et al. (2022)
Value of office (mandates' pay), for the bonus benchmark and the salary equivalent of a ban	2.00 to 5.00	prior	Assumption
Calibrated, not drawn			Re-election of eligible incumbents 66.6% (2024); party switching 15% a year (48% over a mandate; 49% observed); open seats keep the party 24%; amendments R$ 32 billion a year to municipalities; FPM R$ 216 billion; Fundo Partidário R$ 1.23 billion; FEFC R$ 4.96 billion; commissioner R$ 350,000 a year (assumption); mayors' pay by size class (assumption); noise and signal by population class (Table 3 of the article)

Notes: Priors are uniform over the ranges shown. Posterior: median [5th to 95th percentile] after sampling-importance-resampling on the targets of Section S6.5; "prior" where the evidence does not move the parameter. SD: between-municipality standard deviation of the indicator (Section 7.1 of the article).

Table S7.2. The 47 uncertain parameters: uniform priors and calibrated posteriors.

Parameter	Unit	Prior	Posterior median [5th to 95th percentile]
Response of the true indicator to one incentive unit (δ): formal jobs	SD	0 to 0.03	0.015 [0.001 to 0.028] (≈ prior)
… non-public value added	SD	0 to 0.03	0.015 [0.001 to 0.029] (≈ prior)
… literacy	SD	0.02 to 2	0.52 [0.088 to 1.59]
… daycare and preschool	SD	0.02 to 0.3	0.17 [0.036 to 0.29] (≈ prior)
… sewage	SD	0.01 to 0.1	0.055 [0.015 to 0.096] (≈ prior)
… homicides	SD	0 to 0.06	0.03 [0.003 to 0.057] (≈ prior)
Manipulated share of the response: formal jobs	share	0.05 to 0.35	0.2 [0.064 to 0.33] (≈ prior)
… non-public value added	share	0.1 to 0.5	0.3 [0.12 to 0.48] (≈ prior)
… literacy	share	0.15 to 0.5	0.32 [0.17 to 0.48] (≈ prior)

Parameter	Unit	Prior	Posterior median [5th to 95th percentile]
… daycare and preschool	share	0.05 to 0.3	0.17 [0.062 to 0.29] (≈ prior)
… sewage	share	0.2 to 0.6	0.4 [0.22 to 0.58] (≈ prior)
… homicides	share	0.05 to 0.4	0.22 [0.067 to 0.38] (≈ prior)
Weight of tutelage relative to an alert	ratio	1.5 to 3	2.23 [1.57 to 2.92] (≈ prior)
Weight of the ban relative to an alert	ratio	2 to 6	3.98 [2.2 to 5.81] (≈ prior)
Pressure from a 10%-of-FPM tranche at the median FPM per inhabitant	alert units	0.2 to 1	0.56 [0.23 to 0.96] (≈ prior)
Pressure from party funds where the party's stake is high	alert units	0 to 0.5	0.24 [0.022 to 0.47] (≈ prior)
Persistence of real gains	per year	0.5 to 0.95	0.79 [0.54 to 0.94] (≈ prior)
Interaction of competence and effort	per SD	0 to 0.4	0.2 [0.021 to 0.38] (≈ prior)
Neglect of unscored tasks	index	0 to 0.3	0.15 [0.016 to 0.28] (≈ prior)
Scaling of the real response with capacity	slope	0.3 to 1	0.68 [0.34 to 0.97] (≈ prior)
Share of the capacity gap closed by a commissioner	share	0.1 to 0.5	0.3 [0.12 to 0.48] (≈ prior)
Effect of PAIC-like assistance on learning in a weak administration	SD a year	0 to 0.4	0.17 [0.037 to 0.33]
… share of it in daycare and sewage	share	0 to 0.3	0.15 [0.016 to 0.29] (≈ prior)
Share of PAIC's effectiveness reproduced by commissioners	share	0.25 to 1	0.63 [0.29 to 0.96] (≈ prior)
Capacity support during tutelage	share	0 to 0.2	0.099 [0.01 to 0.19] (≈ prior)
Execution drag during tutelage	SD a year	0 to 0.02	0.01 [0.001 to 0.019] (≈ prior)
Unproductive share of amendments today	share	0.4 to 0.7	0.55 [0.42 to 0.68] (≈ prior)
… when tied to programmes	share	0.15 to 0.35	0.25 [0.16 to 0.34] (≈ prior)
Unproductive share of general transfers	share	0.3 to 0.6	0.46 [0.32 to 0.59] (≈ prior)
Cost of connecting a person to sewage	R$	2,500 to 5,000	3,768 [2,629 to 4,875] (≈ prior)
Cost of a daycare or preschool place	R$ a year	7,000 to 14,000	10,834 [7,449 to 13,684] (≈ prior)
Literacy return to programme spending	points per log R$	0.2 to 1	0.59 [0.24 to 0.96] (≈ prior)
Cost of a formal job-year through local spending	R$	60,000 to 140,000	101,047 [64,134 to 135,882] (≈ prior)
Effect of clientelist capacity on re-election	logits	0.2 to 1	0.6 [0.24 to 0.96] (≈ prior)
Elasticity of vulnerability to formal jobs	elasticity	0.2 to 1	0.6 [0.24 to 0.96] (≈ prior)
Additional effect of the published score on re-election	logits per SD	0 to 0.1	0.05 [0.006 to 0.094] (≈ prior)
Competence shift of new mayors from party recruitment (full stake)	SD	0 to 0.3	0.15 [0.015 to 0.29] (≈ prior)
Deterrence of competent entrants by wrongful bans	SD	0 to 0.2	0.1 [0.009 to 0.19] (≈ prior)

Parameter	Unit	Prior	Posterior median [5th to 95th percentile]
Disruption of learning after a change of ruling party	SD	0.02 to 0.5	0.27 [0.055 to 0.48] (≈ prior)
Persistence of that disruption	per year	0.5 to 0.95	0.71 [0.53 to 0.91]
Elasticity of open-seat wins and switching to party-fund growth	elasticity	0 to 0.3	0.15 [0.014 to 0.28] (≈ prior)
Premium in party switching for high-scoring mayors	per point	0 to 1	0.5 [0.047 to 0.95] (≈ prior)
Electoral pressure on mayors who can run again	alert units	0 to 1	0.47 [0.048 to 0.94] (≈ prior)
Voters' reward for measured performance	logits per SD	0 to 0.3	0.14 [0.075 to 0.21]
Competence shift per 10% of revenue in discretionary money (negative)	SD	0 to 0.1	0.049 [0.005 to 0.095] (≈ prior)
Competence shift per 100% of pay (personal bonus)	SD	0 to 0.3	0.15 [0.016 to 0.28] (≈ prior)
Value of office, in mandates' pay (personal bonus)	mandates	2 to 5	3.55 [2.16 to 4.86] (≈ prior)

Notes: "≈ prior": the 90% posterior interval is less than 5% narrower than the prior's. SD: standard deviation of the indicator between municipalities.

S7.7. The chain from municipal outcomes to the Chamber

The illustrative chain of Sections 8.5 and 10.1 of the article links municipal outcomes to the composition of the Chamber under two assumptions. The first is that a share b of all Chamber votes is delivered by mayors, in proportion to the population that each party's mayors govern, and that this brokerage vote scales with national clientelist capacity CC (vulnerability times discretionary money); the rest of each party's vote does not depend on mayors. The second is that campaign money multiplies the vote with elasticity e. Party p's vote is then

$$V_p \propto \left[V_{p0} - b\,M_{p0} + b\,M_p \frac{CC}{CC_0}\right]\left(\frac{F_p}{F_{p0}}\right)^e \tag{S7.4}$$

where V_{p0} is its 2022 Chamber vote share, M_{p0} and M_p the shares of the population governed by its mayors in 2024 and 2036, and F_{p0} and F_p its shares of the electoral fund now and in 2036. The parameters $b \sim U(0.05, 0.25)$ and $e \sim U(0.05, 0.30)$ are drawn independently for each simulation draw, and the effect is the difference between each design and the status quo in 2036. Under these assumptions, tied amendments alone would lower the centrão parties' share of the Chamber vote by 1.1 percentage points [0.4 to 2.2], whereas the full package would change it by −0.1 points [−1.3 to 1.0] on a base of about 69%. Because both parameters are assumptions, the chain is illustrative.

S8. Additional simulation results

Section 8 of the article reports what the simulation says the LRG as proposed would do. This section gives the details and the complete tables behind that summary, in the same order. It shows where the bans fall and how the details of the rules change them (Section S8.1); the full factorial of the instruments, their Shapley values, their channels and the incentive that each of them creates (Section S8.2); the effects indicator by indicator (Section S8.3); where the money of the FPM tranche goes and how its effect scales with its size (Section S8.4); the commissioners (Section S8.5); parties, who governs and clientelism (Section S8.6); what each instrument costs (Section S8.7); and single changes to the proposal (Section S8.8). It closes with a personal bonus for mayors, simulated as a theoretical benchmark (Section S8.9), and with the rules that turn the results into the assessments of Table 10 of the article (Section S8.10).

The conventions are those of Section 7.2 of the article. The LRG as proposed is the only policy evaluated, and every other design switches its instruments on and off, or changes its sanctions, to learn where its effects come from. Unless stated otherwise, results are medians over the 400 calibrated draws, with 5th to 95th percentiles in brackets. The brackets describe what remains uncertain about parameters and shocks once the model has been calibrated on the evidence. They are not confidence intervals of an estimate from data, and they do not cover the assumptions that the evidence cannot test, which Section 9 of the article and Section S9 examine separately. Effects are differences from the status quo in the same draw, averaged over the two full mandates from 2029 to 2036 and over municipalities, and expressed in between-municipality standard deviations (SD) of each indicator's true value, as defined in Section 7.1 of the article. Scenario runs use the first 120 draws with the same random shocks.

S8.1. Where bans fall, and how the rules change them

Section 8.1 of the article reports that the rules as proposed would declare 24.3 [22.7 to 26.0] mayors in every hundred ineligible in each mandate, and that few of these bans would be attributable to the mayor in office. Over the two mandates from 2029 to 2036, 38% of municipalities would see at least one mayor banned. The share of wrongful bans, those that an error-free reading of the indicators would not trigger, is 32.7% [31.0% to 34.4%], and banned mayors are of below-average competence 51% of the time [49% to 54%], against 50% for mayors drawn at random.

This result does not depend on the details of the rules. The "no improvement" rule alone adds 6.6 bans per hundred mayors per mandate; comparing the proposal with and without it, about 96% of the bans it adds are wrongful, and without it bans fall to 17.8 per hundred. Setting the red band at 1.5 instead of one standard deviation below the floor leaves 17.9 bans per hundred, and removing the statistical tolerance altogether leaves 21.4. In every version, banned mayors are of below-average competence in 49% to 51% of cases. The conclusion also survives a much larger role for mayors. With

a mayor component five times larger in variance than estimated, which is within the range that the data cannot exclude for changes (Section S4.3), 4% of bans are attributable and banned mayors are of below-average competence 52% of the time.

Figure S8.1 shows where the bans would fall. Mayors in the North (38 bans per hundred per mandate) and Northeast (36) would be banned 2.4 times as often as those in the South (14) and Southeast (17), because more of their municipalities start far below national floors. By size the rate varies less, from 29 per hundred below 5,000 inhabitants to 24 above 100,000, but the composition of the bans changes: 43% of the bans in the smallest municipalities are wrongful, against 10% in the largest, where bans are mostly structural.

Many bans would also be triggered by indicators that mayors control little. In municipalities up to 20,000 inhabitants, 40% of bans involve formal jobs and 33% non-public value added. Above 100,000 inhabitants, 43% involve homicides, a state responsibility, and 33% involve sewage, which depends on state companies and long investment cycles.

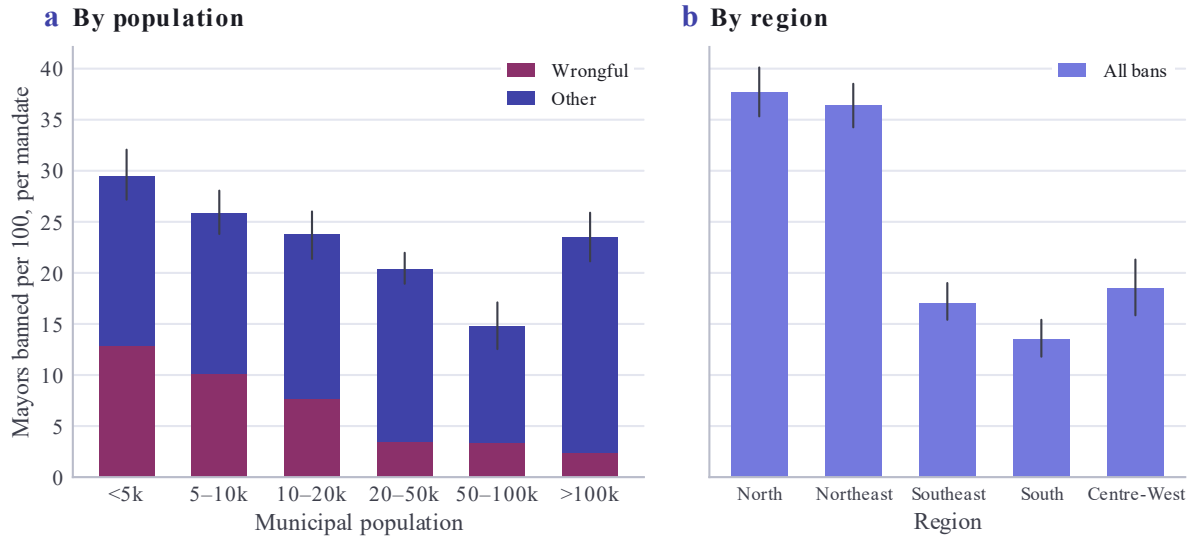


Figure S8.1. Bans per 100 mayors per mandate, by population and region. LRG as proposed, 2029 to 2036; wrongful bans (the error-free reading would not trigger them) in plum. Medians and 5th to 95th percentiles over 400 draws.

S8.2. The instruments: factorial, Shapley values, channels and incentives

Table S8.1 reports the 32 combinations of the five instruments, with the scorecard published in every combination, each measured against the status quo. Two of its columns need explanation. "Classified for tutelage" is the share of municipality-years that the scorecard classifies for tutelage, which is computed whenever the law is in force; "applied" is the share in which tutelage has consequences, which requires the sanctions (S) to be on. The Shapley values of Table 6 of the article and the marginal effects of Section S9.2 are computed from these designs, draw by draw.

Table S8.1. The 2^5 factorial: every combination of the five instruments.

Design (S F A C P)	Real effect (SD)	Pop.-weighted (SD)	Measured (SD)	Effort (SD)	Cosmetic gap (SD)	Bans per 100 per mandate	Classified for tutelage (%)	Tutelage applied (%)
· · · · ·	+0.003	+0.004	+0.006	+0.003	+0.003	0.0	86%	0%
· · · · P	+0.006	+0.025	+0.010	+0.006	+0.004	0.0	86%	0%
· · · C ·	+0.037	+0.018	+0.041	+0.004	+0.001	0.0	85%	0%
· · · C P	+0.041	+0.039	+0.044	+0.007	+0.002	0.0	85%	0%
· · A · ·	+0.082	+0.043	+0.094	+0.008	+0.008	0.0	85%	0%
· · A · P	+0.085	+0.063	+0.096	+0.011	+0.009	0.0	85%	0%
· · A C ·	+0.119	+0.056	+0.124	+0.010	+0.004	0.0	84%	0%
· · A C P	+0.122	+0.077	+0.128	+0.013	+0.005	0.0	84%	0%
· F · · ·	+0.058	+0.016	+0.092	+0.028	+0.032	0.0	84%	0%
· F · · P	+0.058	+0.035	+0.093	+0.030	+0.033	0.0	84%	0%
· F · C ·	+0.098	+0.030	+0.126	+0.033	+0.024	0.0	83%	0%
· F · C P	+0.099	+0.051	+0.127	+0.035	+0.025	0.0	83%	0%
· F A · ·	+0.135	+0.054	+0.176	+0.031	+0.033	0.0	83%	0%
· F A · P	+0.136	+0.074	+0.177	+0.033	+0.034	0.0	83%	0%
· F A C ·	+0.176	+0.068	+0.208	+0.036	+0.023	0.0	82%	0%
· F A C P	+0.176	+0.088	+0.209	+0.038	+0.024	0.0	82%	0%
S · · · ·	+0.036	+0.039	+0.079	+0.037	+0.040	22.3	85%	85%
S · · · P	+0.038	+0.052	+0.083	+0.039	+0.040	22.3	85%	85%
S · · C ·	+0.075	+0.055	+0.112	+0.041	+0.030	21.9	84%	84%
S · · C P	+0.077	+0.068	+0.115	+0.043	+0.030	21.8	84%	84%
S · A · ·	+0.113	+0.074	+0.153	+0.037	+0.036	21.0	84%	84%
S · A · P	+0.114	+0.087	+0.156	+0.039	+0.037	21.0	84%	84%
S · A C ·	+0.150	+0.088	+0.182	+0.040	+0.026	20.7	83%	83%
S · A C P	+0.152	+0.102	+0.184	+0.043	+0.027	20.6	83%	83%
S F · · ·	+0.077	+0.046	+0.135	+0.053	+0.050	26.2	83%	83%
S F · · P	+0.077	+0.058	+0.136	+0.054	+0.051	26.2	83%	83%
S F · C ·	+0.120	+0.061	+0.167	+0.057	+0.039	25.8	83%	83%
S F · C P	+0.120	+0.074	+0.168	+0.058	+0.039	25.7	83%	83%
S F A · ·	+0.153	+0.081	+0.209	+0.053	+0.047	24.8	82%	82%
S F A · P	+0.153	+0.094	+0.210	+0.054	+0.047	24.8	82%	82%
S F A C ·	+0.195	+0.095	+0.241	+0.057	+0.035	24.3	82%	82%
S F A C P	+0.196	+0.108	+0.242	+0.058	+0.036	24.3	82%	82%

Notes: S: sanctions (alerts, tutelage, bans); F: FPM tranche; A: tied amendments; C: commissioners; P: party funds; "·": instrument off. Medians over 400 draws. The cosmetic gap is the median of the per-draw difference between measured and real effect, as in Tables 6 and 7 of the article, and it is smaller than the difference between the medians because manipulation is skewed across draws. The design with all five instruments is the LRG as proposed.

The factorial also gives the effect of each instrument on its own. Added alone to the published scorecard, the instruments raise the indicators by +0.078 (tied amendments), +0.053 (the tranche),

+0.033 (commissioners), +0.032 (sanctions) and +0.002 SD (party funds). Together they add +0.191 SD, so interactions subtract 0.014 SD. The instruments partly overlap, because sanctions, the tranche and party funds all act on the same effort, which responds less and less to further incentive (equation 7 of the article), and the overlap falls mostly on the sanctions. Alone, the sanctions add +0.032 SD; as a Shapley value, +0.024; and on top of the other four instruments, +0.017 (Section S9.2). Figure S8.2 draws the Shapley values for six outcomes.

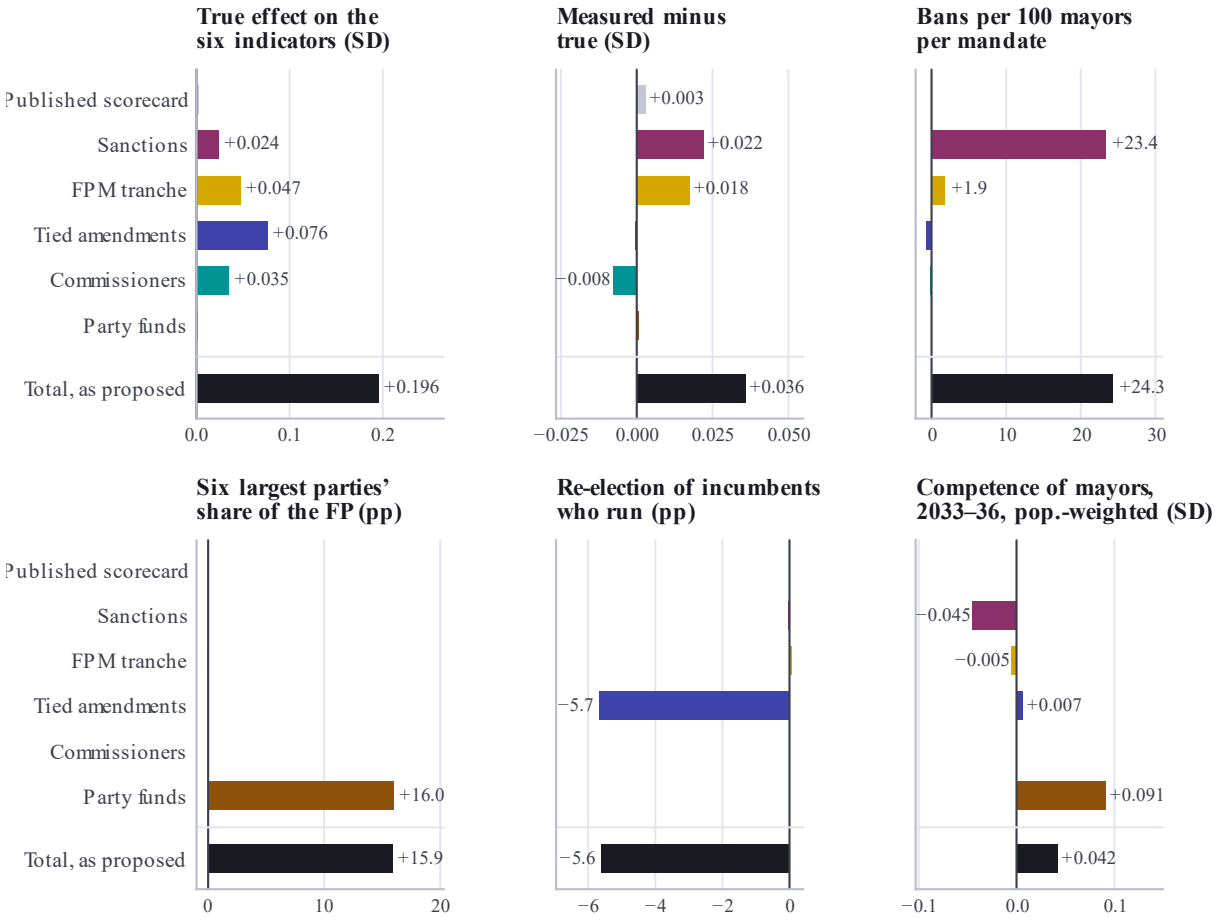


Figure S8.2. What each instrument contributes. Shapley values of each instrument (and of publishing the scorecard alone) for six outcomes; "total" is the effect of the law as proposed against the status quo. Medians over draws.

Two further patterns appear in Table 6 of the article. Weighted by population, the FPM tranche contributes little (+0.008 SD), because its incentive and its money matter only where the FPM is a large share of revenue, whereas sanctions and party funds, whose incentives do not fade with size, contribute +0.027 and +0.015. Sanctions and the tranche, which press on the scores themselves, account for most of the cosmetic gap between measured and real improvement (+0.022 and +0.018 of +0.036 SD, while commissioners reduce it) and of the loss in unscored tasks (−0.023 and −0.017 of −0.050 SD). Sanctions alone, finally, account for almost all the bans.

Two definitions help to read that table. Its row of measured minus true is the median of the per-draw gap, which is smaller than the gap between the medians of Table 5 of the article (+0.242 against +0.196) and than the manipulation stock (+0.045), because manipulation is skewed across draws. Its channels are the stocks that each mechanism of Section 3.4 of the article adds to true quality, before the indicators' natural bounds (no coverage above 100%) are applied; the selection channel is below 0.001 SD for every instrument and is omitted.

The channels of Figure 9 of the article can also be read in SD. Under the rules as proposed, the real improvement in the indicators comes from the allocation of tied money (+0.082 SD), from mayors' effort (+0.058 SD), from technical assistance (+0.037 SD) and from the volume of money that the tranche moves to where it buys more (+0.029 SD). Disruption subtracts 0.003 SD, and manipulation adds +0.045 SD to the measured indicators. Weighted by population, the money-volume channel is negative (−0.003 SD), because the tranche moves money away from the large municipalities where most people live.

Each instrument works through its own channel (Figure 9b of the article). Sanctions act almost only through effort (+0.025 SD) and come with nearly as much manipulation (+0.024); the tranche acts through effort (+0.019) and money (+0.029); tied amendments act almost only through allocation (+0.082); and commissioners act through assistance (+0.037). In all, 71% of the real improvement comes from changing what money buys, where it goes and who helps to spend it, while the effort that the incentives induce comes with substantial manipulation and is the least certain part of the result (+0.058 [0.019 to 0.104] SD).

The marginal expected incentive of equation (4) of the article explains this pattern (Figure S8.3). In the status quo, elections give a mayor who can run again an incentive of 0.19 to 0.26 alert units on each indicator, which is small because voters reward performance only a little (Section 6.2 of the article). The law would add 0.05 to 0.07 through the published score; 0.76 to 0.98 through the sanctions, on average, though concentrated on mayors close to a threshold; 1.39 through the tranche in municipalities under 5,000 inhabitants but only 0.19 above 100,000, where the FPM weighs less in the budget; 0.02 to 0.07 through the conditions on amendments; and from 0.005 to 0.30 through party funds, which reward points in proportion to population. Most of the added incentive would therefore come from the sanctions everywhere and from the tranche in small municipalities, which is where the indicators are noisiest (proposition P2) and where the share of the response that is manipulation is highest (proposition P3).

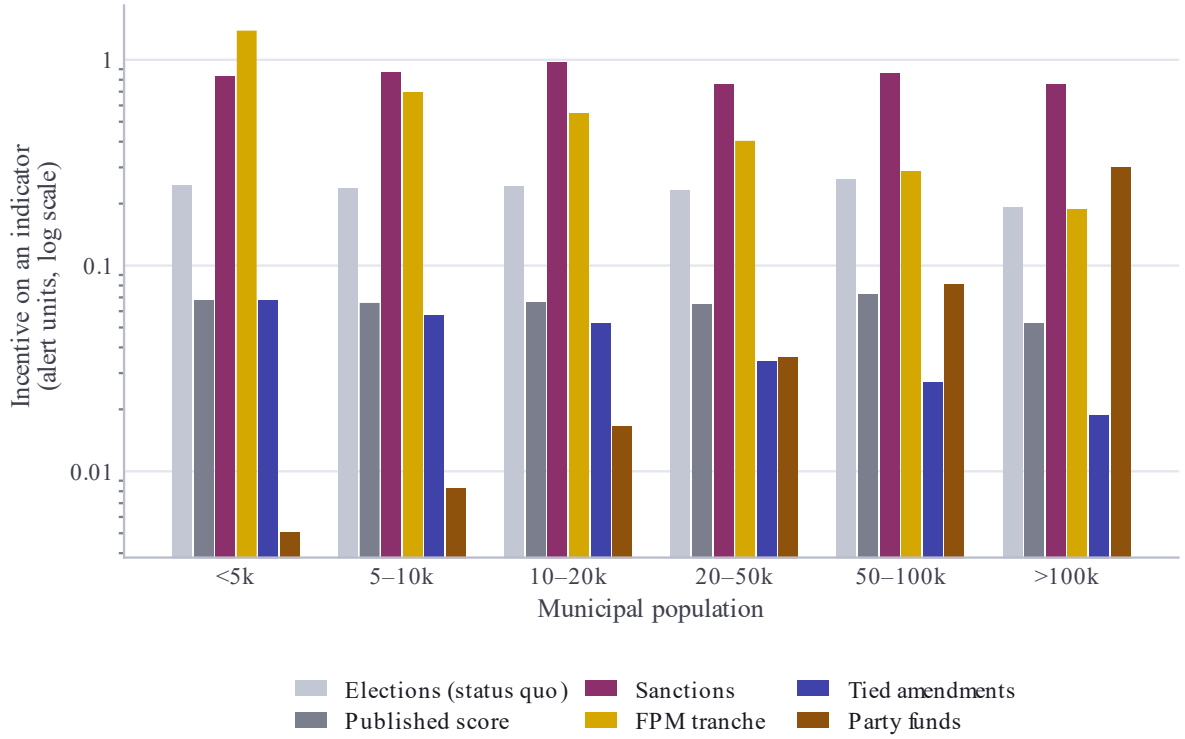


Figure S8.3. Marginal expected incentive by source and municipal size. Incentive on an indicator in alert units (equation 4 of the article), municipal means by population class, LRG as proposed, 2030; log scale. Medians over draws.

S8.3. Effects indicator by indicator

Table 5 of the article reports the effect of the LRG as proposed on each indicator, and its averages hide large differences across municipalities. Gains are largest in small municipalities (+0.328 SD below 5,000 inhabitants against +0.076 SD above 100,000) and in the weakest administrations (+0.296 SD in the fifth with the least capacity against +0.088 in the fifth with the most). These municipalities receive the most amendments and FPM per inhabitant, so tying amendments and paying the tranche change more of their budget, and commissioners close more of their capacity gap.

Relative to the differences between municipalities, the effects also differ by indicator. Literacy rises by 0.58 SD, daycare and preschool coverage by 0.36 SD and sewage coverage by 0.17 SD, whereas formal jobs rise by 0.022 SD, non-public value added by 0.004 SD and homicides fall by 0.015 SD, effects one or two orders of magnitude smaller. The Shapley decomposition by indicator explains this pattern. Daycare is moved by tied amendments (+0.20 SD) and by the tranche (+0.14), sewage by the same two instruments (+0.09 and +0.06), and formal jobs only by the volume of money that the tranche moves (+0.020). Literacy is the one indicator that every instrument moves: commissioners (+0.17 SD), tied amendments (+0.15), sanctions (+0.15) and the tranche (+0.05). It responds both to assistance and to effort, which the calibration on Ceará pins down, and it is the indicator easiest to influence at the moment of measurement. Value added and homicides, finally, depend on the local economy and on state

policing, which no instrument of the law reaches, and their administration component is between zero and 3.3% of the variance (Section 4.3 of the article).

The cosmetic part of the measured gain also varies across draws. In literacy, 22% of the measured improvement is manipulation [3% to 50%], and the services that the scorecard does not measure lose 0.050 SD by 2036 [0.006 to 0.098].

Table S8.2 reports the effect of each instrument added alone to the published scorecard, indicator by indicator, and Table S8.3 the Shapley contribution of each instrument to the LRG's effect on each indicator, in SD and, for the total, in the indicator's own units. Read together, the two tables show that money moves daycare, sewage and formal jobs, that assistance and effort move literacy, and that nothing moves value added or homicides much.

Table S8.2. Each instrument alone, indicator by indicator (SD).

Instrument alone	Formal jobs	Non-public value added	Literacy	Daycare and preschool	Sewage	Homicides (fewer)
Sanctions	+0.000 [0.000 to 0.001]	+0.000 [0.000 to 0.002]	+0.184 [0.039 to 0.368]	−0.005 [−0.013 to 0.002]	+0.003 [−0.004 to 0.012]	+0.012 [0.004 to 0.026]
FPM tranche	+0.019 [0.012 to 0.029]	+0.002 [0.000 to 0.007]	+0.074 [−0.002 to 0.195]	+0.142 [0.094 to 0.207]	+0.065 [0.041 to 0.099]	+0.004 [0.001 to 0.013]
Tied amendments	+0.000 [0.000 to 0.000]	+0.000 [0.000 to 0.000]	+0.165 [0.077 to 0.293]	+0.197 [0.084 to 0.349]	+0.093 [0.039 to 0.158]	0.000 [−0.002 to 0.002]
Commissioners	+0.000 [0.000 to 0.001]	+0.000 [0.000 to 0.001]	+0.158 [0.038 to 0.394]	+0.021 [0.002 to 0.080]	+0.016 [0.002 to 0.066]	+0.000 [0.000 to 0.001]
Party funds	+0.000 [0.000 to 0.001]	+0.000 [0.000 to 0.001]	+0.009 [0.002 to 0.027]	+0.003 [0.001 to 0.009]	+0.001 [0.000 to 0.003]	−0.002 [−0.004 to 0.000]
Published scorecard alone	+0.000	+0.000	+0.010	+0.003	+0.001	+0.001

Notes: Each instrument added to the published scorecard with no other instrument, against the scorecard alone; the scorecard alone against the status quo. Medians and 5th to 95th percentiles over 400 draws.

Table S8.3. Shapley decomposition of the LRG's effect, indicator by indicator.

Indicator	Sanc-tions	FPM tranche	Tied amend-ments	Com-mis-sioners	Party funds	Score-card	Total (SD)	Total (unit)
Formal jobs	−0.001	+0.020	+0.001	+0.001	0.000	+0.000	+0.022 [0.013 to 0.034]	+1.9 [1.1 to 3.0] %
Non-public value added	+0.000	+0.002	+0.000	+0.000	+0.000	+0.000	+0.004 [0.000 to 0.010]	+0.04 [0.00 to 0.11] pp of GDP
Literacy	+0.148	+0.055	+0.153	+0.170	+0.006	+0.010	+0.582 [0.294 to 0.949]	+5.6 [2.8 to 9.1] pp
Daycare and preschool	−0.010	+0.138	+0.198	+0.023	+0.001	+0.003	+0.364 [0.233 to 0.533]	+4.5 [2.9 to 6.5] pp
Sewage	−0.002	+0.058	+0.092	+0.017	0.000	+0.001	+0.171 [0.110 to 0.259]	+4.0 [2.6 to 6.1] pp
Homicides (fewer)	+0.011	+0.004	0.000	+0.001	−0.002	+0.001	+0.015 [0.001 to 0.036]	+0.16 [0.01 to 0.39] per 100,000

Notes: Shapley values over the 32 combinations of instruments (SD), medians over draws; they need not add up to the total median exactly. Units: formal jobs in percent, non-public value added in points of GDP, homicides per 100,000 (positive = fewer), the others in percentage points.

S8.4. The FPM tranche: where the money goes, and its size

Figure 10 of the article shows where the money of the tranche would go. Paid in proportion to points and to each municipality's own FPM, as in the baseline codification, the tranche moves money between regions, with the Northeast gaining R$ 0.65 billion a year and the Southeast losing R$ 0.85 billion, and from richer to poorer municipalities, with the poorest fifth gaining R$ 0.64 billion and the richest losing R$ 0.58 billion. Part of this redistribution happens because noisy indicators give small municipalities more chances of a large positive score. Within each group the allocation depends in part on chance, because the points behind it reflect noise and luck as well as improvement.

The variant paid in proportion to points and population, the weighting that the proposal uses for party funds, turns the tranche into a transfer from small and poor municipalities to large and rich ones. The Northeast would lose R$ 0.85 billion a year, the poorest fifth of municipalities would lose R$ 0.94 billion and the richest fifth would gain R$ 1.85 billion. Because the marginal value of money is higher where coverage is lower, this version also lowers the average effect of the package (+0.082 SD, against +0.196 as proposed), as proposition P4 predicts.

Proposition P8 asks how much of the tranche's effect comes from the condition itself, as distinct from the money. Of its Shapley contribution, +0.019 SD comes from the effort that the condition induces and +0.029 SD from the money moving towards municipalities where it buys more, which is 60% of the total. An unconditional transfer of the same money to the same places would obtain the

second part without the noise of the allocation, without the manipulation (+0.020 SD) and without the loss in unscored tasks (−0.017 SD). The condition itself accounts for the first part.

The size of the tranche matters in a simple way (Figure S8.4). Alone, with the scorecard published, tranches of 2.5%, 5%, 10% (as proposed) and 20% of the FPM raise the average indicator by +0.013, +0.026, +0.053 and +0.098 SD, so the effect grows in proportion to the stake over this range without saturating. Weighted by population, however, the effect stays small (+0.018 SD at 20%), because almost all of it occurs in small municipalities, and the share of municipalities that lose money in a typical year stays between 59% and 60% whatever the dose. Manipulation grows with the stake as well, reaching +0.051 SD at 20%. A larger stake therefore scales all three: more real effect, more manipulation and more money moved by chance.

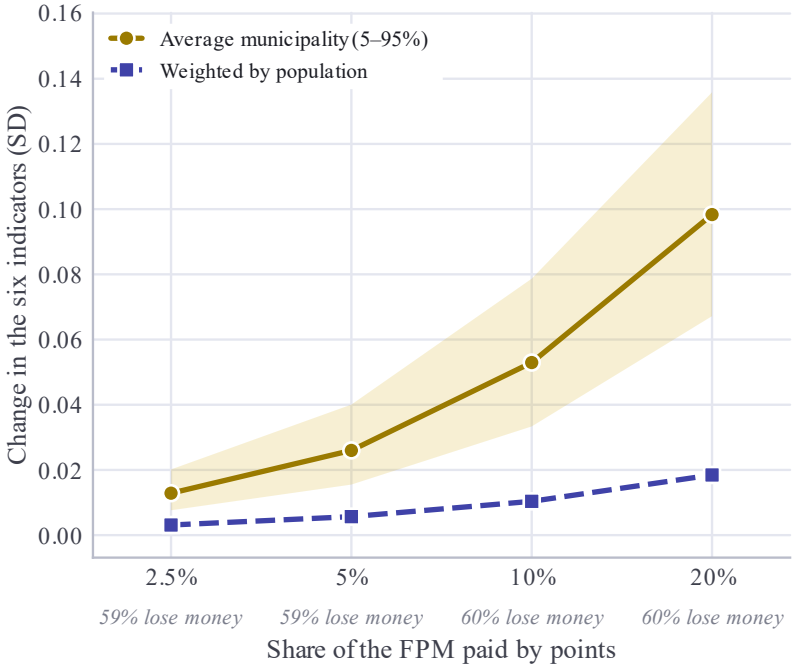


Figure S8.4. Doses of the FPM tranche. FPM tranche alone, with the scorecard published, at 2.5%, 5%, 10% (as proposed) and 20% of the FPM: effect on the six indicators, 2029 to 2036, for the average municipality (with 5th to 95th percentiles) and weighted by population, and the share of municipalities losing money in a typical year. Medians over 400 calibrated draws.

S8.5. Commissioners

Section 8.4 of the article notes that the commissioners are limited by people more than by money. One commissioner per municipality would cost about R$ 1.95 billion a year, and teams of three R$ 5.85 billion, less than 1% of the FPM. The federal career closest to the role, the Especialistas em Políticas Públicas e Gestão Governamental, had about 930 active members in 2023, and the 2024 national civil-service exam offered 150 places in it. Deploying 5,570 commissioners at once, as

proposed, would dilute their quality, whereas at 1,500 hires a year the deployment would take almost four years.

The effect of the commissioners rests on a mix of estimate and assumption. The calibration says that technical assistance can be powerful, with 0.17 SD a year in a weak administration [0.04 to 0.33] (Section S6.6). How much of that a national corps of generalists would reproduce, between 0.25 and 1.00 of the effectiveness of Ceará's PAIC, is an assumption (Table S7.1). Under that assumption, commissioners add +0.033 SD on their own [0.008 to 0.086] and +0.035 SD on average over all combinations, mostly in literacy and in the weakest administrations, and they reduce the cosmetic gap instead of raising it (Table 6 of the article), as proposition P3 predicts. Their effect builds up slowly, reaching +0.025 SD on the four indicators that a pilot could measure after two years and +0.042 after four, which matters for how they could be evaluated (Section 11.2 of the article and Section S11). Hiring 1,500 a year changes the average effect of the package only from +0.196 to +0.199 SD, because slower hiring delays commissioners but preserves their quality.

S8.6. Parties, who governs and clientelism

By 2036 the party-fund rule would have concentrated the Fundo Partidário further (Figure 11a of the article). PSD would receive 15.7% of it (7.6% under the status quo), MDB 15.0% (7.0%) and PP 12.6% (7.4%), whereas PL would receive 8.2% (17.3%), PT 3.8% (12.6%) and PSOL less than 0.2% (3.4%). The new party of Figure 11b of the article, with 3% of the Chamber vote and a 3% chance of winning each open mayoral seat, would receive 1.5% of the fund a year on average over 2027 to 2036 [1.1 to 2.9] and 2.2% in 2036 [1.3 to 4.8], a range that depends on whether it happens to win a large city.

Does the reform select better mayors? The population-weighted competence of mayors in office in 2033 to 2036 is 0.04 [−0.07 to 0.17] SD higher than in the status quo. This is the net of a gain attributable to party funds (Shapley value +0.09) and a loss attributable to sanctions (−0.05), which deter competent candidates and force turnover. In the typical municipality the change is negligible (−0.021 SD), the gain rests on the assumption that parties can identify competence before elections (Table S7.1), and selection adds less than 0.001 SD to services (Table 6 of the article), because the mayor's component of the indicators is small. Figure S8.5 draws the effects of the main designs on competence, turnover and re-election.

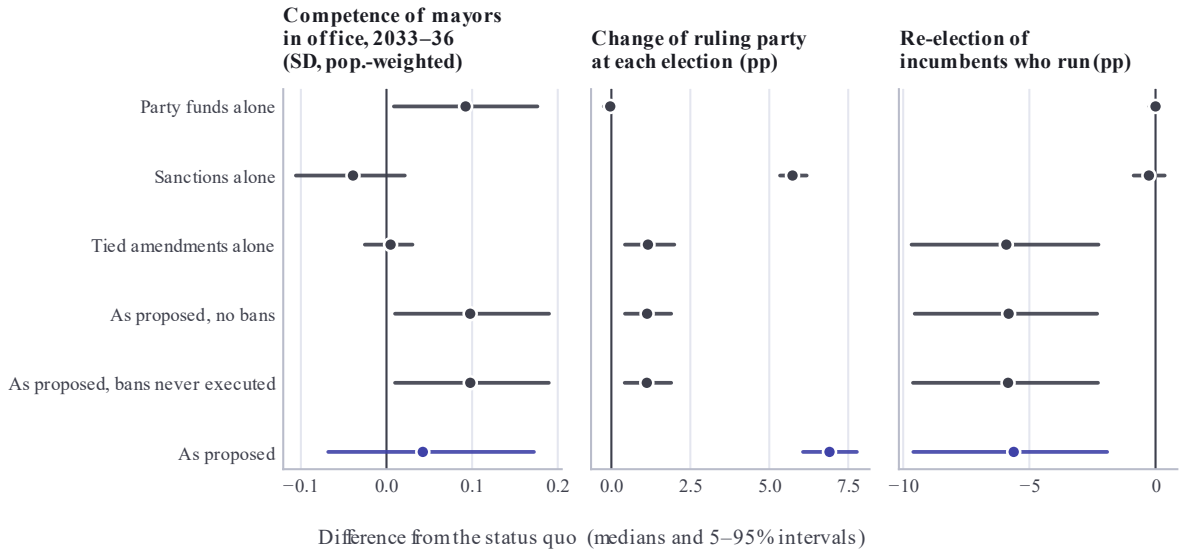


Figure S8.5. Competence, turnover and re-election under the main designs. Differences from the status quo; medians and 5th to 95th percentiles over 400 draws. Competence: population-weighted mean of the competence draw θ of mayors in office, 2033 to 2036.

The law would also raise turnover. Changes of ruling party at each election rise by 6.9 percentage points, mostly because of the bans (Shapley value of sanctions +5.8 points), and the re-election rate of incumbents who run falls by 5.6 points, because tied amendments take away clientelist resources. Each additional change of party disrupts municipal schools (Akhtari, Moreira and Trucco, 2022), by an amount that the calibration bounds only loosely (Section S6.6), and this disruption is one of the channels through which executed bans subtract from the gains (Section 9.1 of the article).

The clientelism channel works through money and hardly at all through emancipation, as proposition P6 anticipated. Tying 70% of amendments to programmes, and routing those of municipalities in tutelage through consortia, cuts the discretionary resources of mayors to 43% of today's, while the share of vulnerable households changes by 0.02 percentage points, because formal employment changes little. In the illustrative chain from municipal outcomes to the Chamber (Section S7.7), tied amendments alone would lower the centrão parties' share of the Chamber vote by 1.1 percentage points [0.4 to 2.2], whereas the full package would change it by −0.1 points [−1.3 to 1.0], because the party-fund rule hands back to those parties, in campaign money, much of what the conditions on amendments take from their brokers.

S8.7. What each instrument costs

An instrument can be judged by what it achieves per real and by what else it costs (Table S8.4). Only one of the LRG's instruments costs the Treasury money: commissioners cost R$ 1.95 billion a year and buy 0.017 SD per billion [0.004 to 0.044]. The FPM tranche and tied amendments cost the Treasury nothing, because they reallocate money that municipalities already receive. Per R$ 10 billion

a year, tying amendments buys 0.035 SD and the tranche 0.025. Each of them has a cost of another kind: the tranche makes 60% of municipalities lose money in a typical year, and tying amendments takes discretion from deputies and senators, which is why it needs their votes (Section 10.1 of the article).

Sanctions cost no money, but they ban 24 mayors in every hundred per mandate, 98% of them for reasons other than their management, and they add about as much cosmetic as real improvement. Party funds cost nothing and add little to services (+0.002 SD on average, +0.019 weighted by population), while moving 16 percentage points of the Fundo Partidário to the six largest parties. None of these figures includes the cost of the agency that would compute the scorecard or audit the data, which any version of the law would need. A personal bonus for mayors, which the proposal does not contain, is compared with these instruments in Section S8.9.

Table S8.4. Effect and cost of each instrument.

Instrument (alone)	True effect (SD)	Cos-metic gap (SD)	Fiscal cost (R$ bn a year)	Money real-located (R$ bn a year)	SD per R$ 1 bn of fiscal cost	SD per R$ 10 bn real-located	Other costs
Sanctions	+0.032 [0.008 to 0.064]	+0.035	0	n.a.	n.a.	n.a.	24 bans per 100 mayors per mandate (as proposed); 1.5% attributable
FPM tranche (10%)	+0.053 [0.033 to 0.079]	+0.028	0	21.6	n.a.	0.025	60% of municipalities lose money in a typical year
Tied amendments (70%)	+0.078 [0.039 to 0.119]	+0.003	0	22.4	n.a.	0.035	Discretion of deputies and senators; the conditions need a constitutional amendment
Commissioners (5,570)	+0.033 [0.008 to 0.086]	−0.002	1.95	n.a.	0.017	n.a.	Scarce qualified staff (930 federal specialists in 2023)
Party funds	+0.002 [0.000 to 0.006]	+0.001	0	n.a.	n.a.	n.a.	+16 pp of the Fundo Partidário to the six largest parties

Notes: Each instrument added alone to the published scorecard, against the same world without it; 2029 to 2036, six indicators, average over municipalities. Fiscal cost: commissioners at R$ 350,000 a year each. Money reallocated: 10% of the FPM (tranche) and 70% of the amendments to municipalities (R$ 32 billion a year), which are tied to programmes instead of being moved between municipalities. Medians [5th to 95th percentiles] over 400 calibrated draws. n.a.: not applicable.

S8.8. The proposal and its variants

Table S8.5 compares the proposal with single changes to it, the basis of the three lessons of Section 8.6 of the article. The first lesson, that the sanctioning rules can be made stricter or looser without making them accurate, appears in the last columns: every variant of the proposal's rules bans between 18 and 27 mayors in every hundred per mandate, and in none of them are banned mayors much more likely than a random mayor to be of below-average competence. The second lesson concerns the money rules. Paying the tranche by population erases more than half of the average gain. The leakage variant mimics what happened after the Supreme Court ended the rapporteur's amendments in 2022, when

discretion moved to committee amendments, which reached R$ 11 to 17 billion a year in the 2024 to 2026 budgets; if half of the tied amendment money leaked back into discretionary uses in this way, the effect would fall to 83% of the proposal's. Redirecting the amendments of municipalities in tutelage to the others raises the unweighted average (+0.239 SD) while lowering the population-weighted one (+0.108 SD), because with 82% of municipality-years in tutelage it concentrates the money in a few places. The third lesson is that distributing party funds by colour bands instead of points changes little, because the formula is driven by population far more than by performance.

Table S8.5. The proposal and single changes to it.

Design	Effect (SD)	Effect, pop.-weighted (SD)	Tutelage (% of years)	Bans per 100 per mandate	Banned mayors below average	Six largest parties' FP share
As proposed	+0.196	+0.108	82%	24.3	51%	73%
No statistical tolerance	+0.195	+0.107	94%	21.4	50%	72%
Red band at 1.5 SD below the floor	+0.199	+0.112	82%	17.9	51%	73%
Without the "no improvement" rule	+0.197	+0.108	82%	17.8	49%	72%
Commissioners hired at 1,500 a year	+0.199	+0.110	82%	24.3	51%	73%
FPM tranche by points × population	+0.082	+0.107	84%	24.7	51%	73%
Amendments redirected away from municipalities in tutelage	+0.239	+0.108	81%	26.9	51%	73%
Half of the tied money leaks back to discretionary uses	+0.162	+0.092	82%	25.0	52%	73%
Party funds by colour bands	+0.196	+0.108	82%	24.3	51%	73%

Notes: Medians over 400 calibrated draws. Effects: differences from the status quo, 2029 to 2036, in SD (Section 7.1 of the article). Tutelage: share of municipality-years, 2027 to 2036. Bans: per hundred mayors per mandate, 2029 to 2036. Banned mayors below average: share of banned mayors whose competence is below the mean (50% for a random draw). FP share: 2036.

S8.9. A personal bonus for mayors: a theoretical benchmark

The LRG contains no personal reward for mayors, and the Constitution forbids one: art. 39 §4 pays elected officials a single fixed *subsídio*, with no bonus or prize (Section 10 of the article). Section S2.1 nevertheless uses a personal bonus as the simplest benchmark of performance pay, and simulating it shows what pure personal pay could do with the LRG's own scorecard. The benchmark pays a bonus of a quarter, a half and one annual salary per SD of the published points, only for above-average points, with the scorecard published and no other instrument (Figure S8.6); Section S7.3 describes how the bonus is computed.

Against the scorecard alone, half a salary per SD raises the indicators by +0.057 [0.009 to 0.135] SD at a fiscal cost of R$ 0.25 billion a year, which is 0.23 SD per billion, against 0.017 for

commissioners. Unlike the tranche, the bonus works as well in large cities as in small towns, because a real paid to the mayor enters her payoff one for one (Section 3.2 of the article). It moves only effort, however, so its effect is less than a third of the whole law's. In addition, 51% of the measured gain that it produces is manipulation, as the cosmetic share derived in Section S2.1 predicts when the reward is paid on the reported number, and unscored tasks lose 0.049 SD. A quarter of a salary buys +0.034 SD and one salary +0.088, at R$ 0.12 and 0.50 billion a year.

The evidence on personal performance pay points in the same direction: it raised the revenue of Punjab's tax collectors, but also the bribes they took (Khan, Khwaja and Olken, 2016; Section S6.4). The benchmark therefore serves only as a point of comparison. It shows that personal pay would be cheap per unit of effort, and that the effort it would buy is the part of the response most exposed to manipulation.



Figure S8.6. A personal bonus for mayors, as a benchmark. Personal bonus of a quarter, a half and one annual salary per SD of the LRG's published points, paid to the mayor, with the scorecard published and no other instrument: real effect (solid) and cosmetic gap (hatched) on the six indicators, 2029 to 2036, against the scorecard alone, and the annual cost; dashed lines: the sanctions and the FPM tranche alone. Medians over 400 calibrated draws.

S8.10. From the results to the assessments

The assessments of Table 10 of the article are assigned by explicit rules applied to the numbers of the simulation and of the real data, so that the same numbers always lead to the same assessment. They describe how far the evidence supports each causal link of the proposal; how its effects should be valued is a separate question, which the rules leave aside. A link is rated *supported, sizeable* if the instrument accounts for at least 35% of the average effect and its contribution is positive in at least 95% of draws, or if the rule mechanically cuts discretionary money by at least 30% and lowers incumbents'

re-election by at least two points. A link is rated *supported, small* if its contribution is positive in at least 95% of draws but smaller; this applies to commissioners, to the effect on services (below 0.25 SD), to formal jobs (above 0.02 SD) and to the chain to the Chamber, if tied amendments lower the centrão's vote in at least 95% of draws.

A link is rated *weak or unproven* in any of the following cases: if the contribution is below 10% of the effect (the published scorecard); if most of it comes from the cosmetic gap instead of real improvement, or if it is below 25% of the effect (sanctions); if it reaches less than 15% of the population-weighted effect or allocates money with a correlation between points and true improvement below 0.8 (the FPM tranche); if it rests on an assumption that the evidence does not support (voters using the scorecard; parties screening candidates); if Brazilian quasi-experimental evidence finds no joint effect and only bounds the incentive (careers, Section 6.1 of the article); or if the measured change is negligible (the mayor-term component below 5% of the variance; vulnerability changing by less than one percentage point). A link is rated *contradicted*, finally, if the evidence runs against the claim: if fewer than 10% of bans are attributable to the mayor or if sanctions lower the competence of mayors. These rules are implemented in the script 15_verdict.py of the replication package (Section S3.3 and Table S3.3).

S9. Tests of the sanctions, robustness and sensitivity

Section 9 of the article tests the sanctions and the assumptions of the simulation, and this section reports those tests in full. It decomposes what the ban for results adds into deterrence and execution and varies the intensity of the threat (Section S9.1); reports the marginal effect of each instrument in every configuration of the others, and the role of the sanctions when implementation depends on incentives (Section S9.2); and gives the procedure and the results of the test of whether the behavioural response is underestimated (Section S9.3). It then reports every member of every scenario family (Section S9.4) and the variance-based sensitivity analysis (Section S9.5). A technical check closes the section: the no-sanction counterfactual and the classification for tutelage (Section S9.6). The conventions are those of Section S8, and the scenario families use the first 120 draws with the same random shocks as the main run.

S9.1. The ban for results: deterrence, execution and intensity

Table 7 of the article isolates the sanctions with designs that change only them. Three details help to read it. Bans for results are those triggered by the red band and by the "no improvement" rule. In the design with bans threatened but never executed, mayors weigh the ban as in the proposal, but no ban is ever applied; in the design with bans executed but not anticipated, bans are applied but carry no weight in mayors' incentives. And the effort that mayors put in moves with the real effect: 0.058 SD as

proposed, 0.056 without bans for results and 0.038 without any performance sanction. Without bans for results, the LRG keeps 99% [95% to 102%] of its effect.

A ban does two things. Its threat raises the stakes for a mayor close to a threshold, which is deterrence; its execution removes a mayor, ends her incentive to deliver for the rest of the mandate and changes who governs. Four designs separate the two, with every other instrument as proposed. The first combines threat and execution, which is the LRG as proposed. The second has the threat without execution, the third has execution without the threat, and the fourth has neither, which is the LRG without bans for results, with alerts and tutelage kept. Deterrence is then the second design against the fourth, execution the third against the fourth, and their interaction the remainder (Table S9.1). The same decomposition is repeated with the sanctions alone, on the published scorecard with no other instrument.

Deterrence raises the real effect by +0.010 [0.003 to 0.020] SD, almost all of it through effort (+0.010 SD), and execution lowers it by 0.007 [0.003 to 0.012] SD. The two combine almost additively (interaction −0.001 SD), so that the ban as proposed adds +0.002 [−0.003 to 0.010] SD. With the sanctions alone, deterrence adds +0.014 SD and execution −0.005 SD.

The decomposition also shows how execution imposes its costs. Against the threat alone, execution costs 0.008 SD of effort through lame-duck mandates and raises the turnover of ruling parties by 5.8 points at each election. Because wrongful bans deter competent candidates, it also changes the population-weighted competence of mayors in office by −0.05 SD, which moves services by less than 0.001 SD because the mayor's component of the indicators is small. Meanwhile, 51% of banned mayors are of below-average competence.

Table S9.1. What the ban adds: deterrence, execution and their interaction.

Outcome	Deterrence: threat alone	Execution: bans alone	Inter-action	Ban as proposed	Deterrence, sanctions alone	Execution, sanctions alone
Real effect (SD)	+0.010 [0.003 to 0.020]	−0.007 [−0.012 to −0.003]	−0.001	+0.002 [−0.003 to 0.010]	+0.014 [0.004 to 0.026]	−0.005 [−0.009 to −0.002]
Real effect, pop.-weighted (SD)	+0.010 [0.003 to 0.017]	−0.011 [−0.018 to −0.005]	−0.001	−0.002 [−0.009 to 0.004]	+0.012 [0.004 to 0.021]	−0.008 [−0.014 to −0.003]
Effort channel (SD)	+0.010 [0.002 to 0.020]	−0.007 [−0.013 to −0.003]	−0.001	+0.002 [−0.004 to 0.009]	+0.013 [0.003 to 0.026]	−0.004 [−0.008 to −0.001]
Cosmetic gap (SD)	+0.008 [0.002 to 0.018]	−0.007 [−0.018 to −0.003]	−0.001	−0.001 [−0.006 to 0.004]	+0.014 [0.004 to 0.036]	−0.006 [−0.016 to −0.002]
Selection channel (SD)	0.0000 [0.0000 to 0.0000]	+0.0001 [−0.0001 to 0.0002]	0.0000	+0.0001 [−0.0001 to 0.0002]	0.0000 [0.0000 to 0.0000]	+0.0000 [−0.0001 to 0.0002]
Disruption channel (SD)	+0.0000 [0.0000 to 0.0000]	−0.0020 [−0.0036 to −0.0004]	+0.0000	−0.0020 [−0.0036 to −0.0004]	0.0000 [0.0000 to 0.0000]	−0.0020 [−0.0035 to −0.0004]
Change of ruling party (pp)	0.0 [−0.1 to 0.1]	+5.8 [5.2 to 6.4]	0.0	+5.8 [5.3 to 6.3]	0.0 [0.0 to 0.0]	+5.8 [5.4 to 6.3]

Outcome	Deterrence: threat alone	Execution: bans alone	Inter-action	Ban as proposed	Deterrence, sanctions alone	Execution, sanctions alone
Re-election of incumbents who run (pp)	0.0 [−0.1 to 0.1]	+0.2 [−0.4 to 0.9]	+0.0	+0.2 [−0.4 to 0.9]	0.0 [−0.1 to 0.1]	−0.2 [−0.7 to 0.4]
Competence of mayors, pop.-weighted (SD)	0.000 [−0.002 to 0.001]	−0.048 [−0.120 to 0.016]	+0.001	−0.048 [−0.119 to 0.016]	0.000 [−0.001 to 0.001]	−0.043 [−0.107 to 0.021]
Literacy (pp)	+0.53 [0.14 to 1.10]	−0.34 [−0.59 to −0.12]	−0.05	+0.13 [−0.08 to 0.52]	+0.70 [0.18 to 1.36]	−0.26 [−0.45 to −0.09]

Notes: Paired differences over 400 draws, 2029 to 2036: deterrence = threat without execution minus no bans; execution = execution without threat minus no bans; interaction = proposed − threat only − execution only + no bans. Sanctions alone: the same designs with no instrument other than the sanctions and the published scorecard. Medians [5th to 95th percentiles].

How much a mayor fears a ban is not identified by any known evidence: the calibration leaves its weight, 4.0 alert units [2.3 to 5.8], at its prior range. Table S9.2 therefore varies it from zero to eight times its calibrated value. To give the weight a concrete meaning, the salary equivalent converts it into annual salaries per SD of the published score, using the model's own conversion of alert units into a personal bonus (equation 4 of the article): a mayor exactly at a threshold weighs a ban as she would weigh losing that bonus. At the calibrated weight the threat is worth 0.9 annual salaries for each SD by which she falls short, and at eight times 7.4, that is, 1.9 mandates' pay.

The response of the real effect is small and roughly linear. A threat twice as heavy raises the LRG's effect by +0.006 SD, four times as heavy by +0.013 and eight times as heavy by +0.022 [0.008 to 0.039], while removing the threat lowers it by 0.009. The number of bans and the share of wrongful bans barely move, and the cosmetic gap grows with the threat (0.051 against 0.036 SD).

Table S9.2. The intensity of the threat.

Weight of a ban (× calibrated)	Alert units	× electoral incentive	Salary equiv-alent at the threshold	LRG: real effect against ×1	LRG: effort against ×1	LRG: cos-metic gap	LRG: bans per 100	LRG: wrong-ful (%)	Sanctions alone: real effect
×0	0.0	0.0	0.00	−0.009	−0.009	0.029	24.5	32.7%	+0.019 [0.004 to 0.042]
×0.5	2.0	4.6	0.46	−0.004	−0.004	0.033	24.4	32.6%	+0.027 [0.006 to 0.054]
×1	4.0	9.1	0.93	0.000	0.000	0.036	24.3	32.7%	+0.032 [0.008 to 0.064]
×2	8.1	18.2	1.86	+0.006	+0.006	0.040	24.2	32.6%	+0.039 [0.011 to 0.076]
×4	16.2	36.4	3.71	+0.013	+0.013	0.045	24.1	32.6%	+0.048 [0.014 to 0.092]
×8	32.3	72.8	7.43	+0.022	+0.022	0.051	24.0	32.7%	+0.059 [0.019 to 0.110]

Notes: ×0 is the design with bans executed but not anticipated. Medians over 400 draws; alert units: median of the calibrated weight S_3 times the factor; × electoral incentive: relative to the electoral incentive of an eligible mayor in the status quo; sanctions alone: real effect against the published scorecard alone, with 5th to 95th percentiles. n.a.: not simulated.

S9.2. Complementarity and implementation

Section 9.1 of the article asks whether sanctions matter more where financial rewards or supervision are absent. The factorial answers this directly, because it contains the sanctions' marginal effect in each of the 16 configurations of the other four instruments (Figure S9.1a). Alone on the published scorecard, sanctions add +0.032 [0.008 to 0.064] SD; on top of the other four instruments, +0.017 [0.003 to 0.040]. With the FPM tranche in place, their average marginal effect is 0.018 SD, against 0.031 without it; with commissioners in place, 0.025 against 0.023; with tied amendments, 0.022 against 0.026; and with party funds, 0.024 against 0.024. Sanctions therefore matter more without financial rewards, and about as much with supervision, unless implementation depends on incentives (scenario D). In that case they also matter more without commissioners (0.030 against 0.026 SD), because supervision and the threat then do the same job. Their Shapley value, +0.024 SD or 12% of the effect, is the average of these marginal effects over all orders of entry, which is why it lies between their effect alone and their effect when added last.

Two mechanisms explain why the marginal effect of the sanctions falls. First, effort responds less and less to further incentive (equation 7 of the article), so where the tranche and the amendments already press, the incentive that the sanctions add buys about 73% of the effort it buys alone. Second, the tranche presses on the same indicators and municipalities as the sanctions, namely the small and noisy ones, so the two instruments compete for the same response instead of adding to each other.

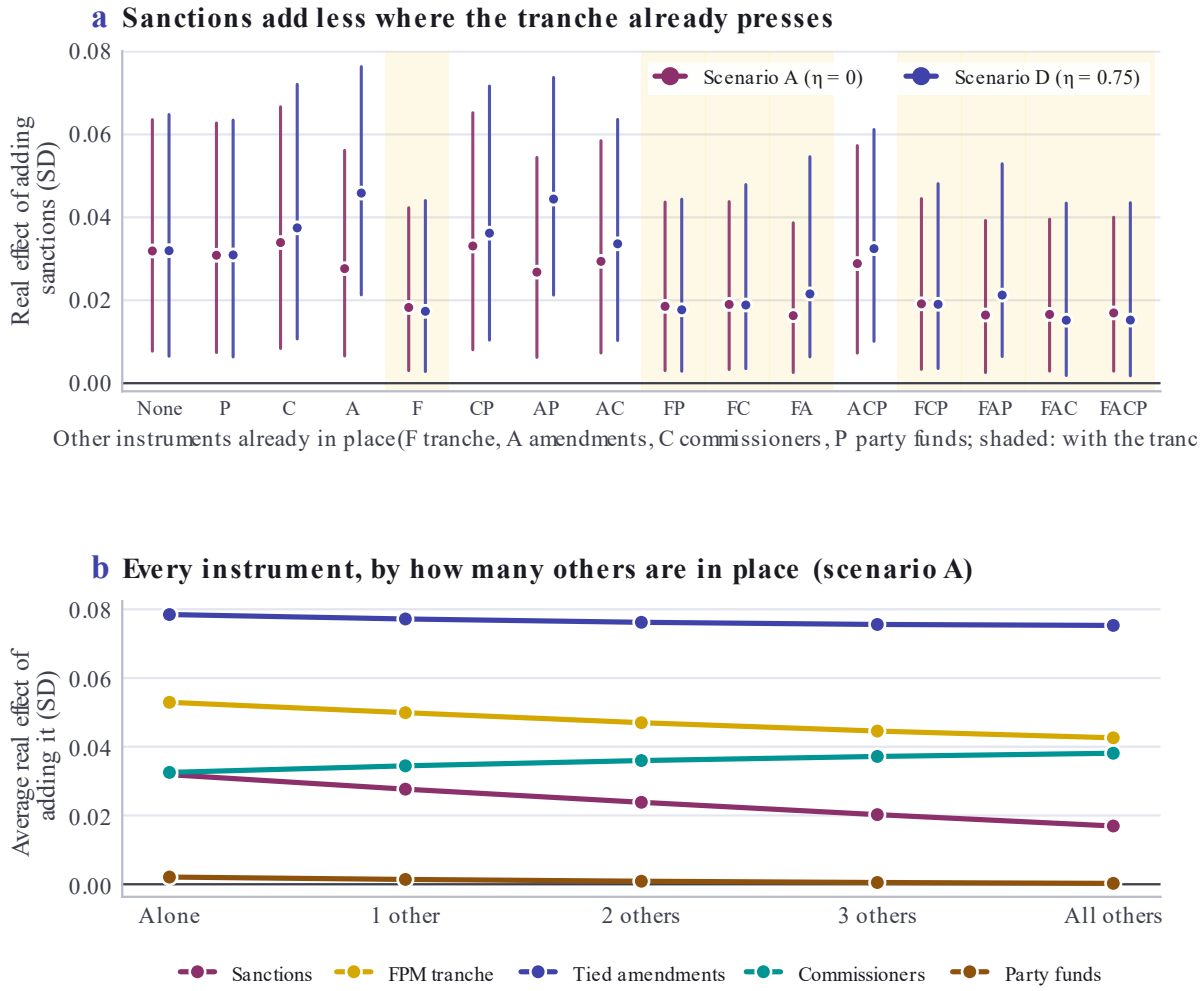


Figure S9.1. The sanctions add less where the other instruments already press. **(a)** Real effect of adding the sanctions in each of the 16 configurations of the other four instruments (F: FPM tranche; A: tied amendments; C: commissioners; P: party funds; shaded: configurations with the tranche), with implementation independent of incentives (scenario A, 400 draws) and dependent on them (scenario D, η = 0.75, first 120 draws); medians and 5th to 95th percentiles. **(b)** Average real effect of adding each instrument, by the number of other instruments already in place, scenario A; medians. SD: Section 7.1 of the article.

Table S9.3 extends the comparison to every instrument. It reports each instrument's marginal real effect when it is added to the published scorecard alone, averaged over the configurations with one, two or three other instruments, and when it is added last, under implementation scenario A (independent of incentives, 400 draws) and scenario D (η = 0.75, first 120 draws; the factorial was run in full under D). Under scenario A, the instruments that work through effort, namely sanctions, the tranche and party funds, show diminishing returns (Figure S9.1b), because effort responds less and less to further incentive and these instruments press on the same effort. Tied amendments keep 96% of their effect when added last, and commissioners gain from the others, because the capacity they build turns effort and money into results. Under scenario D, in which implementation depends on incentives, tied amendments and

commissioners gain from the other instruments, which supply the incentives that implementation needs, whereas the sanctions and the tranche still lose effect as more instruments are added.

Table S9.3. Marginal real effect of each instrument, by how many others are in place (SD).

Instrument	Scenario	Alone	With 1 other (mean)	With 2 others (mean)	With 3 others (mean)	With all others	Shapley value	Shapley share
Sanctions	A (η = 0)	+0.032	+0.028	+0.024	+0.020	+0.017	+0.024	13%
Sanctions	D (η = 0.75)	+0.032	+0.033	+0.029	+0.022	+0.015	+0.027	14%
FPM tranche	A (η = 0)	+0.053	+0.050	+0.047	+0.045	+0.043	+0.047	25%
FPM tranche	D (η = 0.75)	+0.049	+0.050	+0.049	+0.046	+0.045	+0.047	26%
Tied amendments	A (η = 0)	+0.078	+0.077	+0.076	+0.076	+0.075	+0.076	40%
Tied amendments	D (η = 0.75)	+0.051	+0.063	+0.069	+0.071	+0.072	+0.066	36%
Commissioners	A (η = 0)	+0.033	+0.034	+0.036	+0.037	+0.038	+0.035	19%
Commissioners	D (η = 0.75)	+0.027	+0.036	+0.039	+0.039	+0.039	+0.036	21%
Party funds	A (η = 0)	+0.002	+0.001	+0.001	+0.001	+0.000	+0.001	1%
Party funds	D (η = 0.75)	+0.002	+0.001	+0.001	+0.001	+0.000	+0.001	1%

Notes: Marginal effect: the real effect of the design with the instrument minus that of the same design without it, medians over draws, averaged over the configurations of the other instruments with the stated number switched on. Shapley share: share of the sum of the absolute Shapley values.

The second test of Section 9.1 of the article asks whether implementation depends on incentives, which is the case in which sanctions would be expected to matter most. If money and assistance are carried out worse by administrations with little at stake and nobody supervising them (scenarios B to D, Section 7.2 of the article), the role of the sanctions barely changes. Under scenario D their Shapley share is 14%, against 13% under scenario A on the same draws, and the LRG keeps 92% of its effect without any performance sanction across scenarios B to D (91% under A) and 99% to 100% without bans. The ban's two roles still offset each other: under scenario D, deterrence adds +0.010 SD and execution −0.008, and removing the bans changes the realised share of the potential benefit of money from 96% to 98%.

S9.3. Is the behavioural response underestimated?

Section 9.2 of the article asks what parameter values would make the LRG's behavioural response larger and whether the evidence on which the model is calibrated allows them (Figure S9.2). Four parameters govern the behavioural response: how strongly effort responds to incentives (δ), how much weight mayors give to money (κ), how quickly effort saturates as incentives grow, and how much mayors fear a ban, which Section S9.1 already varied. For the responsiveness of effort (every δ) and the weight of money (κ), each calibrated draw is scaled, and the fifteen calibration targets are recomputed with the equations of the calibration and with the nuisance draws that each posterior draw had in it,

which are recovered exactly from the calibration's seeds. Table S9.4 reports, for each scaled model, the likelihood of the evidence relative to the calibrated model (the mean likelihood over the draws), the share of draws whose fifteen implied moments lie jointly inside the 95% region of the estimates, three implied moments, and the LRG simulated with the scaled parameters on the first 120 draws.

A more responsive effort raises the LRG's effect, but less than proportionally, because effort is a minority of it: with responses 1.5, 2 and 3 times the calibrated ones, the effort channel would be 0.084, 0.107 and 0.146 SD. Doubling the response keeps 90% of the draws inside the joint 95% region of the fifteen targets, whereas tripling it keeps only 62%. The binding constraint is Ceará's rule alone: a response three times as large implies an effect of the rule on learning of 0.145 pupil SD, against 0.06 estimated (95% interval −0.02 to 0.14; Figure S9.2c). A heavier weight of money is allowed as well (likelihood 0.88 at twice and 0.78 at three times the calibrated weight) and adds less (0.220 SD at three times).

If the response were twice as large, effort would explain 45% of the gain, and the sanctions would add 0.026 SD on top of the other instruments, of which the bans 0.003. A larger response makes both the threat and the execution of bans matter more, and the bans still add little, so the conclusions of Section 9.1 of the article hold.

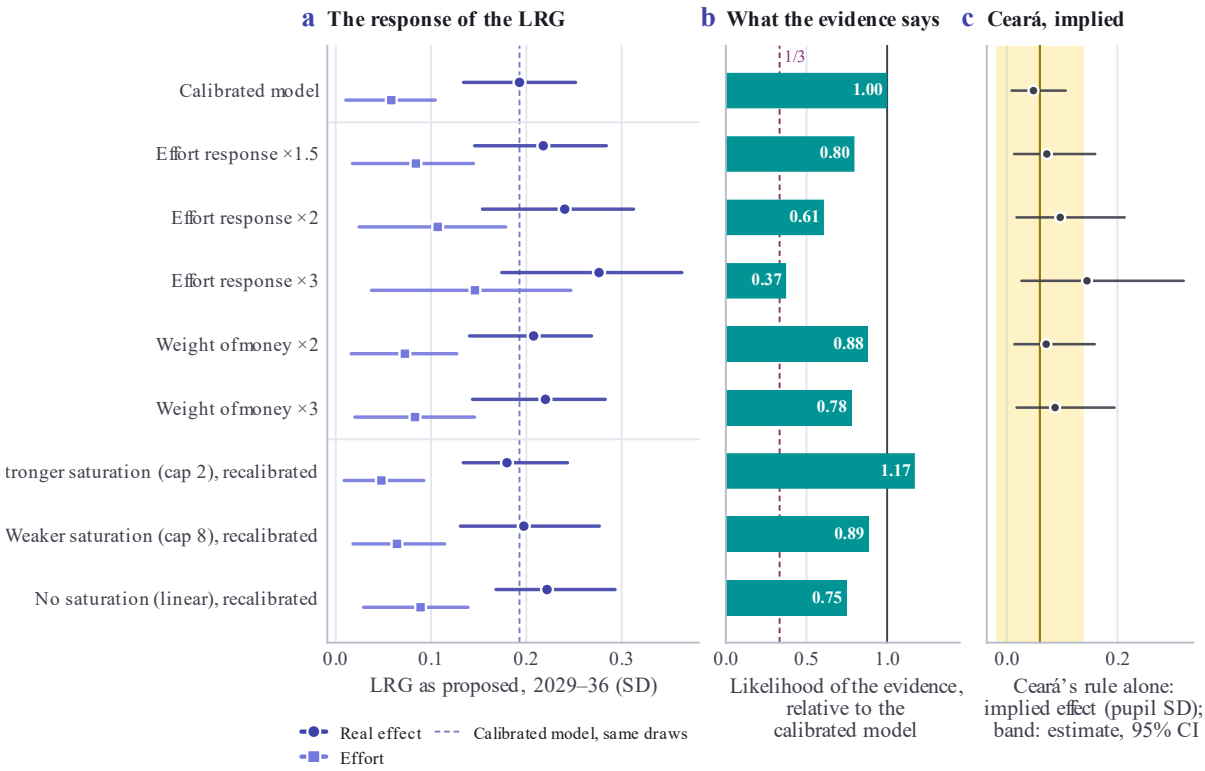


Figure S9.2. What would it take for the behavioural response to be larger? **(a)** Real effect and effort channel of the LRG as proposed with the response of effort to incentives scaled by 1.5, 2 and 3, the weight of money scaled by 2 and 3 (first 120 calibrated draws, not recalibrated), and with three alternative saturations of effort, each recalibrated; dashed line: the calibrated model on the same draws. **(b)** Likelihood of the fifteen calibration targets relative to the calibrated model (scaled parameters: mean likelihood over the draws; saturation: Bayes factor of the recalibrated model); dashed line at one third. **(c)** Effect of Ceará's results-based rule alone on learning implied by each scaled model (pupil SD), against the estimate and its 95% interval (shaded). Medians and 5th to 95th percentiles.

Table S9.4. Larger behavioural responses and what the evidence says about them.

Model	Likeli-hood relative to the cali-brated model	Draws inside the joint 95% region	Ceará, rule alone (pupil SD; target 0.06)	Ceará, rule and PAIC (target 0.42)	RD learning, lame minus first-term (target 0.19)	LRG real effect (SD)	LRG effort (SD)	Sanc-tions added last (SD)	Bans added (SD)
Calibrated	1.00	100%	0.048 (z −0.3)	0.41 (z −0.1)	0.146 (z −0.5)	+0.193 [0.134 to 0.252]	+0.058	+0.017	+0.003
Effort response ×1.5	0.80	98%	0.073 (z 0.3)	0.47 (z 0.3)	0.121 (z −0.7)	+0.218 [0.146 to 0.284]	+0.084	+0.022	+0.003
Effort response ×2	0.61	90%	0.097 (z 0.9)	0.54 (z 0.8)	0.098 (z −1.0)	+0.240 [0.154 to 0.313]	+0.107	+0.026	+0.003
Effort response ×3	0.37	62%	0.145 (z 2.1)	0.68 (z 1.6)	0.053 (z −1.5)	+0.276 [0.174 to 0.363]	+0.146	+0.033	+0.003
Weight of money ×2	0.88	99%	0.071 (z 0.3)	0.48 (z 0.4)	0.146 (z −0.5)	+0.208 [0.140 to 0.268]	+0.073	+0.009	−0.001
Weight of money ×3	0.78	97%	0.087 (z 0.7)	0.52 (z 0.6)	0.146 (z −0.5)	+0.220 [0.144 to 0.283]	+0.083	+0.005	−0.003

Notes: Implied moments: medians over the 400 calibrated draws, with z = (median − target)/SE. LRG: first 120 draws, same shocks; sanctions added last: LRG minus LRG without performance sanctions; bans added: LRG minus LRG without bans for results.

Saturation cannot be scaled in this way, because it changes the shape of the effort rule. For it, the model is recalibrated under each alternative rule and compared with the calibrated model by its log evidence (Table S9.5). With effort twice as quick to saturate, twice as slow or linear, the recalibrated model fits the evidence about equally well (Bayes factors of 1.17, 0.89 and 0.75 against the calibrated rule), and the LRG's effect is 0.180, 0.197 and 0.222 SD. Even a linear response, the most favourable to incentives, raises the effect by only 15%: effort grows from 0.058 to 0.089 SD, while what money and assistance deliver stays the same. Figure S9.3 draws the four effort rules together with the incentives that mayors face, and it shows that the recalibrated models differ mostly above the incentives that Brazil already has, which is why the evidence cannot tell them apart.

Table S9.5. Saturation of effort, recalibrated.

Saturation of effort	Log evi-dence	Bayes factor against cap 4	Effec-tive sample	δ, literacy (posterior median)	κ, money	LRG real effect (SD)	LRG effort (SD)	Deter-rence (SD)	Exe-cution (SD)
Cap 4 (calibrated model)	28.86	1.00	5,394	0.52	0.56	+0.193 [0.134 to 0.252]	+0.058	+0.010	−0.006
Cap 2 (faster)	29.01	1.17	6,713	0.64	0.58	+0.180 [0.134 to 0.243]	+0.048	+0.007	−0.007
Cap 8 (slower)	28.73	0.89	4,645	0.45	0.54	+0.197 [0.131 to 0.277]	+0.064	+0.013	−0.007
None (linear)	28.56	0.75	3,792	0.38	0.51	+0.222 [0.168 to 0.293]	+0.089	+0.025	−0.007

Notes: Effort a = *δI/(1 + I/cap). Log evidence: log of the mean likelihood of the calibration targets over the 120,000 prior draws; Bayes factor: exponential of its difference from the calibrated model. LRG: first 120 draws of each recalibrated posterior, same shocks.*

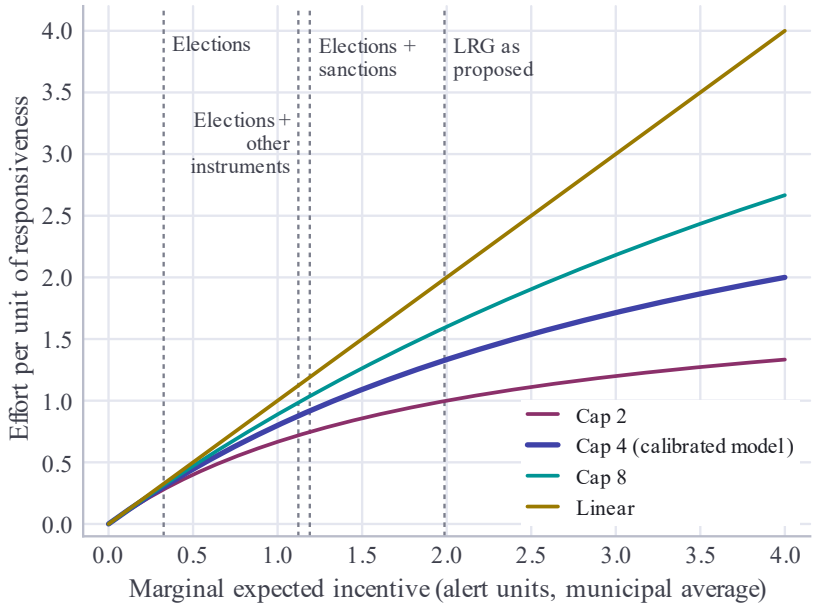


Figure S9.3. The effort rule under four saturations. Effort per unit of responsiveness as a function of the marginal expected incentive; vertical lines: municipal-average incentives from elections, and with the sanctions, the other instruments and all of them (LRG as proposed, 2030, medians). The recalibrated models differ mostly above the incentives that Brazil already has.

S9.4. Every scenario family

Table 8 of the article summarises the scenario families family by family, and Table S9.6 reports every member. Each family varies one assumption that the evidence cannot settle and is simulated on the first 120 calibrated draws, with the same random shocks as the main run, and the main

calibration restricted to the same draws is the reference. The calibration variants use wider priors, leave out each block of targets in turn, make Ceará twice as responsive as the nation, treat the targets as independent and drop the open-seat target (Section S6.5). The saturation variants make effort saturate twice as fast, twice as slowly or not at all, each recalibrated (Section S9.3). Voters' reward is set at none, twice and four times the calibrated value. Manipulation is removed, removed with the effort it absorbs redirected to real improvement, halved, raised by half, or cut by 60% by audits, with and without that redirection. Implementation follows scenarios B to D (Section 7.2 of the article), and in the last family banned mayors keep responding to money and party funds, under scenarios A and D. Figure S9.4 shows, in every family, the shares of the LRG's effect kept without bans and without any performance sanction, together with the two roles of the ban.

Four results of the families add to the summary of Section 9.3 of the article. Money, assistance and allocation explain 64% to 77% of the real effect in every scenario. The ban's two roles keep their signs in every family, with deterrence adding 0.007 to 0.025 SD and execution subtracting 0.003 to 0.011 SD, and the LRG keeps 81% to 94% of its effect without any performance sanction. Sanctions alone add 0.029 to 0.052 SD of real improvement across the manipulation scenarios, and with no manipulation and the effort it absorbed redirected to real improvement they still ban 23 mayors in every hundred per mandate. Across voters' rewards, finally, sanctions alone add 0.026 to 0.037 SD.

Voters' reward barely moves the share of eligible incumbents re-elected in the status quo (66.5% with none and 65.6% with four times the calibrated value, against 66.3% in the main calibration and 66.6% observed in 2024), so aggregate re-election rates cannot pin it down beyond the association of Section 6.2 of the article.

Table S9.6. Every scenario family.

Scenario	Real effect, as proposed (SD)	Effort share (%)	Kept without bans (%)	Kept without sanctions (%)	Deter-rence (SD)	Exe-cution (SD)
Main calibration (first draws)	+0.193 [0.134 to 0.252]	28%	99%	92%	+0.010	−0.006
Calibration: wider priors	+0.208 [0.123 to 0.290]	33%	100%	93%	+0.011	−0.011
Calibration: no Ceará targets	+0.230 [0.145 to 0.333]	33%	99%	90%	+0.011	−0.008
Calibration: no discontinuities	+0.195 [0.133 to 0.279]	34%	99%	88%	+0.012	−0.008
Calibration: no electoral return	+0.183 [0.133 to 0.254]	28%	99%	91%	+0.010	−0.007
Calibration: no FPM targets	+0.198 [0.141 to 0.265]	31%	98%	90%	+0.011	−0.007
Calibration: Ceará twice as responsive	+0.154 [0.110 to 0.207]	25%	99%	92%	+0.008	−0.006
Calibration: independent targets	+0.196 [0.132 to 0.249]	29%	99%	91%	+0.011	−0.007

Scenario	Real effect, as proposed (SD)	Effort share (%)	Kept without bans (%)	Kept without sanctions (%)	Deter-rence (SD)	Exe-cution (SD)
Calibration: no open-seat target	+0.197 [0.141 to 0.254]	29%	99%	90%	+0.011	−0.007
Saturation twice as fast (recalibrated)	+0.180 [0.134 to 0.243]	26%	100%	94%	+0.007	−0.007
Saturation twice as slow (recalibrated)	+0.197 [0.131 to 0.277]	31%	98%	88%	+0.013	−0.007
No saturation (recalibrated)	+0.222 [0.168 to 0.293]	37%	93%	81%	+0.025	−0.007
Voters' reward ×0	+0.201 [0.138 to 0.264]	29%	98%	90%	+0.010	−0.006
Voters' reward ×2	+0.186 [0.131 to 0.248]	27%	99%	92%	+0.009	−0.007
Voters' reward ×4	+0.180 [0.124 to 0.242]	25%	100%	94%	+0.008	−0.007
No manipulation	+0.198 [0.135 to 0.262]	29%	98%	90%	+0.012	−0.007
No manipulation, effort redirected	+0.222 [0.147 to 0.298]	37%	98%	87%	+0.017	−0.011
Manipulation ×0.5	+0.197 [0.135 to 0.256]	28%	99%	90%	+0.011	−0.007
Manipulation ×1.5	+0.187 [0.134 to 0.247]	27%	99%	93%	+0.008	−0.005
Audits remove 60%	+0.197 [0.135 to 0.256]	28%	99%	90%	+0.011	−0.007
Audits, effort redirected	+0.212 [0.142 to 0.282]	33%	98%	89%	+0.013	−0.009
Implementation $\eta = 0.25$	+0.191 [0.132 to 0.250]	28%	99%	92%	+0.010	−0.007
Implementation $\eta = 0.5$	+0.190 [0.130 to 0.248]	28%	99%	92%	+0.010	−0.008
Implementation $\eta = 0.75$	+0.188 [0.128 to 0.246]	28%	100%	92%	+0.010	−0.008
Banned mayors respond to money	+0.195 [0.135 to 0.254]	29%	97%	90%	+0.010	−0.003
Banned mayors respond to money, $\eta = 0.75$	+0.194 [0.130 to 0.253]	30%	97%	89%	+0.010	−0.004
Effort response ×1.5 (not recalibrated)	+0.218 [0.146 to 0.284]	36%	99%	90%	n.a.	n.a.
Effort response ×2 (not recalibrated)	+0.240 [0.154 to 0.313]	43%	99%	89%	n.a.	n.a.
Effort response ×3 (not recalibrated)	+0.276 [0.174 to 0.363]	52%	99%	88%	n.a.	n.a.
Weight of money ×2 (not recalibrated)	+0.208 [0.140 to 0.268]	33%	100%	95%	n.a.	n.a.
Weight of money ×3 (not recalibrated)	+0.220 [0.144 to 0.283]	36%	101%	98%	n.a.	n.a.

Notes: First 120 draws, medians [5th to 95th percentiles]. Kept: real effect of the LRG without bans for results, or without any performance sanction, as a share of the LRG as proposed in the same scenario and draw. Deterrence and execution: against the LRG without bans for results. n.a.: not simulated in that family.

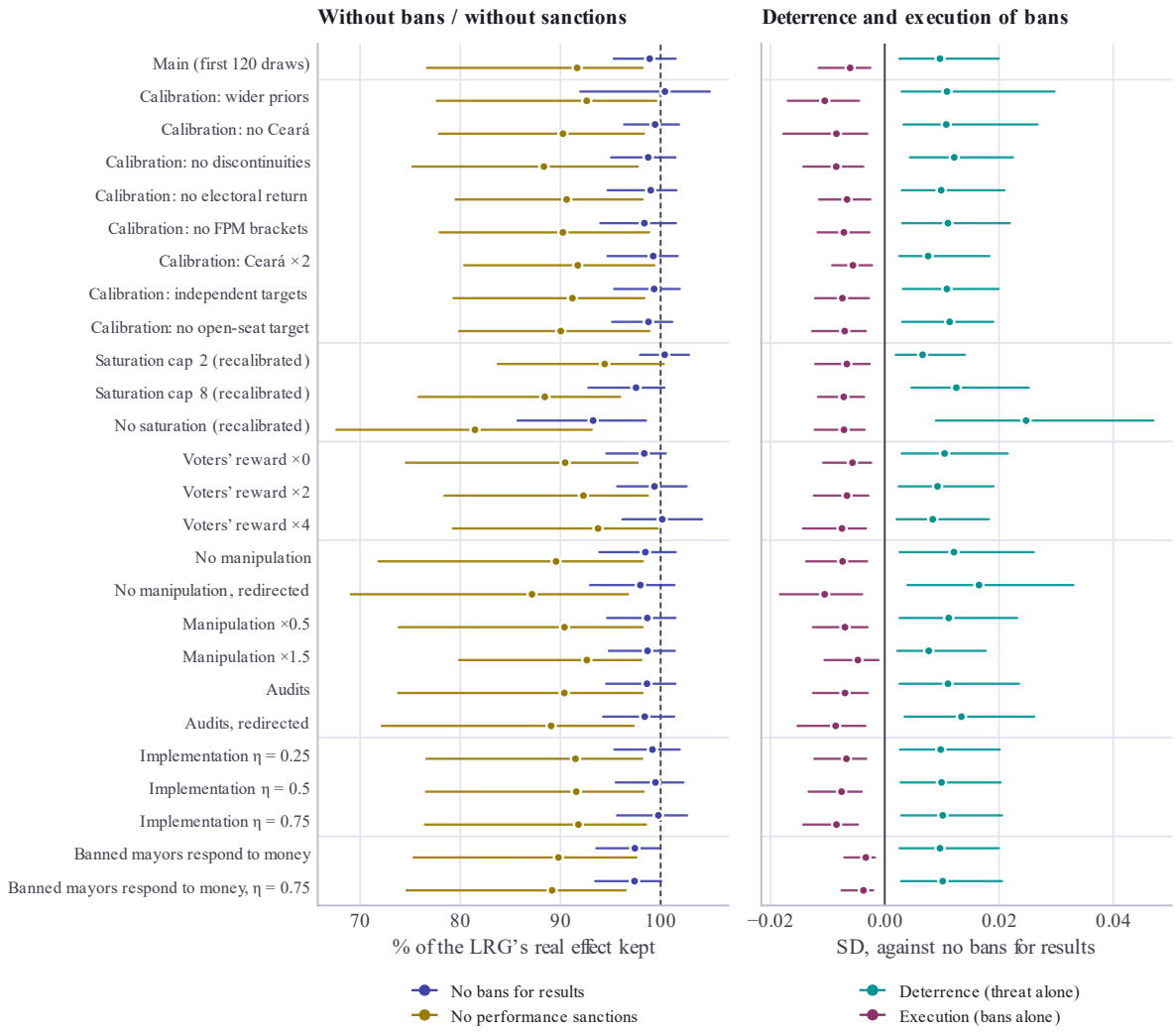


Figure S9.4. The sanctions in every scenario family. **(a)** Share of the LRG's real effect kept without bans for results and without any performance sanction. **(b)** Deterrence (threat without execution) and execution (bans without threat) of bans, against the LRG without bans for results. Medians and 5th to 95th percentiles over the first 120 draws.

S9.5. Variance-based sensitivity analysis

Table S9.7 reports the total-effect Sobol indices of the 17 groups of parameters for every output of the sensitivity analysis, and Figure S9.5 draws eight of them; Section 9.3 of the article discusses the main results. A total-effect index measures the share of the variance of an output that is due to a group of parameters, including its interactions with the others (Saltelli et al., 2008). Each parameter varies uniformly over its calibrated 5th to 95th percentile range, and implementation complementarity η over 0 to 0.75. The designs are the status quo, the LRG as proposed, the LRG without bans for results, the LRG without any performance sanction and the LRG with bans executed but not

anticipated, all with common shocks, and the indices are computed with SALib (Herman and Usher, 2017).

What the sanctions add depends on parameters different from those that drive the size of the effect. The share of the effect kept without bans depends on the weight mayors attach to tutelage and bans (0.37), the strength of the money incentive (0.29) and how strongly learning and daycare respond to incentives (0.14); the share kept without any performance sanction on how strongly learning and daycare respond to incentives (0.43), the strength of the money incentive (0.38) and capacity and the effectiveness of commissioners (0.09); and the deterrent effect of bans on how strongly learning and daycare respond to incentives (0.50), the weight mayors attach to tutelage and bans (0.27) and the strength of the money incentive (0.09). How much mayors fear a ban thus matters for what bans add, which the evidence does not bound, although it barely matters for the size of the effect (0.00). Implementation complementarity has a total-effect index of 0.00 both for the size of the effect and for the share kept without sanctions. The cosmetic gap depends on the manipulation shares (0.46) and how strongly learning and daycare respond to incentives (0.33); the competence of mayors on party pressure and recruitment (0.66); and the loss in unscored tasks on how much mayors neglect unscored tasks (0.81).

The number and accuracy of bans, finally, barely depend on any parameter. Across the calibrated space their standard deviation is 0.8 bans per hundred mayors per mandate around a mean of 24.8, and that of the wrongful share 0.7 points around 33%. Without bans the LRG keeps on average 100% of its effect (standard deviation 2.2 points), and without any performance sanction 89% (5.5 points).

Table S9.7. Total-effect Sobol indices: services and bans.

Group of parameters	Real effect	Pop.-weighted effect	Cosmetic gap	Unscored tasks	Bans	Wrongful share
Real response: other indicators	0.00	0.00	0.01	0.00	0.01	0.05
Real response: education	0.34	0.54	0.33	0.02	0.22	0.24
Manipulation shares	0.00	0.00	0.46	0.00	0.02	0.09
Weight of tutelage and ban	0.00	0.00	0.00	0.01	0.00	0.02
Money incentive	0.03	0.01	0.02	0.04	0.01	0.10
Party pressure and recruitment	0.00	0.05	0.00	0.00	0.00	0.00
Persistence and competence	0.07	0.09	0.02	0.00	0.04	0.10
Neglect of unscored tasks	0.00	0.00	0.00	0.81	0.00	0.00
Capacity and commissioners	0.15	0.03	0.04	0.00	0.11	0.07
Tutelage side effects	0.00	0.00	0.01	0.00	0.01	0.01
Productivity of money	0.20	0.05	0.00	0.00	0.29	0.21
Unit costs	0.04	0.01	0.00	0.00	0.31	0.34
Clientelism	0.00	0.00	0.00	0.00	0.00	0.01
Electoral accountability and information	0.01	0.02	0.01	0.01	0.01	0.05

Group of parameters	Real effect	Pop.-weighted effect	Cosmetic gap	Unscored tasks	Bans	Wrongful share
Turnover and deterrence	0.00	0.00	0.00	0.00	0.10	0.04
Selection by money and pay	0.00	0.00	0.00	0.00	0.00	0.00
Implementation complementarity	0.00	0.00	0.00	0.00	0.00	0.01

Table S9.7 (continued). Total-effect Sobol indices: politics and the sanctions.

Group of parameters	Compe-tence	Six largest parties' FP share	Kept without bans	Kept without sanctions	Deter-rence	Exe-cution
Real response: other indicators	0.00	0.08	0.00	0.00	0.00	0.00
Real response: education	0.01	0.22	0.14	0.43	0.50	0.46
Manipulation shares	0.00	0.10	0.02	0.04	0.05	0.02
Weight of tutelage and ban	0.00	0.03	0.37	0.05	0.27	0.00
Money incentive	0.00	0.07	0.29	0.38	0.09	0.05
Party pressure and recruitment	0.66	0.58	0.00	0.00	0.00	0.00
Persistence and competence	0.00	0.14	0.06	0.04	0.09	0.23
Neglect of unscored tasks	0.00	0.00	0.00	0.00	0.00	0.00
Capacity and commissioners	0.00	0.10	0.02	0.09	0.05	0.06
Tutelage side effects	0.00	0.03	0.00	0.03	0.00	0.00
Productivity of money	0.00	0.13	0.01	0.08	0.00	0.00
Unit costs	0.00	0.13	0.00	0.02	0.00	0.00
Clientelism	0.02	0.24	0.00	0.00	0.00	0.00
Electoral accountability and information	0.00	0.22	0.12	0.07	0.03	0.02
Turnover and deterrence	0.28	0.11	0.02	0.00	0.00	0.02
Selection by money and pay	0.00	0.00	0.00	0.00	0.00	0.00
Implementation complementarity	0.00	0.02	0.02	0.00	0.00	0.02

Notes: Saltelli design with 64 base samples; negative estimates are set to zero. Kept: real effect of the counterfactual as a share of the LRG as proposed; deterrence: LRG as proposed minus LRG with bans executed but not anticipated; execution: LRG with bans executed but not anticipated minus LRG without bans for results.

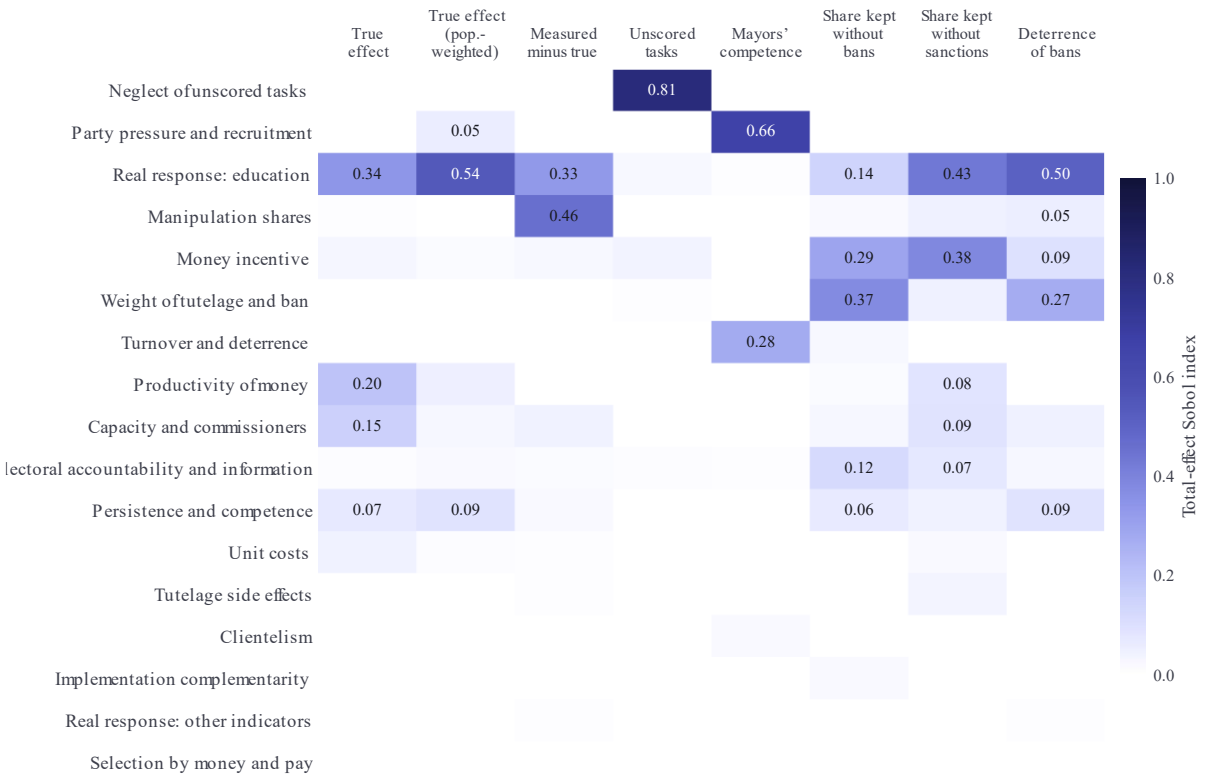


Figure S9.5. Which assumptions drive which results. Total-effect Sobol indices of 17 groups of parameters for eight outputs, each parameter varying uniformly over its calibrated 5th to 95th percentile range and implementation complementarity η over 0 to 0.75 (Saltelli design with 64 base samples, 1,216 runs per design; one realisation of the random shocks for all runs). Share kept: real effect of the LRG without bans for results, or without any performance sanction, as a share of the LRG as proposed; deterrence: the effect of the threat of bans when bans are executed. Indices below 0.05 are not printed.

S9.6. The no-sanction counterfactual and the classification for tutelage

Whenever the law is in force, the published scorecard computes the bands and classifies municipalities for tutelage. This classification belongs to the measurement pillar. The consequences of tutelage (federal co-signature of voluntary transfers, supervision, extra support and slower execution) and the bans follow only where the sanctions are switched on and mayors respond. In the LRG without performance sanctions, the scorecard therefore still classifies 82.3% of municipality-years for tutelage (81.6% as proposed), whereas the share in which tutelage is applied, reported in Table 7 of the article, is zero.

The engine was checked line by line for this distinction. Without sanctions, no incentive from alerts, tutelage or bans enters mayors' effort, no supervision or support is given, no execution drag is applied, nobody is banned and nobody becomes a lame duck. The classification keeps one consequence: the amendment pillar routes the tied amendments of classified municipalities through regional consortia, which lowers mayors' discretion over them. Switching that routing off in the LRG without sanctions

changes the real effect by less than 0.0001 SD (5th to 95th percentiles −0.0002 to 0.0003), as Table S9.8 shows.

Table S9.8. Classification for tutelage and its consequences.

Design	Classified for tutelage (%)	Tutelage applied (%)	Real effect (SD)	Bans per 100
LRG as proposed	81.6%	81.6%	+0.1959	24.3
Without bans for results	81.9%	81.9%	+0.1936	0.0
Without performance sanctions	82.3%	0.0%	+0.1763	0.0
Without performance sanctions, amendments not routed by the classification	82.3%	0.0%	+0.1763	0.0

Notes: Medians over 400 draws, 2027 to 2036 (tutelage) and 2029 to 2036 (effects and bans).

S10. Legal feasibility in detail

Section 10 of the article maps each instrument of the LRG to the legal vehicle it needs and to the constitutional provisions it would meet (Table 9 of the article). This section develops the reasoning behind each assessment, in the same three groups: what ordinary or complementary law can do (Section S10.1), what needs constitutional amendments (Section S10.2) and what is hard or impossible to enact even by amendment (Section S10.3). Section S10.4 then gives the details of the votes that the law would need and of the proponents' answers to the objections. Like the article, this section offers no legal opinion. It reads the text of the Constitution and the decisions of the Supreme Court listed among the references.

S10.1. What ordinary and complementary law can do

The Comissariado can be created as an autarchy by specific law (art. 37 XIX), and the federal government can attach conditions to voluntary transfers, as the Fiscal Responsibility Law already does (art. 25), including co-signature by a commissioner. Because these conditions attach to the Union's own voluntary transfers, the oversight that an ordinary law can give a commissioner reaches federal money, and the municipality's own revenue stays outside it (Section S10.3).

The party funds are distributed by ordinary law (Laws 9,096/1995 and 9,504/1997), so the formula can be changed by simple majority, within the constitutional rule that only parties above the performance threshold in Chamber elections receive the Fundo Partidário at all (art. 17 §3). A new party with mayors but below that threshold would therefore still receive nothing. A formula that left a party above the threshold with nothing because it has no mayors would, for its part, be exposed to challenge, since the Supreme Court struck down the first barrier clause and later restrictions on new parties' access to funds and airtime on grounds of political pluralism (ADI 1351 and 1354, 2006; ADI 5105, 2015).

A performance tranche of the FPM can also be created by complementary law, because the Constitution leaves the criteria of distribution to complementary law (art. 161 II). The same article states their purpose, "to promote socio-economic balance between municipalities", which a tranche that moves money towards better-off places would contradict. The tranche as proposed, paid in proportion to points and to each municipality's own FPM, moves money towards poorer municipalities, whereas the variant paid in proportion to points and population moves it towards richer ones (Section 8.4 of the article and Section S8.4), which is why Table 9 of the article rates the tranche feasible if it equalises and risky otherwise.

S10.2. What needs constitutional amendments

Ineligibilities can only be created by complementary law and only for the purposes listed in art. 14 §9: to protect administrative probity, the morality of office given the candidate's past, and the normality and legitimacy of elections against economic power and abuse of office. The Supreme Court upheld the Ficha Limpa law, which bans candidates whose accounts were rejected for irremediable irregularities amounting to intentional improbity (ADC 29, ADC 30 and ADI 4578). That ineligibility also passes through a political body, because mayors' accounts are judged by the municipal council, on the prior opinion of the court of accounts, which prevails unless two-thirds of the councillors reject it (RE 848826, 2016). A ban for poor results without intent, triggered by indicators, is a different category, and it would probably need a constitutional amendment adding managerial performance to art. 14 §9.

Removal for "managerial improbity" would require a law defining improbity without intent, which would reverse the 2021 reform of the improbity law (Law 14,230/2021) and run against the Supreme Court's reading that improbity requires intent (ARE 843989, 2022). Any new ineligibility would, in addition, be subject to the annuality rule of art. 16, which delayed the application of the Ficha Limpa law by one election (RE 633703, 2011).

The money pillar and the reform of the federation run into the same requirement. Individual amendments are mandatory by the Constitution (art. 166 §§9 to 20), so conditioning them on the recipient's performance needs a constitutional amendment, although tying them to programmes could in part be done by the complementary law that governs the execution of the budget. Unpaid councillors require changing art. 29 VI. A personal bonus for mayors, which the proposal does not contain and which Section S8.9 examines as a benchmark, would require changing art. 39 §4, which pays elected office holders a single fixed *subsídio* and forbids adding any bonus, premium or prize to it; the amount itself is set by municipal law (art. 29 V). Freeing the minimum shares of revenue for education and health, finally, requires changing arts. 198 and 212.

S10.3. What is hard or impossible to enact

The Constitution allows a state to intervene in its municipalities only in four cases (unpaid debt, accounts not rendered, the constitutional minimum not applied in education or health, and a judicial order), and it allows the Union to intervene only in municipalities of federal territories (art. 35). A commissioner who takes over a failing mayor's functions, vetoes on contracts, hiring and credit financed by the municipality's own revenue, or an appointed administrator (*interventor*) would therefore require an amendment, and one that replaced elected mayors indefinitely would collide with the entrenched clauses that protect the federative form and the direct, periodic vote (art. 60 §4 I and II). The Constitution also forbids withholding or restricting the delivery of constitutional transfers (art. 160), which limits how far the FPM can be made conditional.

Mergers require a state law and a plebiscite of the populations involved, under a federal complementary law that has never been enacted (art. 18 §4; ADO 3682). PEC 188 tried to replace this procedure with incorporation by law and was never voted; Section S5.4 simulates what incorporation by its criterion would imply. Turning a state into a federal territory has no constitutional path and would violate the federative form.

The anti-mafia clause, finally, borrows from Italy, where the national government has dissolved municipal councils infiltrated by organised crime since 1991 (Acconcia, Corsetti and Simonelli, 2014; Daniele and Geys, 2015). Those dissolutions rest on evidence gathered by the police, and performance indicators play no part in them. Brazil's Constitution offers no equivalent federal power over municipalities.

S10.4. Votes, and the proponents' answers in detail

Section 10.1 of the article summarises who would vote for the law. In more detail, the party-fund rule, applied to the mayors elected in 2024, would benefit parties holding 225 of the 513 seats elected in 2022 and harm parties holding 288, including the two largest benches, PL and PT (Section 5.2 of the article and Section S5.3); by 2036 the six largest municipal parties would hold 73% of the Fundo Partidário in the simulation, against 57% under current rules. The conditions on amendments would take from every deputy and senator the discretion that successive constitutional amendments have expanded, namely mandatory individual and state-bench amendments and the transferências especiais that reach mayors without agreements or programmes (art. 166 §§9 to 20 and 166-A; EC 100/2019, 105/2019 and 126/2022). Undoing them requires 308 votes from the same deputies. The FPM tranche requires a complementary law (257 deputies and 41 senators), and the ineligibility at least one and probably a constitutional amendment. The tranche also creates losers among the municipalities that deputies and senators represent, since in a typical year about 58% of municipalities would receive less than they do today.

Section 5.2 of the article recalls the pre-candidate's answer to the objection that the centrão would capture the new funds: the rule would requalify those parties, making them compete for good managers (Santos, 2026). The simulation allows this answer to be examined. Tying amendments to programmes weakens the brokerage on which those parties rely: on its own, it would lower their share of the Chamber vote by 1.1 percentage points in the illustrative chain of Section S7.7 [0.4 to 2.2]. Paying parties by their mayors' scores works in the opposite direction, because it channels money to the parties with the most mayors in the largest municipalities. With all instruments together, the net change is −0.1 points [−1.3 to 1.0] on a base of about 69%. The reform would therefore leave the centrão's weight in the Chamber almost unchanged, while changing the basis on which its parties receive public funds.

Section 10.2 of the article assesses five answers that the pre-candidate has given to the main objections (Santos, 2026), and each of them can be stated more fully. The first answer is that *the indicators would be defined with Congress, and that most of them already exist, since the Federal Court of Accounts (TCU) publishes many of them in a dashboard and lacks only the power to sanction*. Agreement in Congress gives the indicators legitimacy, and the dashboard shows that the data exist. Neither changes the share of the variance that moves with each administration, between 0 and 3.3% whichever indicators are chosen (Section 4.3 of the article), and audits can fight manipulation but not the noise and the inherited conditions behind almost all wrongful and structural bans.

The second answer is that *targets would take into account the size and the socio-economic conditions of each municipality, comparing it with similar ones in its region*. Comparing like with like removes part of the place effect, and the floors simulated here already compare municipalities of the same size group. The bottom of every group, however, is flagged every year, and noise remains: with a red band set further below the floor, 1.7% of bans are attributable to the mayor, and without the "no improvement" rule 1.2% (Section S8.1).

The third answer is that *the Fiscal Responsibility Law is the precedent*. The LRF sanctions acts that the mayor controls and that are recorded in the municipality's own accounts, such as exceeding the limits on personnel spending (arts. 20 to 23) or failing to report. Outcomes produced jointly by the mayor, the state government, the economy and chance lie outside it. A sanction for results is a different category, which is why ineligibility, where it exists, sanctions conduct (Section S10.2).

The fourth answer is that *persecution of politicians already happens, only on a basis that is neither republican nor focused on performance*. The scale would be new. About 24 bans in every 100 mayors per mandate, triggered by indicators computed by a federal agency, however autonomous its board, could give whoever controls that agency a channel to shape who may run, in a country where federal governments already steer discretionary transfers towards aligned mayors (Brollo and Nannicini, 2012). The fifth answer is that *the centrão would be used in the reform's favour*. As the chain above shows, that would require the votes of parties whose brokerage the conditions on amendments would weaken, even though the package as a whole would leave their weight in the Chamber almost unchanged.

S11. Learning whether the law works

Section 11.2 of the article proposes a way to learn whether a version of the law works. This section gives the evidence behind that proposal: how large regression to the mean is in real data (Section S11.1), the design of a pilot and the power it would have (Section S11.2 and Figure S11.1), and other routes to evaluation that the data allow (Section S11.3).

S11.1. Regression to the mean in real data

Whatever version of the law is enacted, its effects will be hard to see, for two reasons. A law applied everywhere at once leaves no comparison group, and the municipalities it flags would improve by themselves. The second problem can be measured in real data, with no law at all, by following the municipalities that a scorecard would have flagged. Among the 1,078 municipalities below the 20th percentile of their population group in literacy in 2023, literacy improved by 11.7 percentage points more than in the others by 2024 (1.22 SD), and 46% of them left the flagged group. In IDEB, the same comparison over the two-year intervals from 2017 to 2025 gives an average relative gain of 0.29 points, with 45% of the flagged municipalities leaving the group. For homicides, the rates of the corresponding flagged group fell by 19 per 100,000 more than the others in one year (1.77 SD), on average over the pairs of years from 2010 to 2016, and 48% of them left the group. These spurious improvements exceed the simulated effect of the whole package (Table 5 of the article), and a before-and-after evaluation would credit the law with them.

S11.2. The design and power of a pilot

Two features of the problem point to a solution. The first is the staffing constraint of Section S8.5: commissioners cannot be deployed everywhere at once, so the order of deployment can be drawn by lot within strata of size, region and capacity. The second is the question that matters most for the design, namely whether conditioning money on results adds anything to the money itself and to assistance, which can be answered only by comparing municipalities that receive the same expected money under different rules. A pilot with four arms of equal expected money would do both: (A) a transfer without conditions; (B) the same expected transfer paid by results; (C) the transfer of A with technical assistance; and (D) the transfer of B with technical assistance. B against A measures the incentive, C against A the assistance, and D against the rest their complementarity, which is also a test of whether implementation depends on incentives (Section 9.1 of the article and Section S9.2). Because all arms receive the same expected money, the incentive measured by B against A is only the effort part of the tranche's effect, the part that an unconditional transfer would not obtain (Section S8.4).

Real data say how large the pilot must be (Figure S11.1). After controlling for the baseline and for strata of state and size, the standard deviation of two-year changes is 0.31 SD for a composite of literacy, daycare, formal jobs and sewage, and 1.25 SD for one-year changes in literacy alone, which are dominated by noise in small municipalities. With 1,200 municipalities, 300 per arm, a pilot detects a main effect of 0.051 SD on the composite and an interaction of 0.101 SD with 80% power at the 5% level; on the components taken one at a time, the detectable main effects are 0.08 SD for daycare, 0.04 for formal jobs and 0.07 for sewage.

The calibrated simulation predicts much smaller effects after two years: 0.031 SD for the incentive [0.009 to 0.060] and 0.025 for assistance [0.006 to 0.065], which builds up slowly and reaches 0.042 after four years. A two-year pilot of 1,200 municipalities would therefore detect neither. One of 2,400 municipalities, followed for a full mandate, would detect the assistance (minimum detectable effect 0.036 SD) but still not, with confidence, an incentive effect of the predicted size, and the interaction only if it were several times larger than the model predicts.

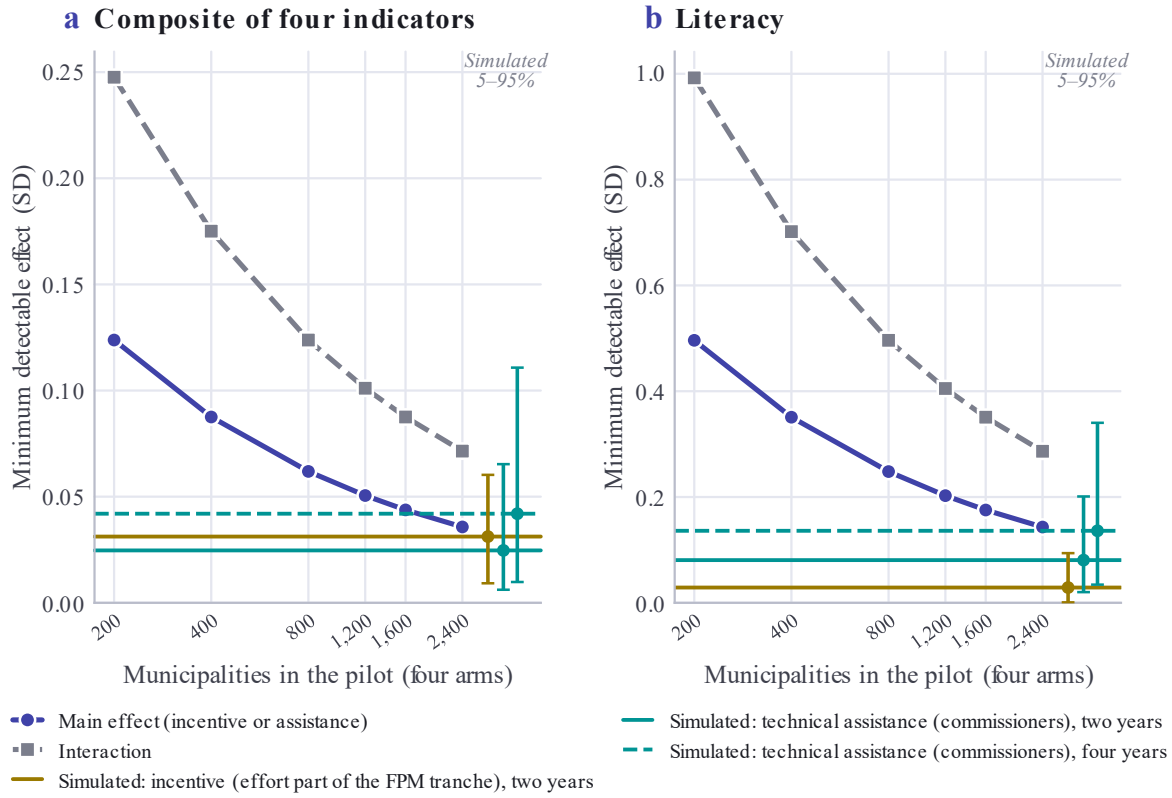


Figure S11.1. How large a pilot would have to be. Minimum detectable effect (80% power, 5% two-sided) of a pilot with four arms of equal size, by number of municipalities: main effects (incentive or assistance) and their interaction, for a composite of literacy, daycare, formal jobs and sewage **(a)** and for literacy alone **(b)**. Residual standard deviations of two-year changes (one-year for literacy) after baseline and state-by-size strata, real data. Horizontal lines: simulated effects of the incentive (the effort part of the FPM tranche) and of technical assistance after two years, and of assistance after four (medians over 400 calibrated draws); bars on the right: their 5th to 95th percentiles.

S11.3. Other routes to evaluation

Three further consequences follow for the evaluation. For literacy alone the detectable effect is 0.20 SD with 1,200 municipalities, so a pilot focused on learning should measure it with an audited test of its own instead of the national indicator. Because states adopt results-based ICMS rules at different dates, the evaluation can also use staggered difference-in-differences with not-yet-treated states as controls (Callaway and Sant'Anna, 2021), although with 26 states inference is fragile. And a pre-registered evaluation, run by an institution independent of the Comissariado and with an audited data pipeline, would let the law learn from its own first mandate, producing the kind of evidence that Brazil's audit lotteries produced for anti-corruption policy (Ferraz and Finan, 2008; Avis, Ferraz and Finan, 2018).

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